MICROSOFT EXCEL:

PRO TIPS FOR POWER USERS

75+ EXCEL TIPS TO HELP YOU WORK SMARTER, BOOST PRODUCTIVITY, AND BECOME A CERTIFIED POWER USER

★★★★★ *With Best-Selling Excel instructor* ***Chris Dutton***



EXCEL MAVEN

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COURSE STRUCTURE & OUTLINE

PRODUCTIVITY TIPS

- CTRL & ALT Shortcuts
- Data Validation
- Autofill & Flash Fill
- Cell Protection
- Advanced Sorting & Filtering
- Etc.

FORMATTING TIPS

- Frozen Panes
- Invisible Text
- Custom Number Formats
- Snap to Grid
- Formula Formatting
- Etc.

FORMULA TIPS

- Formula Auditing Tools
- Calculation Modes
- Fuzzy-Match Lookups
- Uniques & Duplicates
- INDEX & MATCH
- Etc.

VISUALIZATION TIPS

- Filled Maps
- Sparklines
- Dynamic Ranges
- Form Controls
- Custom Templates
- Etc.

PIVOT TABLE TIPS

- Date & Value Grouping
- Custom Sort Lists
- Slicers & Timelines
- Cached Source Data
- Conditional Formatting
- Etc.

ANALYTICS TIPS

- Goal Seek & Solver
- Data Modeling
- CUBE Functions
- Outlier Detection
- Monte Carlo Simulation
- Etc.

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SETTING EXPECTATIONS

1 I'm using **Microsoft Office 365 ProPlus**, for PC

- What you see on your screen **will not always match** what you see on mine, especially if you're using a different operating system or an older version of Excel

2 This course is **not designed** to serve as an intro to Excel

- Pro Tips are intended to showcase **specific tools and techniques**; for deeper dives, check out my other Excel & Power BI courses (Formulas, Charts, Pivots, Power Query, etc.)

3 This is a **non-linear** course, and can be taken in any order

- Each demo is **self-contained**, so feel free to skip ahead or focus on specific areas

4 Questions? Comments? **Let's talk.**

- If you need support along the way, feel free to post a question or reach out directly

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PRODUCTIVITY TIPS

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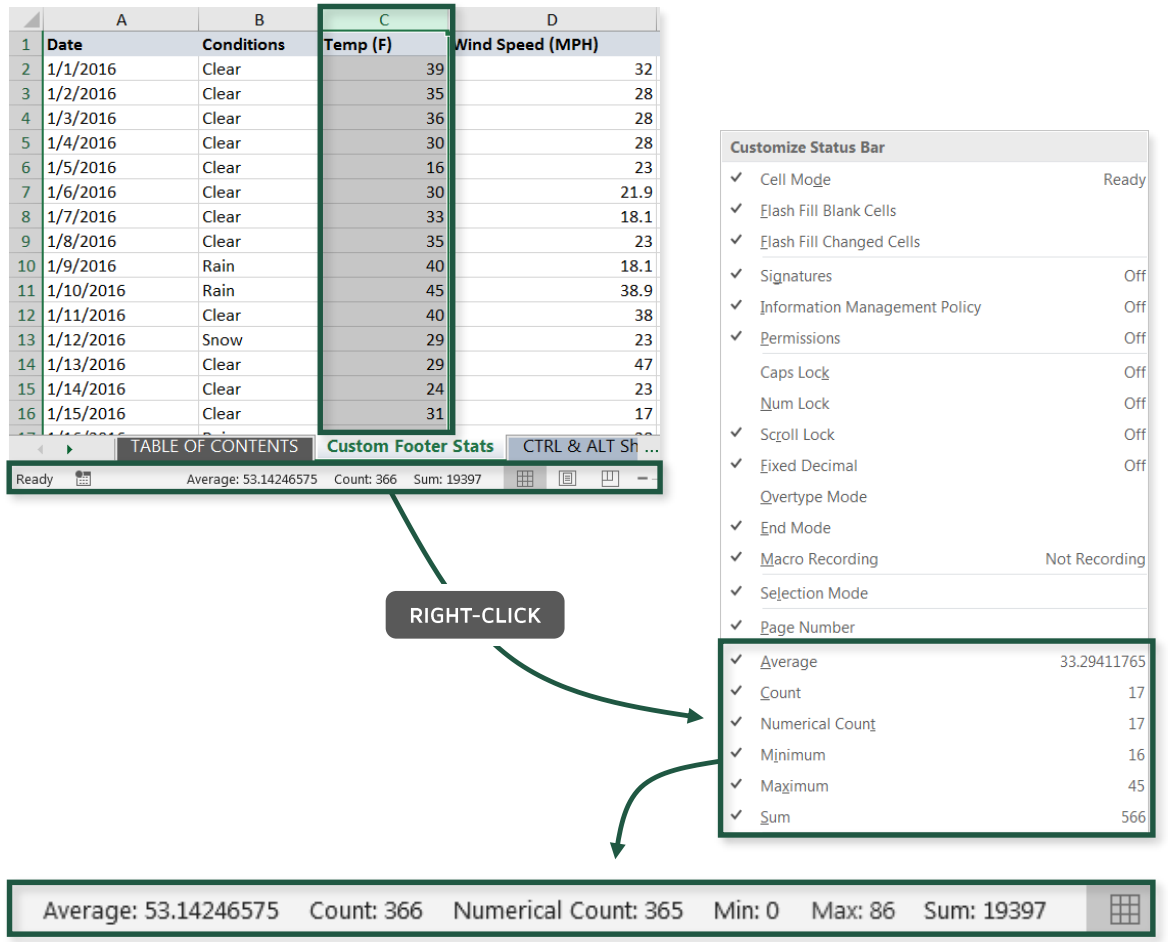
PRO TIP

CUSTOMIZE YOUR WORKBOOK FOOTER STATS

- The Excel footer bar typically shows the **Sum**, **Average**, and **Count** of a selected range of values, by default
- **Right-click** the footer to customize your options, and display additional stats like **Numerical Count**, **Max**, and **Min**, as well as other notifications *(Flash Fill results, Caps Lock, Macro status, etc.)*

PRODUCTIVITY

1 STAR (VERY BASIC)



COMMON USE CASES:

- *Quickly generating summary statistics without using formulas*
- *Adding a status indicator to display while macros are in recording mode*

PRO TIP

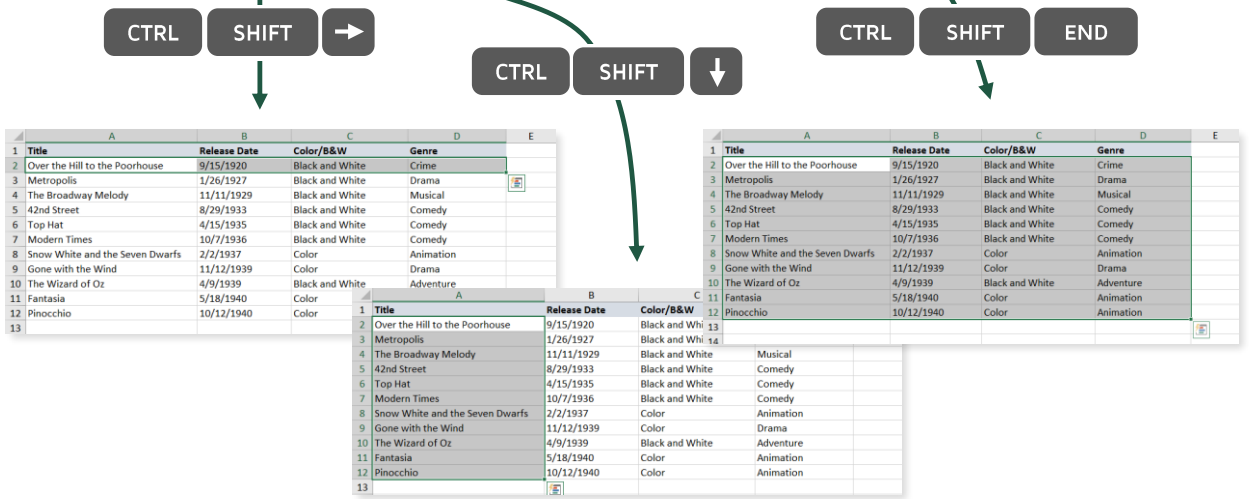
QUICKLY NAVIGATE SHEETS WITH CTRL SHORTCUTS

- **CTRL-ARROW:** Jump to the edge of a contiguous range (***command+arrow*** *on a Mac*)
  - Hold **SHIFT** to select cells at the same time
- **CTRL-HOME** & **CTRL-END:** Jumps to the top-left or bottom-right cell in a range (***control+fn+arrow*** *on a Mac*)
- **CTRL-PgUp** & **CTRL-PgDn:** Switches between tabs (***option+arrow*** *on a Mac*)
- **CTRL-G:** Launches the "**Go-To**" menu, containing any named cell, ranges or tables

PRODUCTIVITY

1 STAR (VERY BASIC)

	A	B	C	D	E
1	**Title**	**Release Date**	**Color/B&W**	**Genre**	
2	Over the Hill to the Poorhouse	9/15/1920	Black and White	Crime	
3	Metropolis	1/26/1927	Black and White	Drama	
4	The Broadway Melody	11/11/1929	Black and White	Musical	
5	42nd Street	8/29/1933	Black and White	Comedy	
6	Top Hat	4/15/1935	Black and White	Comedy	
7	Modern Times	10/7/1936	Black and White	Comedy	
8	Snow White and the Seven Dwarfs	2/2/1937	Color	Animation	
9	Gone with the Wind	11/12/1939	Color	Drama	
10	The Wizard of Oz	4/9/1939	Black and White	Adventure	
11	Fantasia	5/18/1940	Color	Animation	
12	Pinocchio	10/12/1940	Color	Animation	
13					



COMMON USE CASES:

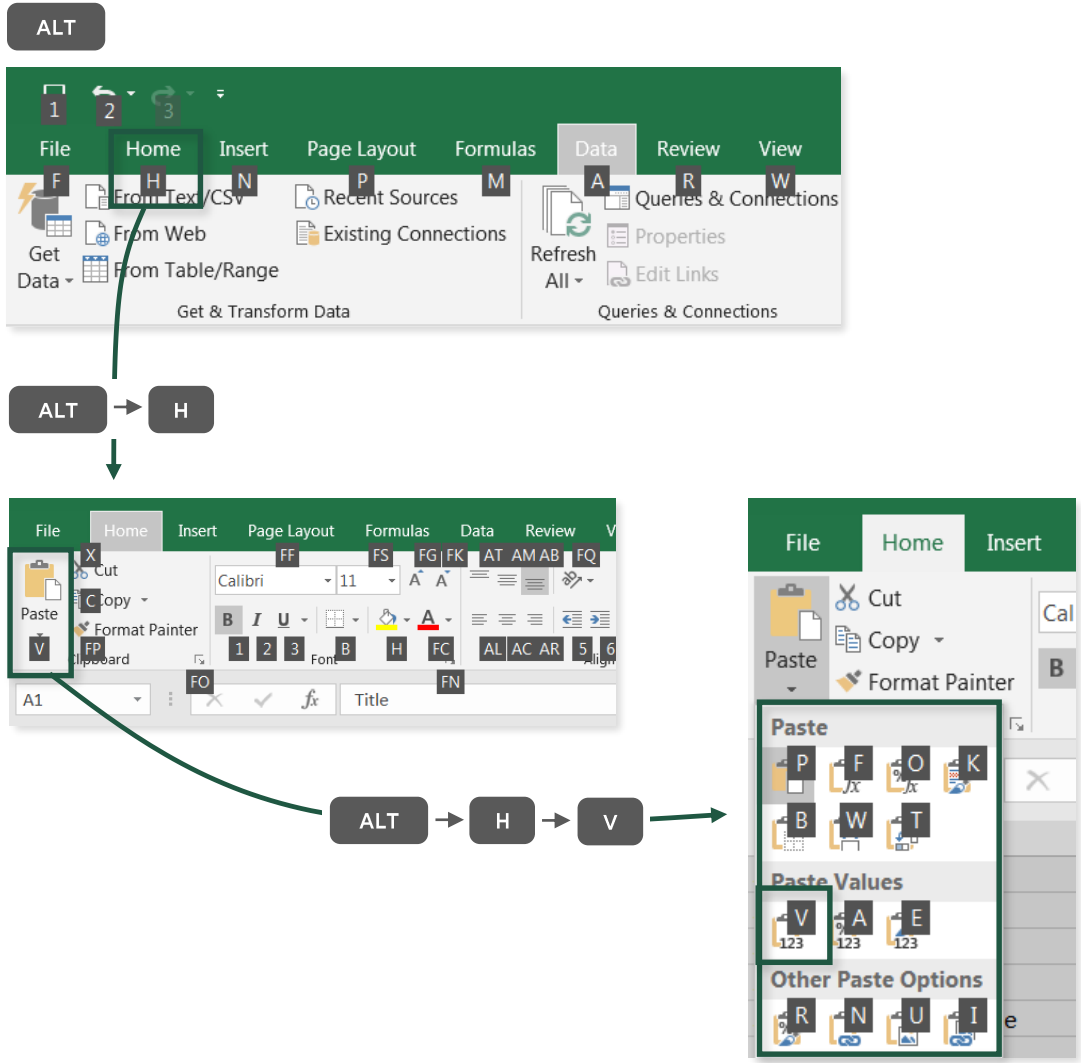
- *Selecting large tables or cell ranges without manually scrolling, or identifying the last active cell in a worksheet*

Windows *shortcuts:* http://bit.ly/2Cx0gCa
Mac *shortcuts:* http://bit.ly/2QwyqZQ

PRO TIP

ACCESS RIBBON CONTROLS WITH ALT KEY TIPS

- Press and release the **ALT key** to activate ribbon shortcuts, which allow you to navigate through menu options with simple keystrokes *(no mouse required!)*
  - **ALT → H → V → V**: Paste special as values
  - **ALT → A → T**: Add or remove filters
  - **ALT → H → E → F**: Clear formats
  - **ALT → N → V:** Insert a PivotTable
- **NOTE**: Alt key tips are not available for Mac



PRODUCTIVITY

1 STAR (VERY BASIC)

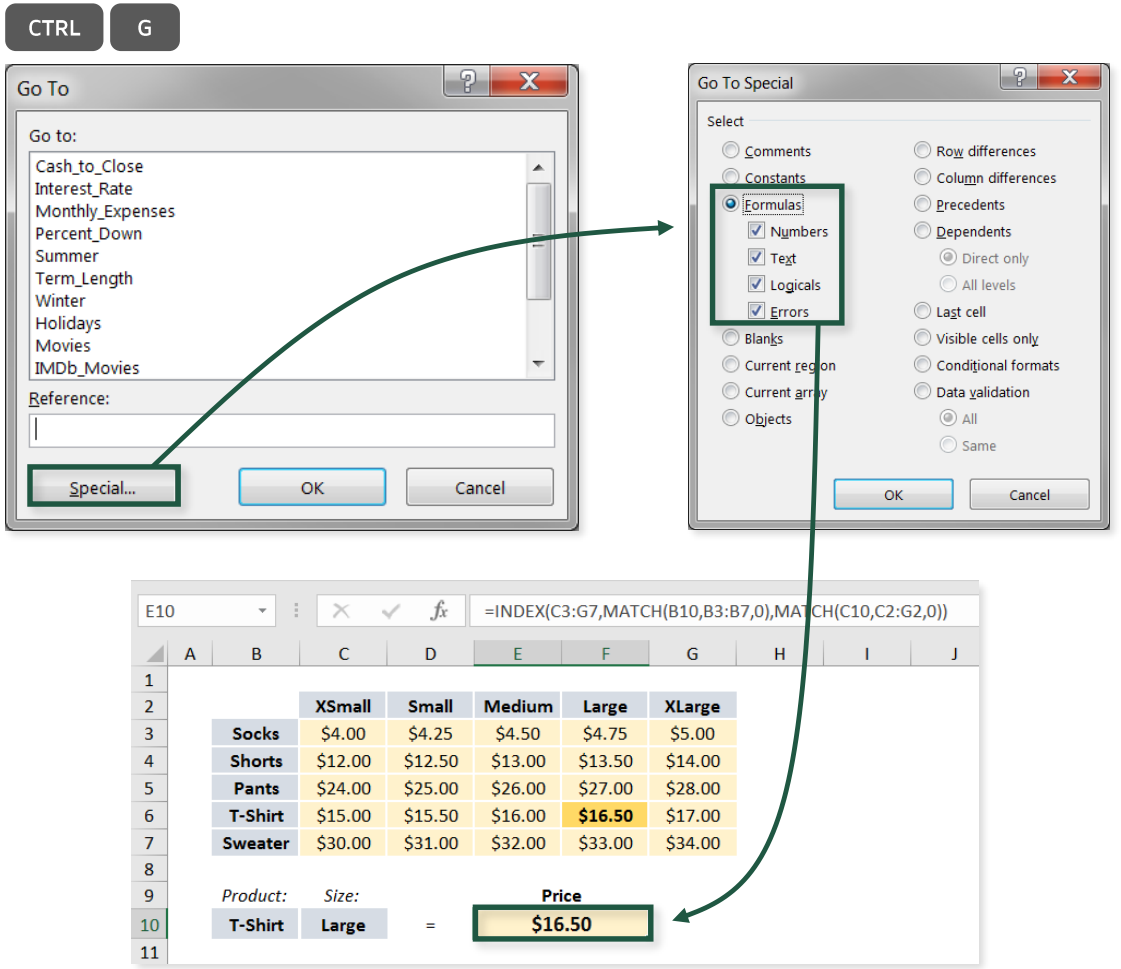
COMMON USE CASES:

- *Quickly accessing tools or commands that would normally require multiple clicks (i.e. paste special options)*

PRO TIP

SELECT SPECIAL CELL TYPES WITH "GO TO" OPTIONS

- **CTRL-G:** Launches the default "**Go To**" options, which allow you to jump to specific tables or named ranges in your workbook
- The **SPECIAL** menu includes additional options to select specific *types* of cells or objects in the sheet *(blanks, formulas, conditional formats, validation cells, etc.)*



COMMON USE CASES:

- *Quickly identifying or highlighting all cells containing formulas*
- *Selecting and deleting all objects in a worksheet with one click*
- *Identifying cells that have data validation rules applied (i.e. drop-downs)*

PRODUCTIVITY

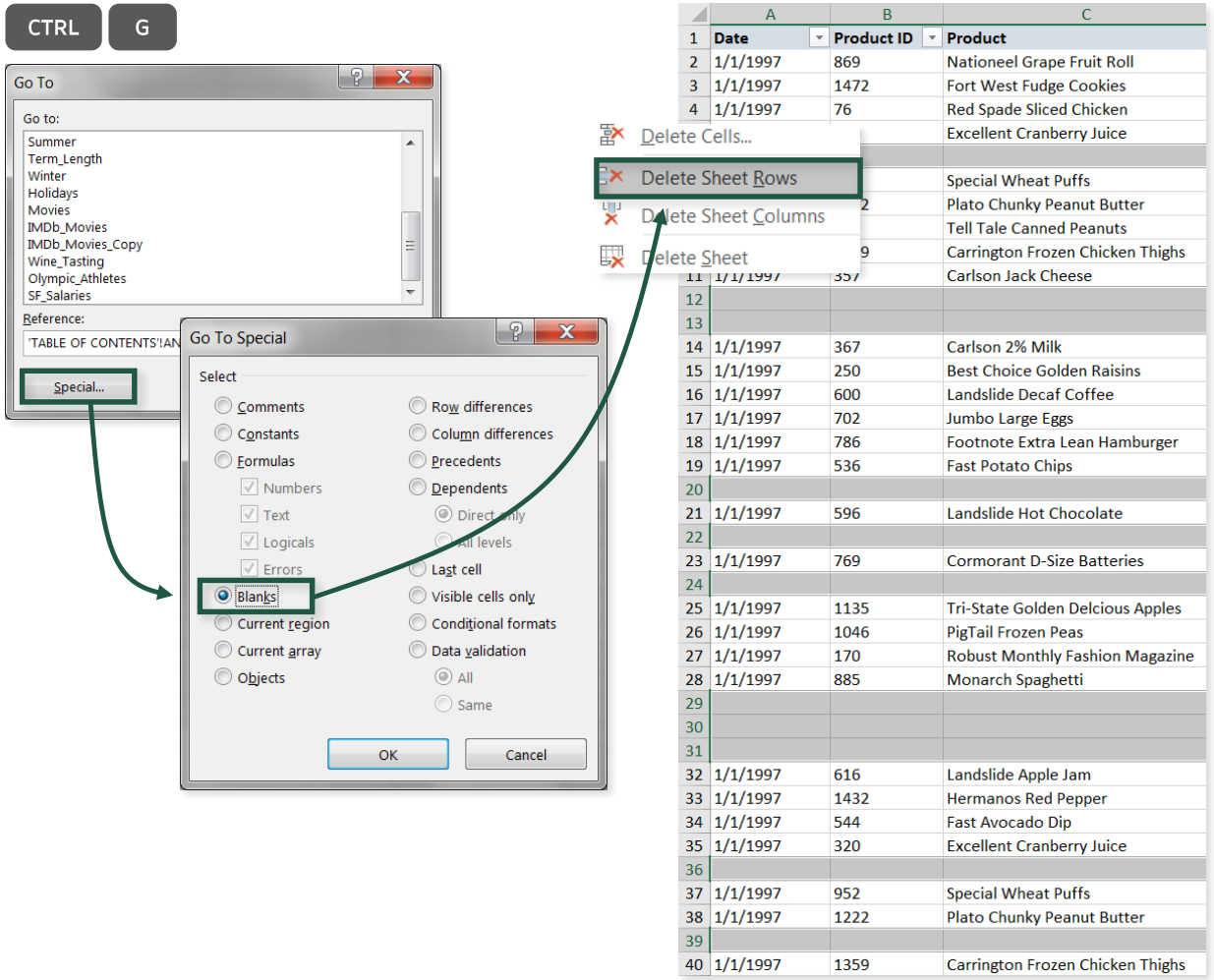
1 STAR (VERY BASIC)

PRO TIP

REMOVE ALL BLANK ROWS IN A WORKSHEET

- **STEP 1**: Use **CTRL-G** to launch the "**Go To**" menu, click **Special**, and select **Blanks** to select all blank rows in the sheet

- **STEP 2**: Delete the selected rows, using any of the following options:
  - ***Home > Delete > Delete Sheet Rows***
  - ***ALT-H-D-R***
  - ***CTRL+- > Shift Cells Up***



	A	B	C
1	Date	Product ID	Product
2	1/1/1997	869	Nationeel Grape Fruit Roll
3	1/1/1997	1472	Fort West Fudge Cookies
4	1/1/1997	76	Red Spade Sliced Chicken
5			Excellent Cranberry Juice
6			
7			Special Wheat Puffs
8		2	Plato Chunky Peanut Butter
9			Tell Tale Canned Peanuts
10		9	Carrington Frozen Chicken Thighs
11	1/1/1997	357	Carlson Jack Cheese
12			
13			
14	1/1/1997	367	Carlson 2% Milk
15	1/1/1997	250	Best Choice Golden Raisins
16	1/1/1997	600	Landslide Decaf Coffee
17	1/1/1997	702	Jumbo Large Eggs
18	1/1/1997	786	Footnote Extra Lean Hamburger
19	1/1/1997	536	Fast Potato Chips
20			
21	1/1/1997	596	Landslide Hot Chocolate
22			
23	1/1/1997	769	Cormorant D-Size Batteries
24			
25	1/1/1997	1135	Tri-State Golden Delcious Apples
26	1/1/1997	1046	PigTail Frozen Peas
27	1/1/1997	170	Robust Monthly Fashion Magazine
28	1/1/1997	885	Monarch Spaghetti
29			
30			
31			
32	1/1/1997	616	Landslide Apple Jam
33	1/1/1997	1432	Hermanos Red Pepper
34	1/1/1997	544	Fast Avocado Dip
35	1/1/1997	320	Excellent Cranberry Juice
36			
37	1/1/1997	952	Special Wheat Puffs
38	1/1/1997	1222	Plato Chunky Peanut Butter
39			
40	1/1/1997	1359	Carrington Frozen Chicken Thighs

COMMON USE CASES:

- *Cleaning up raw data by quickly eliminating extra/blank rows, without having to manually select or use query editing tools*

PRODUCTIVITY

2 STARS (BASIC)

PRO TIP

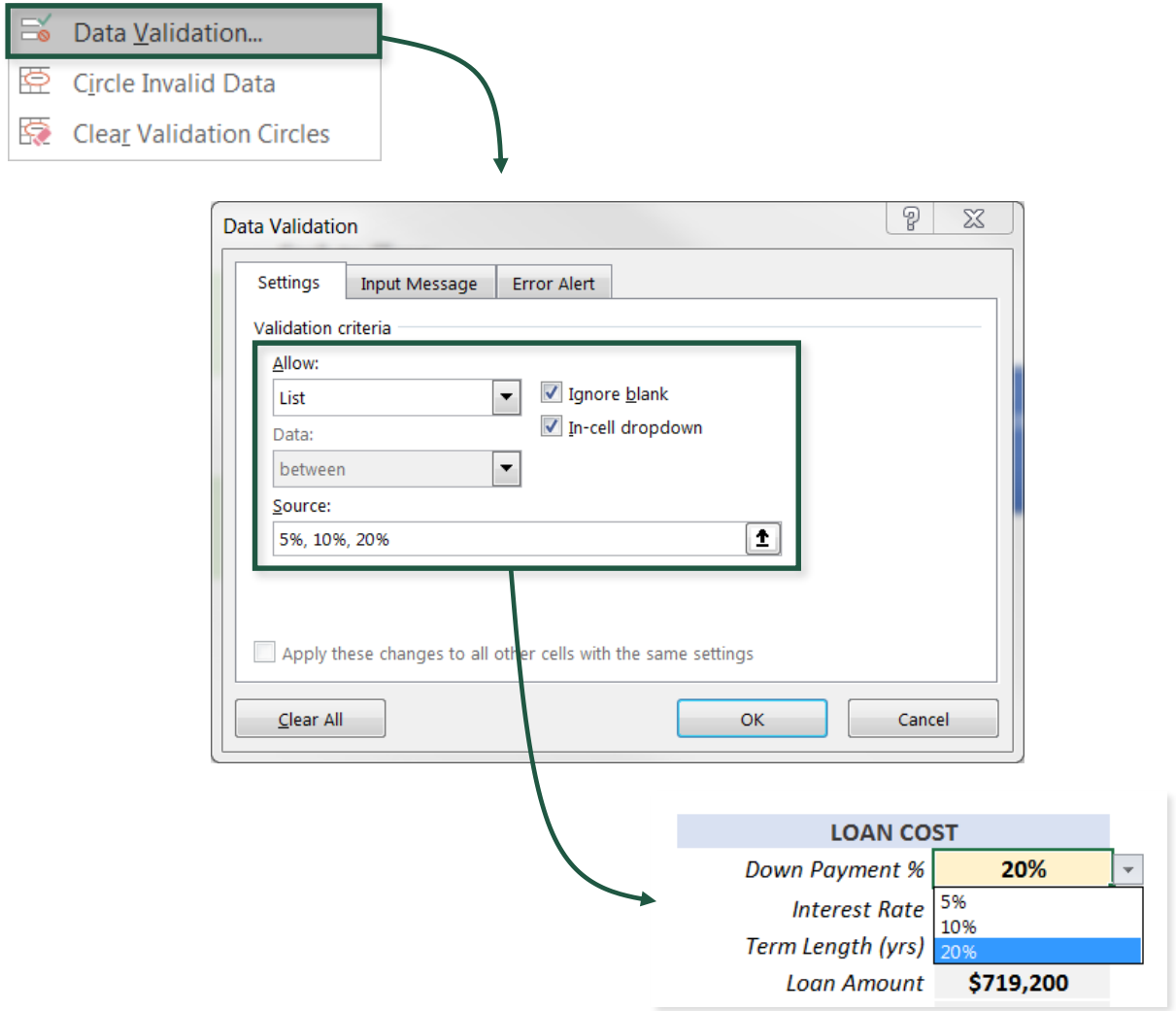
CREATE DROP-DOWN LISTS WITH DATA VALIDATION

- **Data Validation** allows you to limit the values that a particular cell will accept *(whole numbers, ranges, dates, text, etc.)*
- The **List** option allows you to create a drop-down menu containing specific items
  - ***Note:*** *Items can be typed directly or referenced as a cell range (i.e. A1:A10)*
- Add **Input Messages** or **Error Alerts** to customize what users see when they select the cell or enter invalid values

PRODUCTIVITY

2 STARS (BASIC)

(Data > Data Validation)



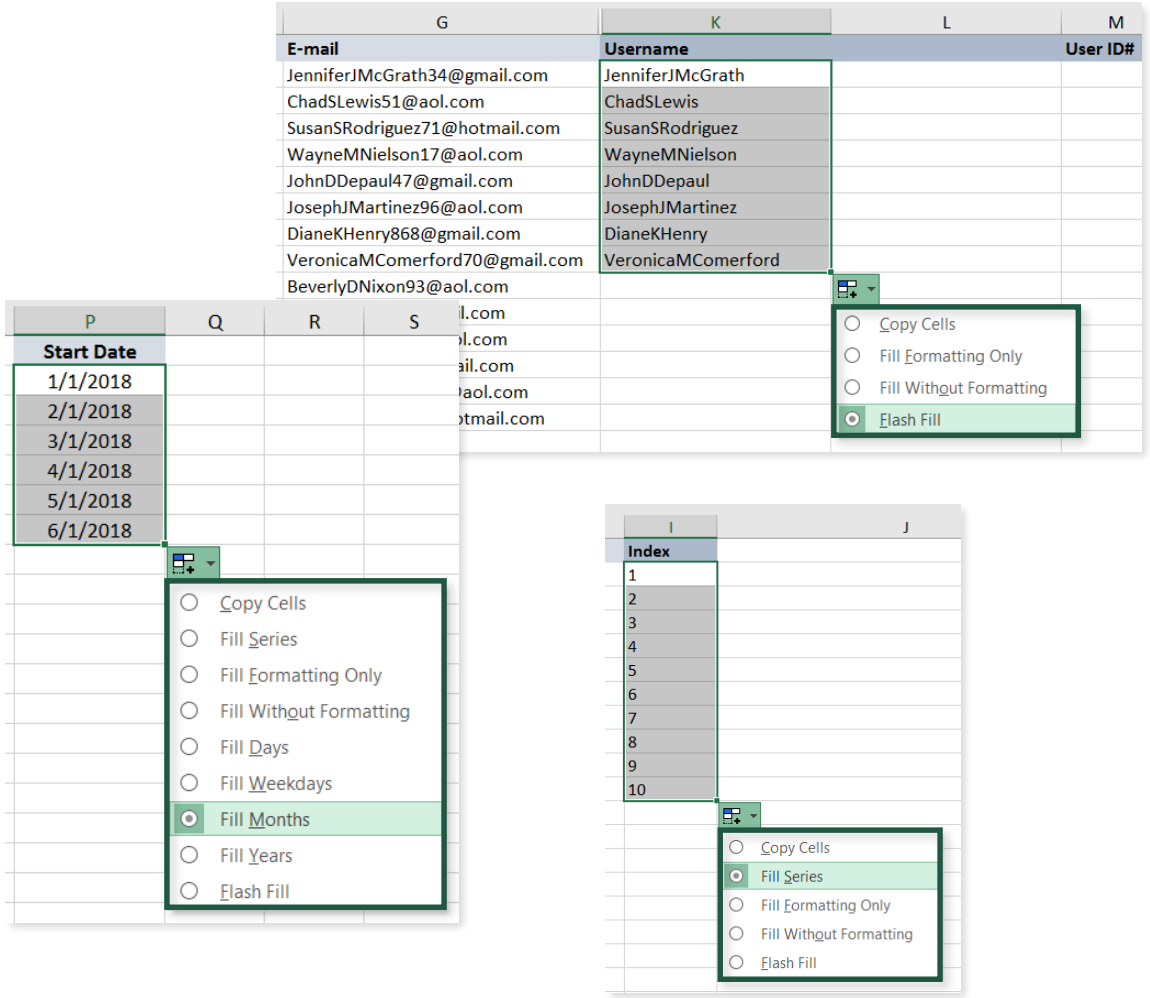
COMMON USE CASES:

- *Creating formula-based models with variable inputs*
- *Preventing users from entering invalid values (decimals, negatives, etc.)*

PRO TIP

POPULATE VALUES WITH AUTO FILL & FLASH FILL

- Basic **Auto Fill** options allow you to copy cells, add a sequential series, fill with or without formatting, or apply date intervals
  - Drag or double-click the **lower-right corner** of a cell to access auto fill options and apply values to new rows
- **Flash Fill** identifies patterns based on a sample set of given values, and uses those patterns to populate the whole column



COMMON USE CASES:

- *Applying values to thousands of rows without dragging or copy/pasting*
- *Filling formulas down to new rows without overwriting existing formats*
- *Extracting text from strings that would be difficult to isolate using formulas*
- *Quickly creating sequential index columns or calendar tables*

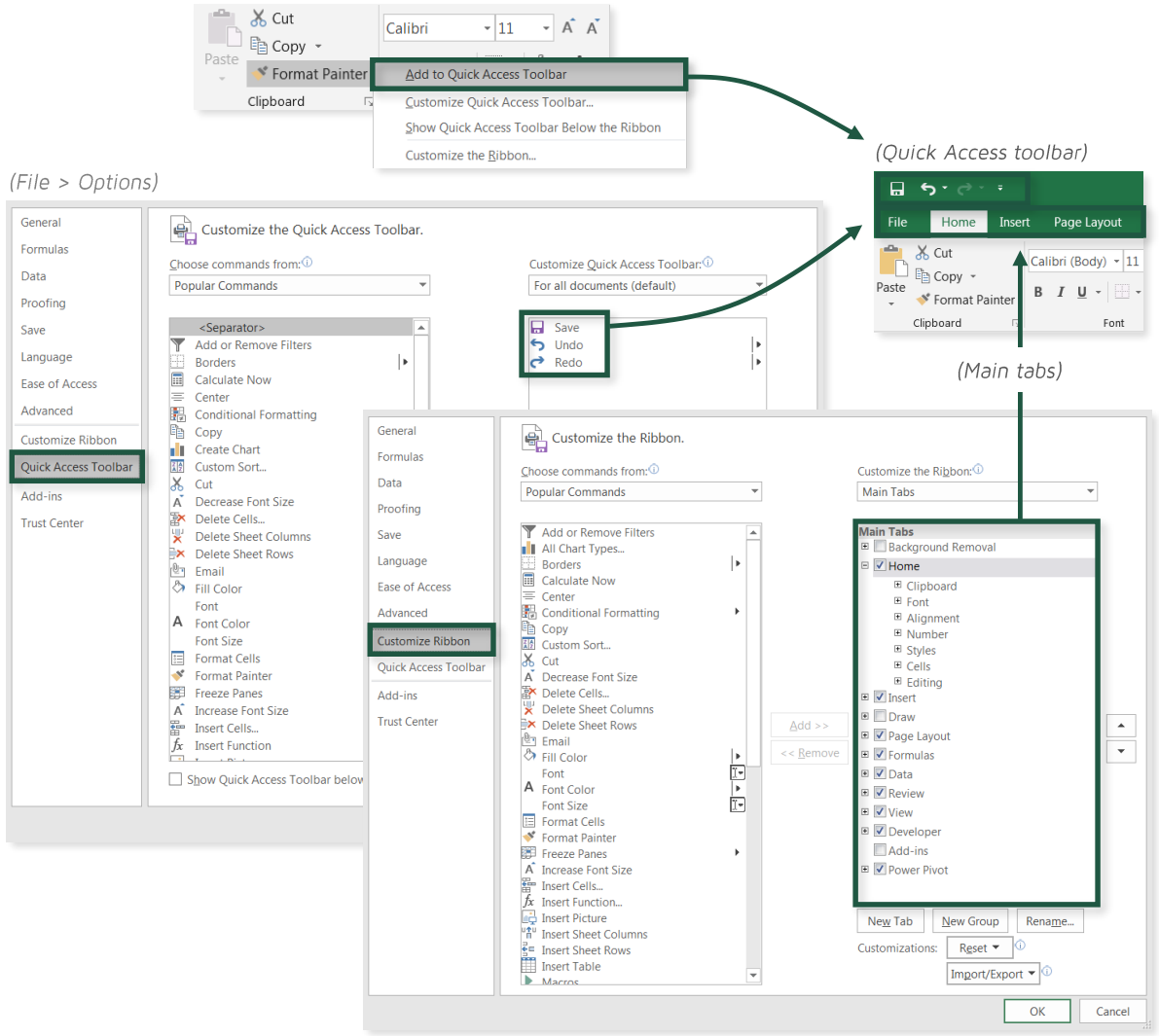
PRODUCTIVITY

2 STARS (BASIC)

PRO TIP

CUSTOMIZE THE RIBBON WITH YOUR OWN TABS

- Excel provides options to add, remove, or rearrange tabs and tools in the workbook ribbon (including the **Quick Access** toolbar)
  - **Quick Access** tools live above the ribbon, and include *Save*, *Undo* & *Redo* by default
- Use ***File*** > ***Options*** > ***Customize Ribbon*** to show, hide, or rearrange tabs in the ribbon, or to create your own custom tabs



PRODUCTIVITY

2 STARS (BASIC)

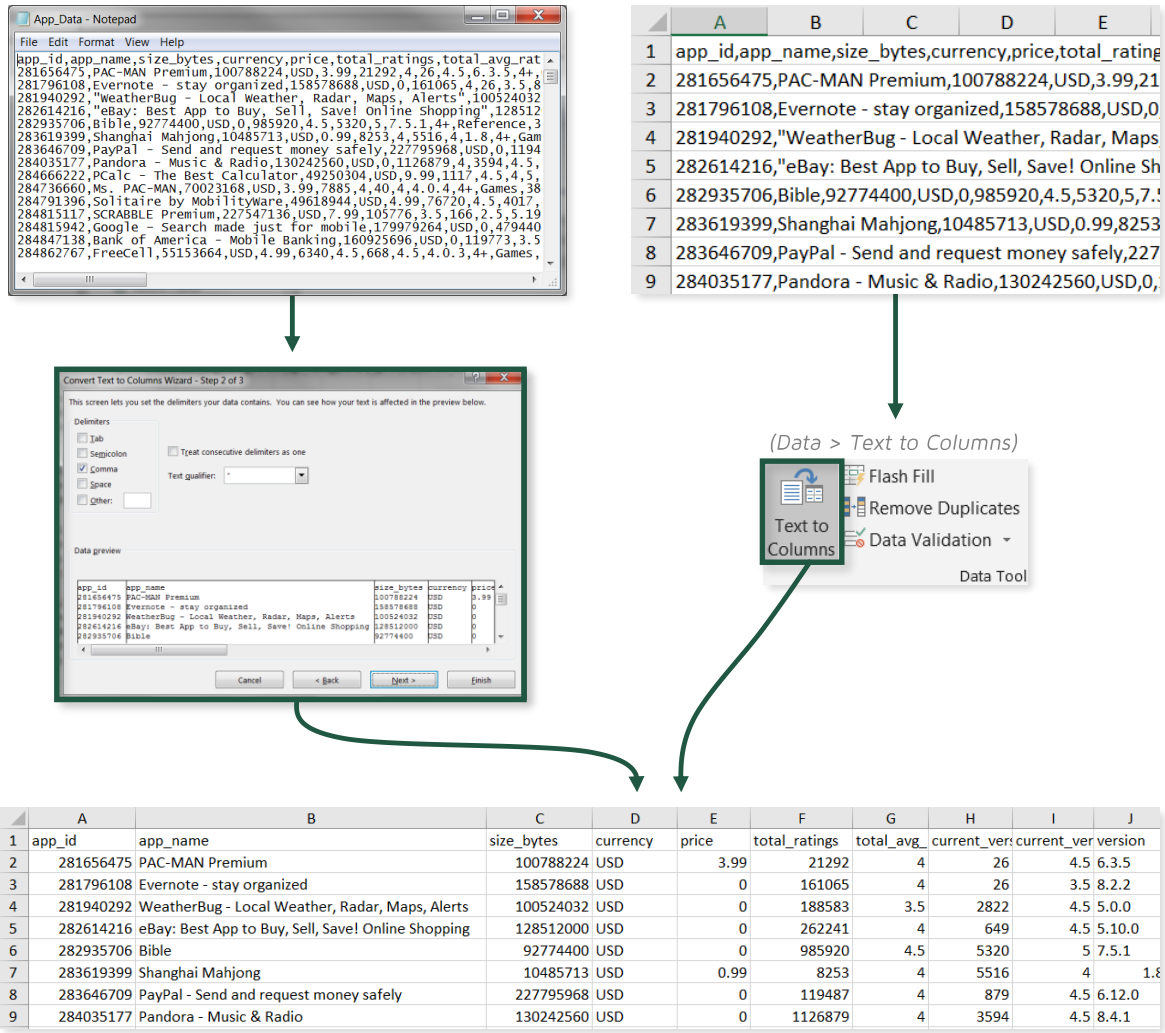
COMMON USE CASES:

- *Consolidating commonly used tools into a single, custom tab*
- *Exposing the **Developer** tab to access form controls or macros*

PRO TIP

SPLIT TEXT STRINGS USING TEXT TO COLUMNS

- If your data is stored as text or separated values (i.e. **.txt** or **.csv** files), use **Text to Column** to split values into columns & rows
  - **OPTION 1:** Open a txt file from Excel to launch the Text to Column wizard
  - **OPTION 2:** Open a csv file or copy data into one column, and use ***Data > Text to Column***
- **NOTE:** Both methods include options to specify delimiters, define text qualifiers, and format or exclude columns



COMMON USE CASES:

- *Transforming text-based data into tabular formats for analysis*
- *Splitting text strings without using formulas or functions*

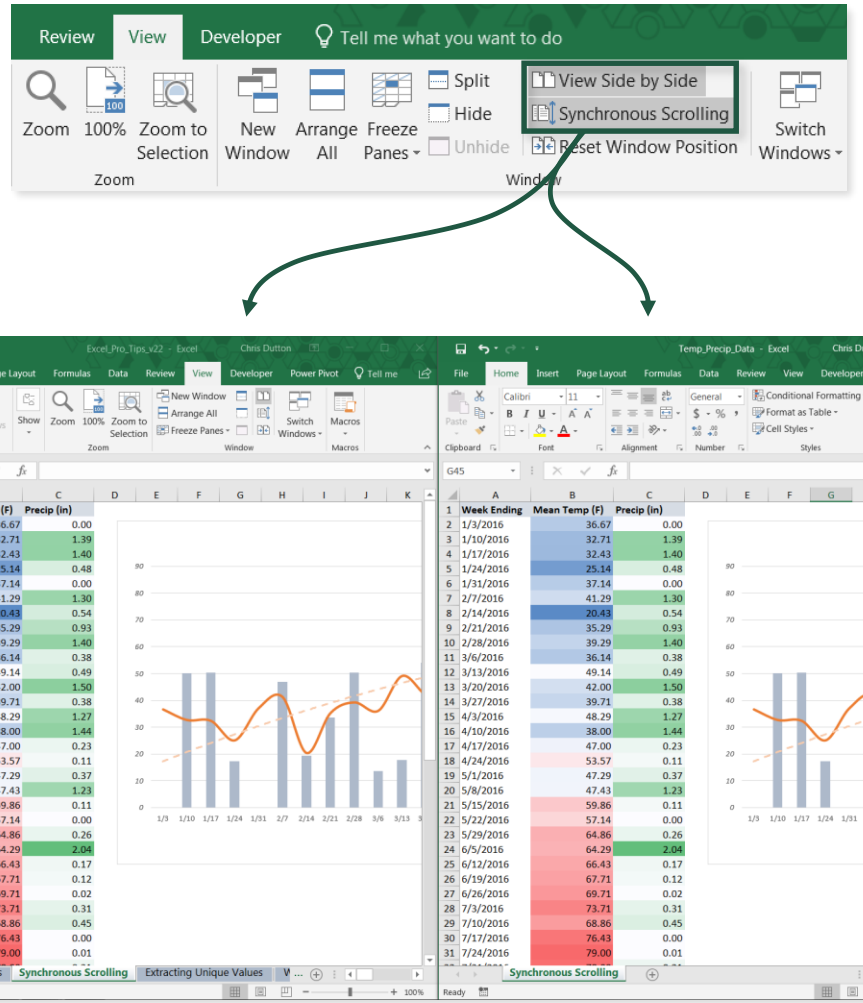
PRODUCTIVITY

2 STARS (BASIC)

PRO TIP

COMPARE FILES WITH SYNCHRONOUS SCROLLING

- **Synchronous scrolling** allows you to arrange Excel workbooks side-by-side and scroll through them simultaneously
  - **STEP 1**: Open both workbooks that you want to compare
  - **STEP 2**: Click ***View*** > ***View Side by Side*** in order to "stack" your windows
  - **STEP 3**: Activate ***View*** > ***Synchronous Scrolling*** to allow windows to scroll simultaneously



PRODUCTIVITY

2 STARS (BASIC)

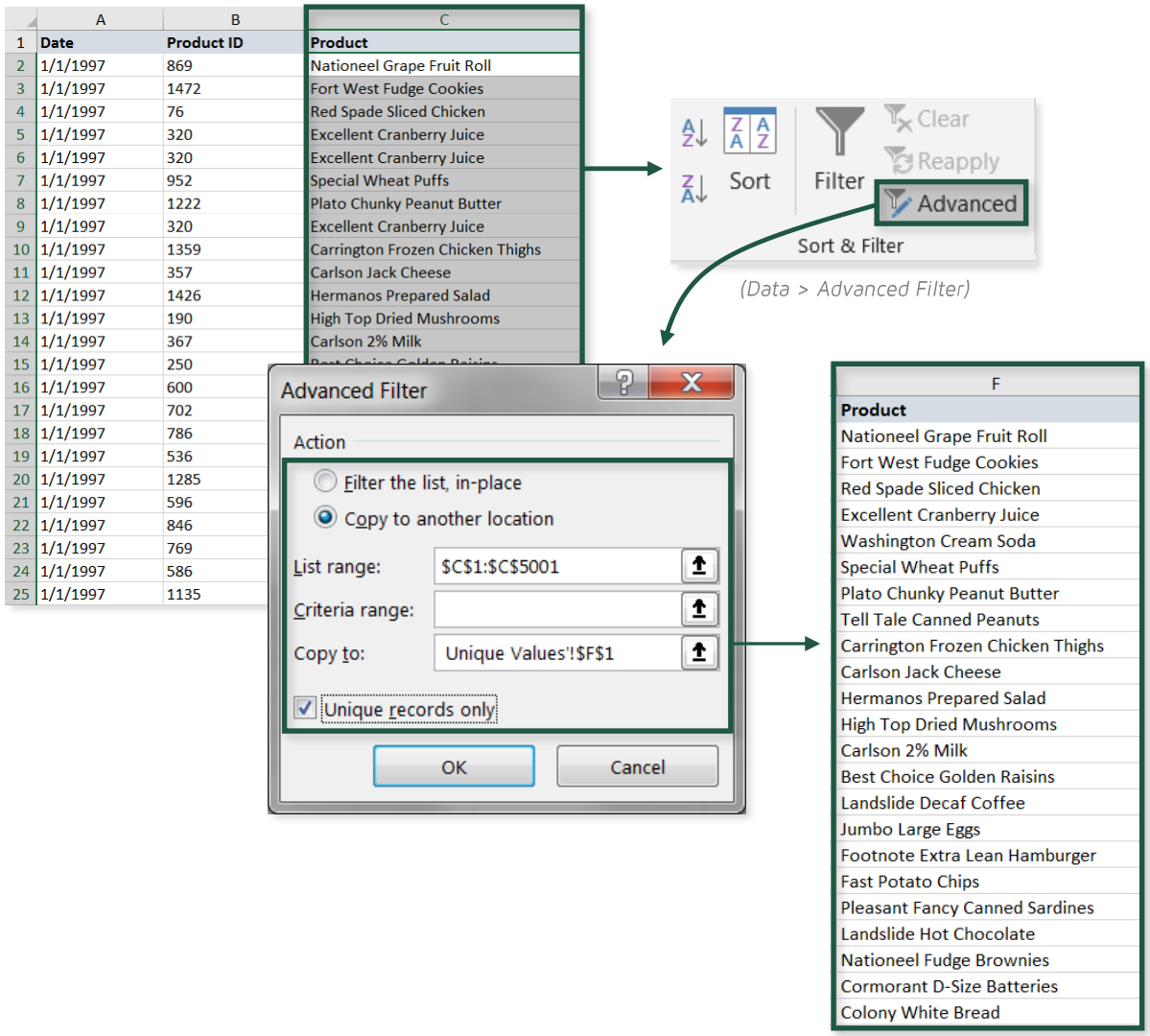
COMMON USE CASES:

- *Spot-checking workbooks for version control issues*
- *Making quick visual comparisons between similar files*

PRO TIP

EXTRACT UNIQUE VALUES WITH ADVANCED FILTERS

- There are several ways to remove duplicate values from a list in Excel, including the **Remove Duplicates** tool, **Advanced Filters**, **PivotTables** and **Power Query**
- The **Advanced Filter** is the most flexible option, allowing you to either filter in place or copy unique values to a new location
  - **NOTE:** To extract uniques into a new list, select "***Copy to another location***" and "***Unique records only***"



Date	Product ID	Product
1/1/1997	869	Nationeel Grape Fruit Roll
1/1/1997	1472	Fort West Fudge Cookies
1/1/1997	76	Red Spade Sliced Chicken
1/1/1997	320	Excellent Cranberry Juice
1/1/1997	320	Excellent Cranberry Juice
1/1/1997	952	Special Wheat Puffs
1/1/1997	1222	Plato Chunky Peanut Butter
1/1/1997	320	Excellent Cranberry Juice
1/1/1997	1359	Carrington Frozen Chicken Thighs
1/1/1997	357	Carlson Jack Cheese
1/1/1997	1426	Hermanos Prepared Salad
1/1/1997	190	High Top Dried Mushrooms
1/1/1997	367	Carlson 2% Milk
1/1/1997	250	Best Choice Golden Raisins
1/1/1997	600	Landslide Decaf Coffee
1/1/1997	702	Jumbo Large Eggs
1/1/1997	786	Footnote Extra Lean Hamburger
1/1/1997	536	Fast Potato Chips
1/1/1997	1285	Pleasant Fancy Canned Sardines
1/1/1997	596	Landslide Hot Chocolate
1/1/1997	846	Nationeel Fudge Brownies
1/1/1997	769	Cormorant D-Size Batteries
1/1/1997	586	Colony White Bread
1/1/1997	1135	Tell Tale Canned Peanuts

Product
Nationeel Grape Fruit Roll
Fort West Fudge Cookies
Red Spade Sliced Chicken
Excellent Cranberry Juice
Washington Cream Soda
Special Wheat Puffs
Plato Chunky Peanut Butter
Tell Tale Canned Peanuts
Carrington Frozen Chicken Thighs
Carlson Jack Cheese
Hermanos Prepared Salad
High Top Dried Mushrooms
Carlson 2% Milk
Best Choice Golden Raisins
Landslide Decaf Coffee
Jumbo Large Eggs
Footnote Extra Lean Hamburger
Fast Potato Chips
Pleasant Fancy Canned Sardines
Landslide Hot Chocolate
Nationeel Fudge Brownies
Cormorant D-Size Batteries
Colony White Bread

PRODUCTIVITY

2 STARS (BASIC)

COMMON USE CASES:

- *Identifying unique values while preserving the original list*
- *Creating a new lookup or dimension table containing unique primary keys*

PRO TIP

SIMPLIFY FORMULAS WITH NAMED RANGES & TABLES

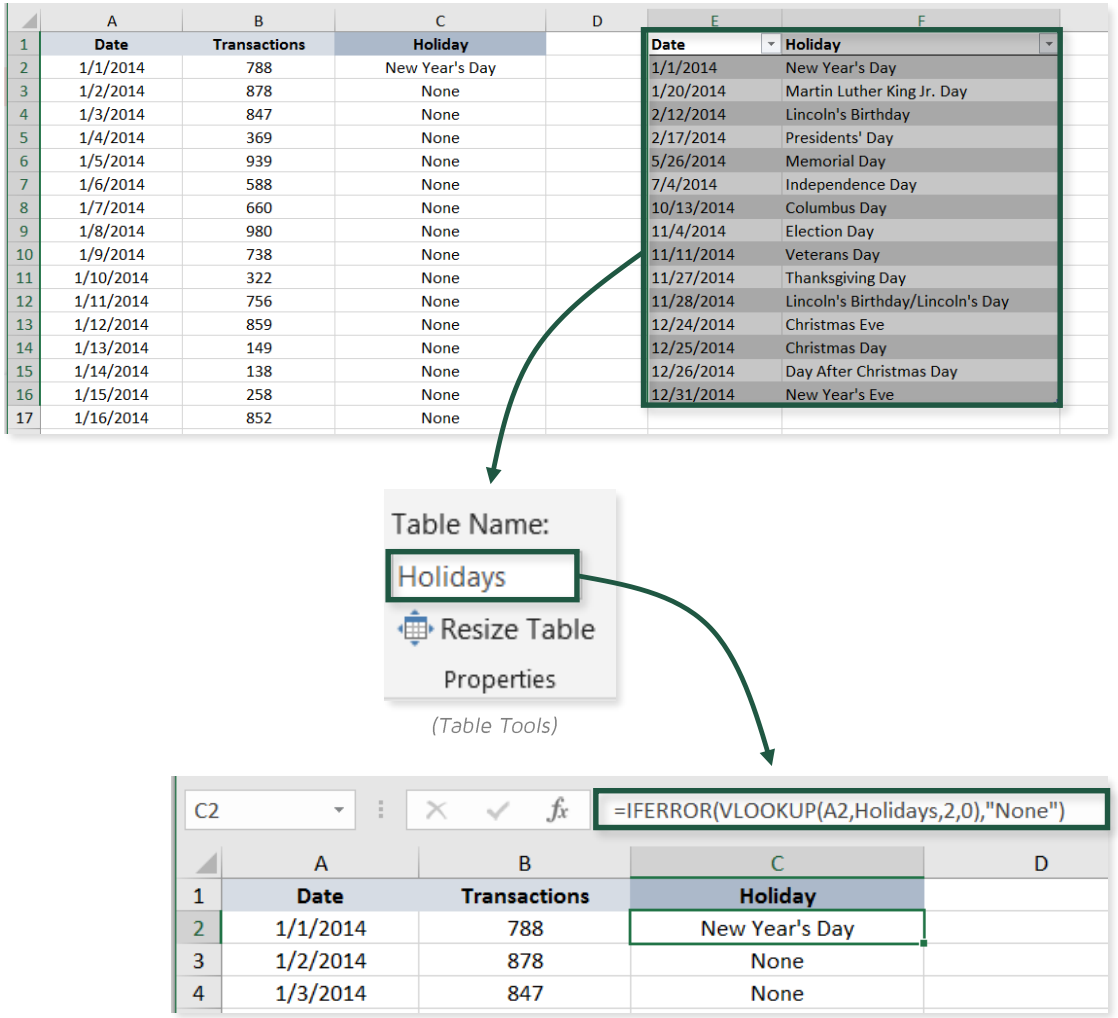
- Use **Named Ranges** or **Tables** (*vs. raw cell ranges*) to simplify formula references
- Compared to Named Ranges, formatting as a **Table** ***(CTRL-T)*** allows you to:
  1. Automatically ingest new rows of data
  2. Access quick formatting & filtering options (*fixed headers, banded rows, etc.*)
  3. Write efficient calculated columns vs. traditional (A1-style) cell formulas



PRODUCTIVITY



3 STARS (MODERATE)



COMMON USE CASES:

- *Replacing cell references with tables to make formulas easier to interpret*
- *Converting chart source data to enable automatic updates*
- *Preparing data for analysis with Excel BI tools (data model, Power Pivot)*

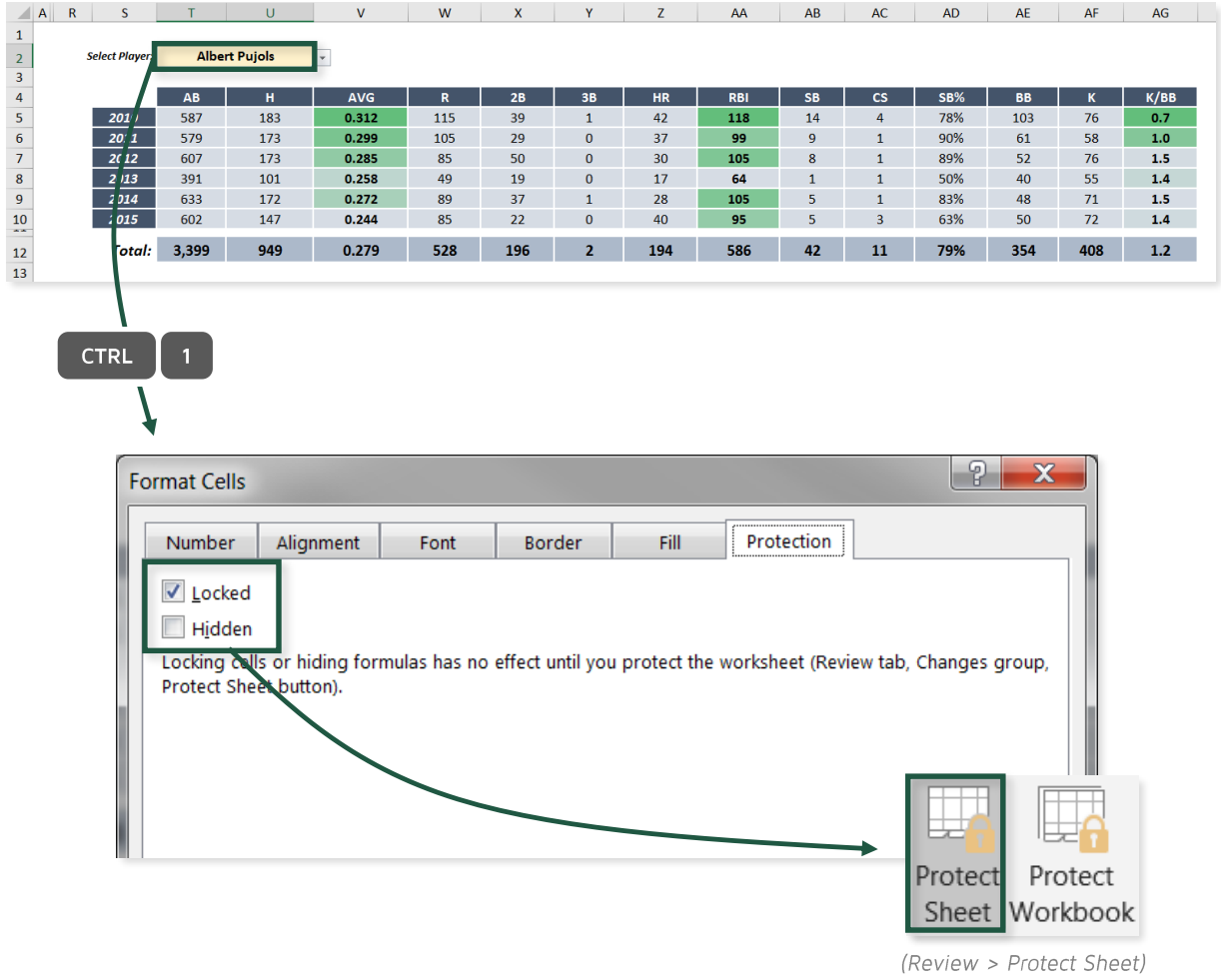
PRO TIP

ADD CUSTOM WORKBOOK PROTECTION SETTINGS

- The **Protection** tab in the **Format Cells** dialog box allows you to specify how cells behave once the worksheet is protected:
  - **Locked** means that users can view the cell contents but not edit (*default*)
  - **Hidden** means that users cannot see any underlying formulas or references
- **NOTE**: All cells are **locked** by default; you must actively *unprotect* the specific cells that you want to remain editable

PRODUCTIVITY

3 STARS (MODERATE)



(Review > Protect Sheet)

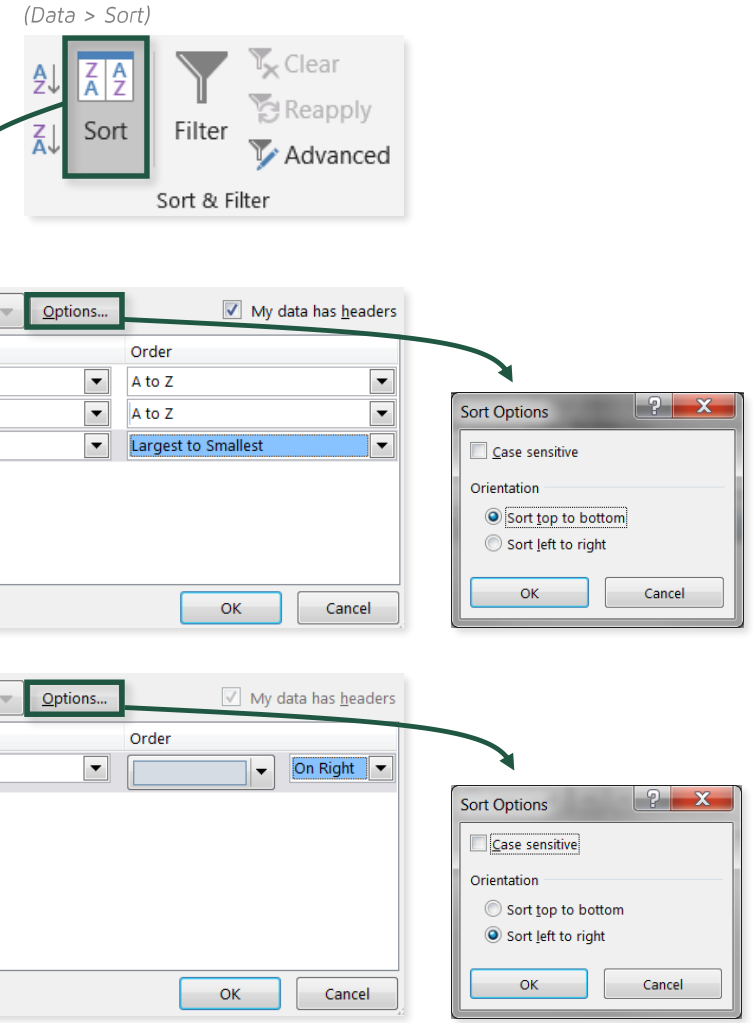
COMMON USE CASES:

- *Preventing users from accidentially modifying sensitive content*
- *Obscuring underlying formulas or cell references from view*
- *Limiting user interaction to a specific set of inputs or cells*

PRO TIP

APPLY MULTI-LEVEL ROW & COLUMN SORTING

- Instead of using column header options, use the **Sort** tool to apply **multi-level sorting**
  - **Example:** Sort a table alphabetically by *Country*, then *Province*, and finally by *Price*
- Select entire columns and use the ***Options*** menu to sort **Left to Right**, which allows you to **sort columns** instead of rows
  - **NOTE:** In addition to values, you can sort based on **cell color**, **font color**, or **icon**



PRODUCTIVITY

3 STARS (MODERATE)

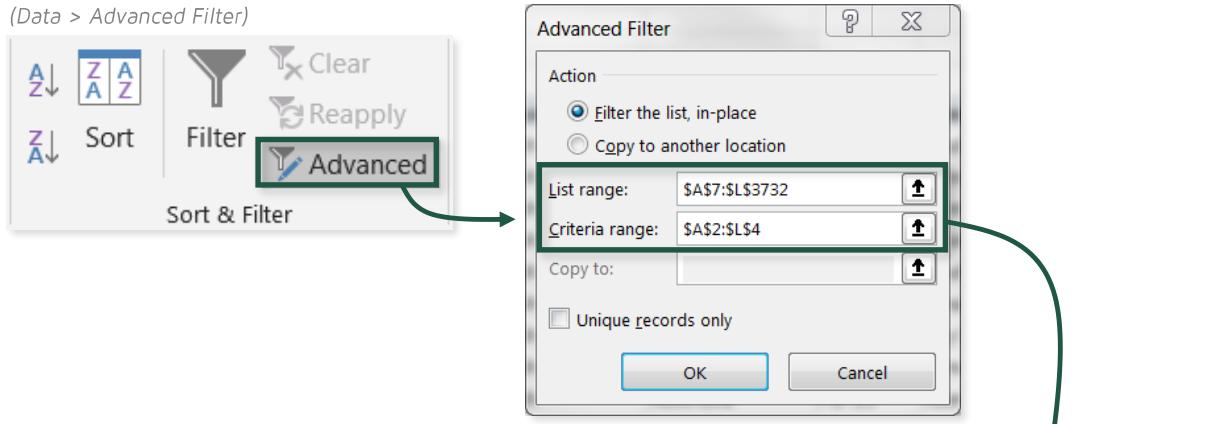
COMMON USE CASES:

- *Applying complex or custom sorting rules to a table or range*
- *Rearranging columns (alphabetically or by color) to organize a large table*

PRO TIP

EXPLORE TABLES WITH ADVANCED FILTERING

- **Advanced Filters** allow you to define a custom **criteria range**, which can contain complex combinations of filtering rules
  - Use advanced filtering to define rules that would be impossible with standard filters or PivotTables (*i.e. filtering on Action films rated 8.5+ plus Biography films rated 8+*)
- **NOTE**: Advanced filters iterate through all rows of the table, and can be *very slow* for large tables or very broad criteria



	A	B	C	D	E	F	G	H	I
1	*Filter Criteria:*								
2	**Title**	**Release Year**	**Color/B&W**	**Genre**	**Language**	**Country**	**Rating**	**Lead Actor**	**IMDb Score**
3	The*			Biography					>8
4				Action					>8.5
5									
6									
7	**Title**	**Release Year**	**Color/B&W**	**Genre**	**Language**	**Country**	**Rating**	**Lead Actor**	**IMDb Score**
118	Seven Samurai	1954	B&W	Action	Japanese	Japan	Unrated	Takashi Shimura	8.7
119	The Lord of the Rings: The Fellowship of th	2001	Color	Action	English	New Zealand	PG-13	Christopher Lee	8.8
209	The Dark Knight	2008	Color	Action	English	USA	PG-13	Christian Bale	9
211	Star Wars: Episode IV - A New Hope	1977	Color	Action	English	USA	PG	Harrison Ford	8.7
219	The Lord of the Rings: The Return of the K	2003	Color	Action	English	USA	PG-13	Orlando Bloom	8.9
224	The Lord of the Rings: The Two Towers	2002	Color	Action	English	USA	PG-13	Christopher Lee	8.7
237	Inception	2010	Color	Action	English	USA	PG-13	Leonardo DiCaprio	8.8
239	Star Wars: Episode V - The Empire Strikes I	1980	Color	Action	English	USA	PG	Harrison Ford	8.8
260	Saving Private Ryan	1998	Color	Action	English	USA	R	Tom Hanks	8.6
310	The Matrix	1999	Color	Action	English	USA	R	Keanu Reeves	8.7
1323	The Pianist	2002	B&W	Biography	English	France	R	Emilia Fox	8.5
1339	The Sea Inside	2004	Color	Biography	Spanish	Spain	PG-13	BelÃ©n Rueda	8.1
1342	The Imitation Game	2014	Color	Biography	English	UK	PG-13	Benedict Cumberbatch	8.1
1379	The Act of Killing	2012	Color	Biography	Indonesian	UK	Not Rated	Anwar Congo	8.2
1395	The Wolf of Wall Street	2013	Color	Biography	English	USA	R	Leonardo DiCaprio	8.2

COMMON USE CASES:

- *Applying custom or complex filtering rules that cannot be replicated with standard filter tools or PivotTables*
- *Adding a user-facing criteria range to provide transparency into the filter settings that have been applied to a table*

PRODUCTIVITY

★★★★★ **5 STARS** (EXPERT)

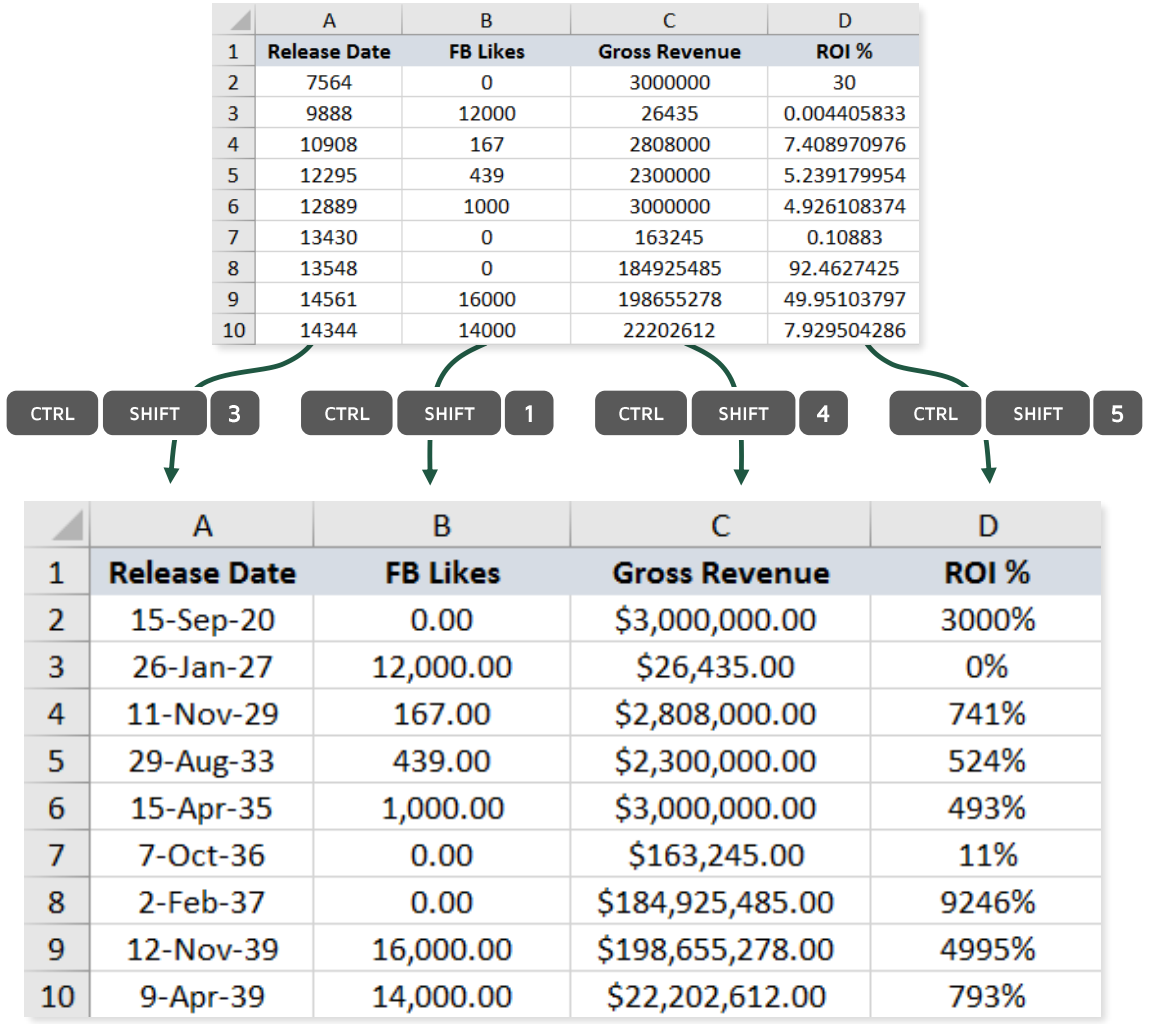
FORMATTING TIPS

EXCEL MAVEN

PRO TIP

USE CTRL TO APPLY FORMATTING SHORTCUTS

- Access common number formats using **CTRL shortcuts:**
  - **CTRL-SHIFT-1** (*CTRL-!*): **Number**
  - **CTRL-SHIFT-2** (*CTRL-@*): **Time**
  - **CTRL-SHIFT-3** (*CTRL-#*): **Date**
  - **CTRL-SHIFT-4** (*CTRL-$*): **Currency**
  - **CTRL-SHIFT-5** (*CTRL-%*): **Percentage**
  - **CTRL-SHIFT-6** (*CTRL-^*): **Scientific**
- **CTRL-SHIFT-~** (*tilde*) reverts to General



	A	B	C	D
1	Release Date	FB Likes	Gross Revenue	ROI %
2	7564	0	3000000	30
3	9888	12000	26435	0.004405833
4	10908	167	2808000	7.408970976
5	12295	439	2300000	5.239179954
6	12889	1000	3000000	4.926108374
7	13430	0	163245	0.10883
8	13548	0	184925485	92.4627425
9	14561	16000	198655278	49.95103797
10	14344	14000	22202612	7.929504286

	A	B	C	D
1	Release Date	FB Likes	Gross Revenue	ROI %
2	15-Sep-20	0.00	$3,000,000.00	3000%
3	26-Jan-27	12,000.00	$26,435.00	0%
4	11-Nov-29	167.00	$2,808,000.00	741%
5	29-Aug-33	439.00	$2,300,000.00	524%
6	15-Apr-35	1,000.00	$3,000,000.00	493%
7	7-Oct-36	0.00	$163,245.00	11%
8	2-Feb-37	0.00	$184,925,485.00	9246%
9	12-Nov-39	16,000.00	$198,655,278.00	4995%
10	9-Apr-39	14,000.00	$22,202,612.00	793%

COMMON USE CASES:

- *Quickly applying common number formats without using the mouse or accessing the **Format Cells** dialog box*

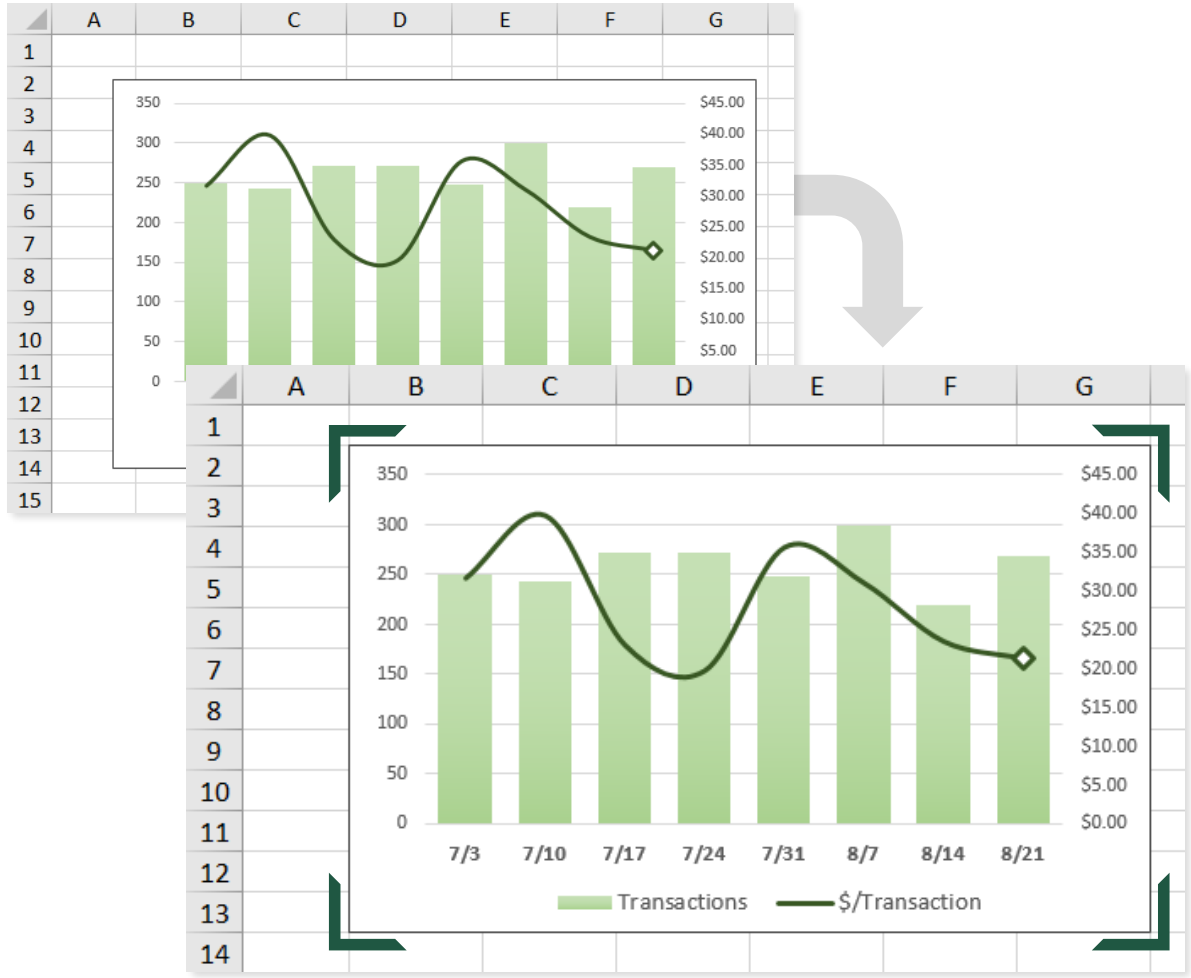
FORMATTING

1 STAR (VERY BASIC)

PRO TIP

ALIGN WORKBOOK OBJECTS USING SNAP TO GRID

- Hold the **ALT key** as you drag or resize objects to "*snap*" them to cell borders
- **BONUS TIPS:**
  - Hold **CTRL** as you drag to create a copy of the object at the same time
  - Hold **SHIFT** as you drag to limit movement to the horizontal or vertical plane



FORMATTING

1 STAR (VERY BASIC)

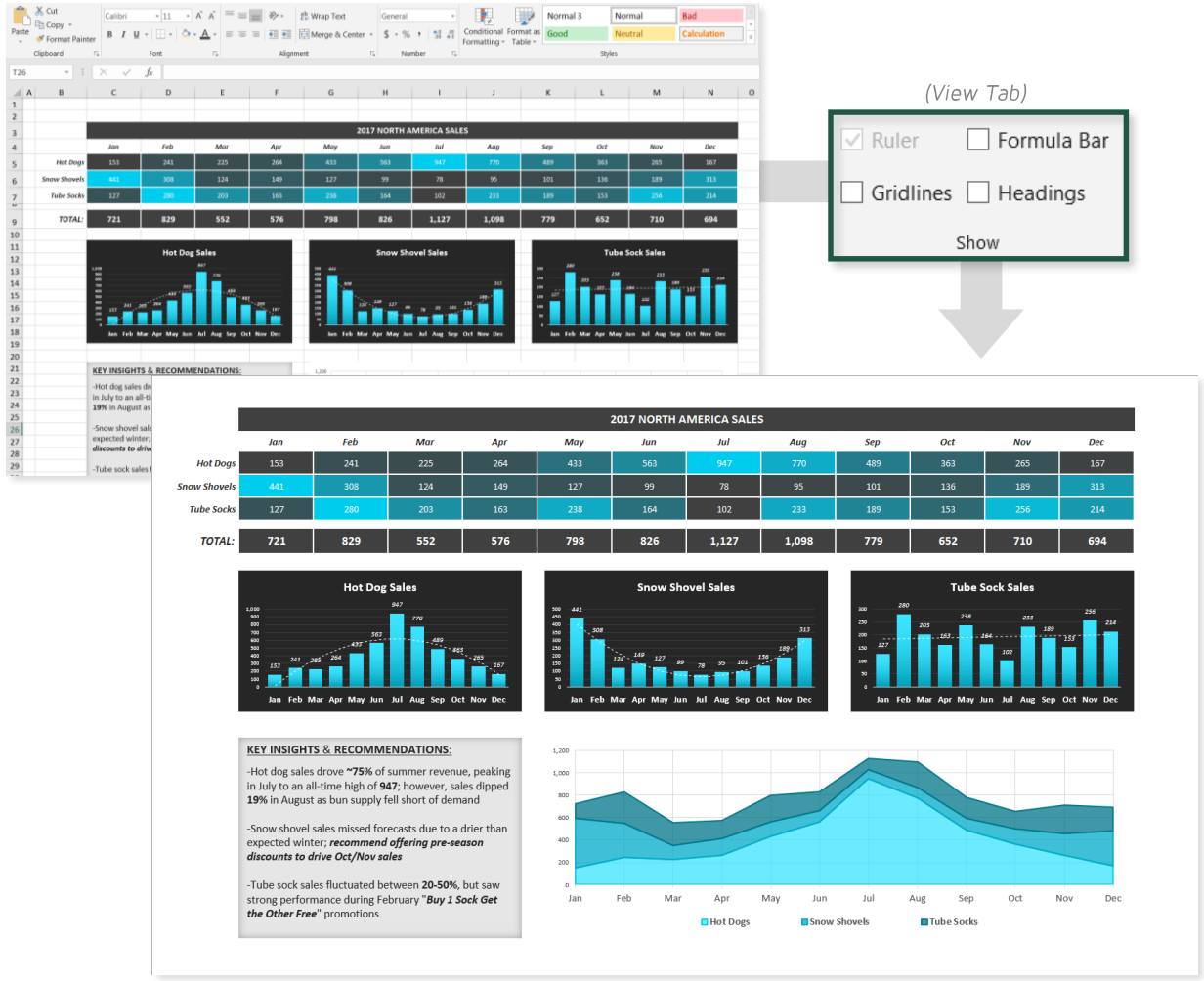
COMMON USE CASES:

- *Ensuring that all worksheet objects are aligned to a common grid*
- *Designing clean and polished user-facing tools or dashboards*

PRO TIP

HIDE WORKBOOK ELEMENTS TO REDUCE CLUTTER

- Use the ***View*** > ***Show*** checkboxes to hide workbook elements from view, including **Gridlines**, **Headings**, or the **Formula Bar**
- To collapse the ribbon from view, click the **caret icon (^)** in the lower right corner
  - To anchor the ribbon, select a tab and click the **"pin" icon** (*where the caret used to be*)
- **NOTE:** Gridlines and headings are specific to individual sheets, but formula bar and ribbon settings impact the entire workbook



2017 NORTH AMERICA SALES	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Hot Dogs	153	241	225	264	433	563	947	770	489	363	265	167
Snow Shovels	441	308	124	149	127	99	78	95	101	136	189	313
Tube Socks	127	280	203	163	238	164	102	233	189	153	256	214
TOTAL:	721	829	552	576	798	826	1,127	1,098	779	652	710	694

FORMATTING

1 STAR (VERY BASIC)

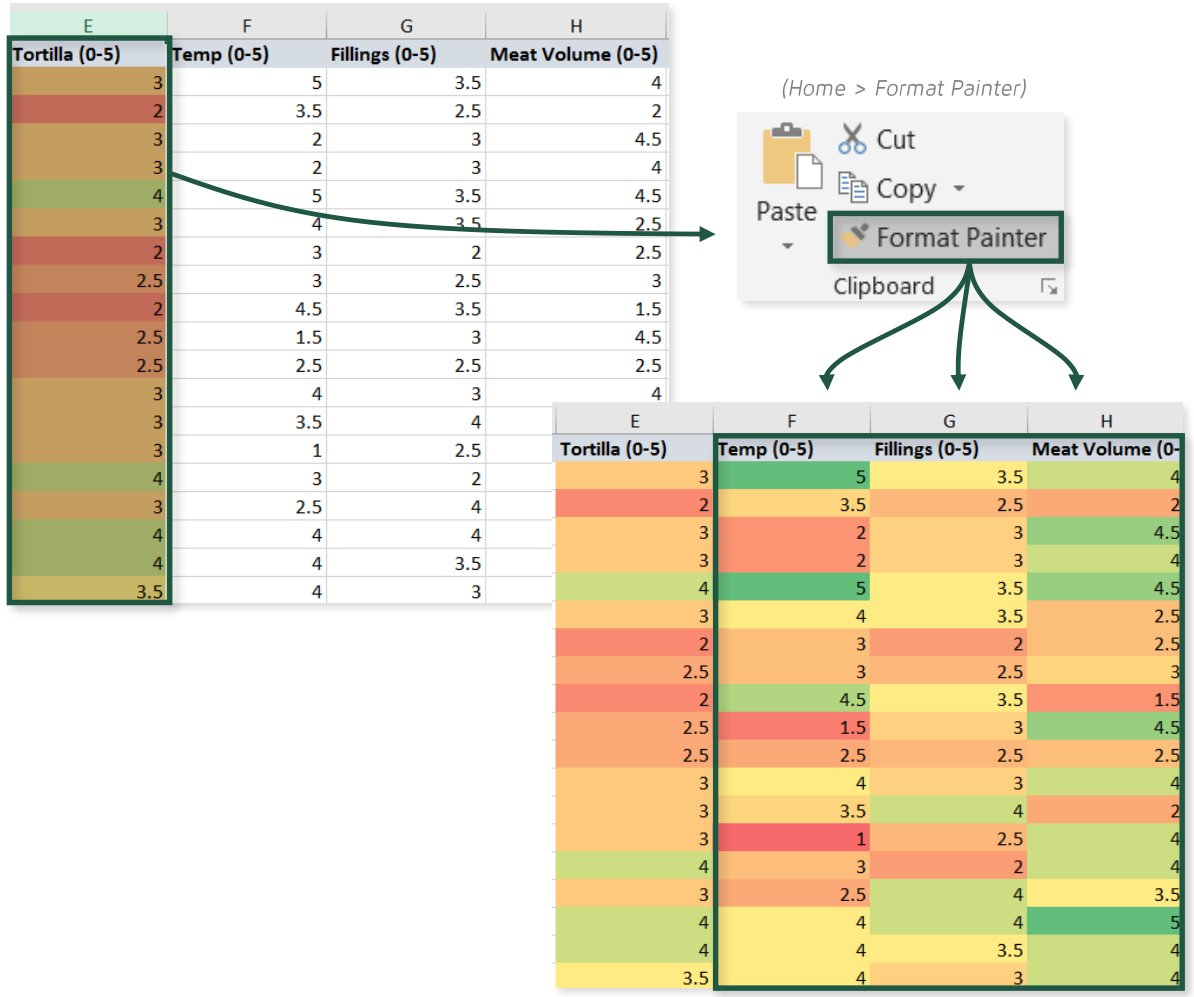
COMMON USE CASES:

- *Creating clean and polished "dashboard-style" views*
- *Preventing users from modifying rows, columns, or formulas*

PRO TIP

REPLICATE FORMATS WITH THE FORMAT PAINTER

- Use the **Format Painter** to copy all formatting settings (*fill, number format, borders, font styles, alignment, etc.*) from one cell or range of cells to another
- Click the format painter once to copy formats one time; **double click** if you want to copy formats multiple times (*until you click the format painter again to deactivate*)



E	F	G	H
Tortilla (0-5)	Temp (0-5)	Fillings (0-5)	Meat Volume (0-5)
3	5	3.5	4
2	3.5	2.5	2
3	2	3	4.5
3	2	3	4
4	5	3.5	4.5
3	4	3.5	2.5
2	3	2	2.5
2.5	3	2.5	3
2	4.5	3.5	1.5
2.5	1.5	3	4.5
2.5	2.5	2.5	2.5
3	4	3	4
3	3.5	4	
3	1	2.5	
4	3	2	
3	2.5	4	
4	4	4	
4	4	3.5	
3.5	4	3	

E	F	G	H
Tortilla (0-5)	Temp (0-5)	Fillings (0-5)	Meat Volume (0-
3	5	3.5	4
2	3.5	2.5	2
3	2	3	4.5
3	2	3	4
4	5	3.5	4.5
3	4	3.5	2.5
2	3	2	2.5
2.5	3	2.5	3
2	4.5	3.5	1.5
2.5	1.5	3	4.5
2.5	2.5	2.5	2.5
3	4	3	4
3	3.5	4	2
3	1	2.5	4
4	3	2	4
3	2.5	4	3.5
4	4	4	5
4	4	3.5	4
3.5	4	3	4

FORMATTING

1 STAR (VERY BASIC)

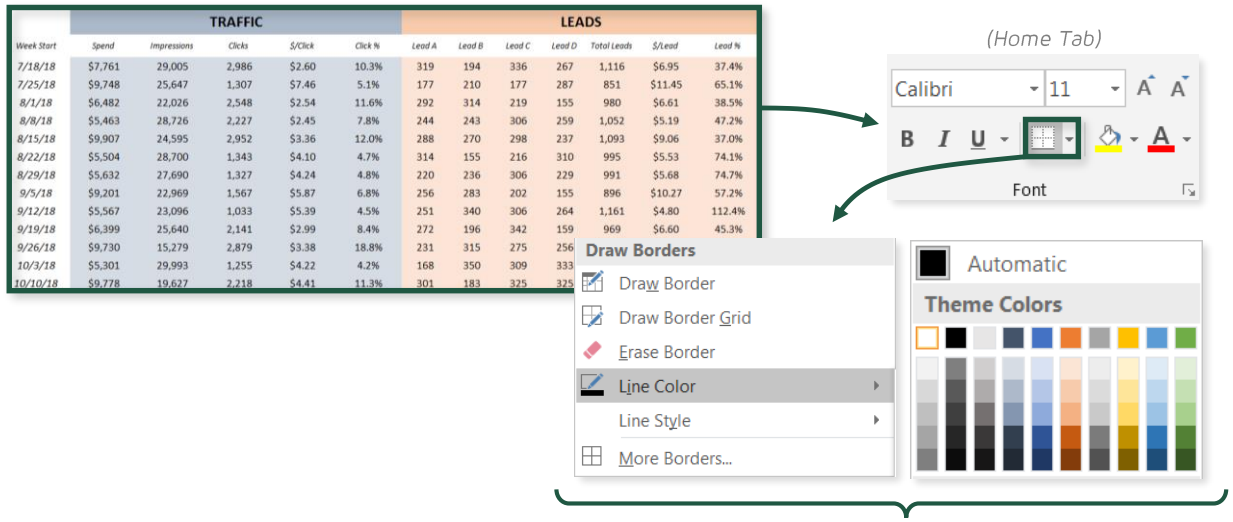
COMMON USE CASES:

- *Quickly applying formats across multiple columns without having to manually replicate each setting*
- *Ensuring that cell formats are exactly the same (borders, fill, number format, etc.)*

PRO TIP

USE COLOR & BORDERS TO CREATE POLISHED REPORTS

- **Fill colors** and **borders** aren't just stylistic tools; they can be used strategically to improve readability, create spacing, and draw attention to key metrics or trends
- Hide worksheet gridlines and add **white borders** to create a clean and polished look
  - Under the border options in the *Home* tab, use **Line Color** to set the border color and **Line Style** to determine the thickness



	TRAFFIC					LEADS						
Week Start	Spend	Impressions	Clicks	$/Click	Click %	Lead A	Lead B	Lead C	Lead D	Total Leads	$/Lead	Lead %
6/30/18	$7,761	29,005	2,986	$2.60	10.3%	319	194	336	267	1,116	$6.95	37.4%
7/7/18	$9,748	25,647	1,307	$7.46	5.1%	177	210	177	287	851	$11.45	65.1%
7/14/18	$6,482	22,026	2,548	$2.54	11.6%	292	314	219	155	980	$6.61	38.5%
7/21/18	$5,463	28,726	2,227	$2.45	7.8%	244	243	306	259	1,052	$5.19	47.2%
7/28/18	$9,907	24,595	2,952	$3.36	12.0%	288	270	298	237	1,093	$9.06	37.0%
8/4/18	$5,504	28,700	1,343	$4.10	4.7%	314	155	216	310	995	$5.53	74.1%
8/11/18	$5,632	27,690	1,327	$4.24	4.8%	220	236	306	229	991	$5.68	74.7%
8/18/18	$9,201	22,969	1,567	$5.87	6.8%	256	283	202	155	896	$10.27	57.2%
8/25/18	$5,567	23,096	1,033	$5.39	4.5%	251	340	306	264	1,161	$4.80	112.4%
9/1/18	$6,399	25,640	2,141	$2.99	8.4%	272	196	342	159	969	$6.60	45.3%
9/8/18	$9,730	15,279	2,879	$3.38	18.8%	231	315	275	256	1,077	$9.03	37.4%
9/15/18	$5,301	29,993	1,255	$4.22	4.2%	168	350	309	333	1,160	$4.57	92.4%
9/22/18	$9,778	19,627	2,218	$4.41	11.3%	301	183	325	325	1,134	$8.62	51.1%

FORMATTING

1 STAR (VERY BASIC)

COMMON USE CASES:

- *Designing tools or dashboards that don't have a "spreadsheet" feel*
- *Creating separation between values without adding extra rows or columns*

PRO TIP

FREEZE PANES TO KEEP ROWS & COLUMNS IN VIEW

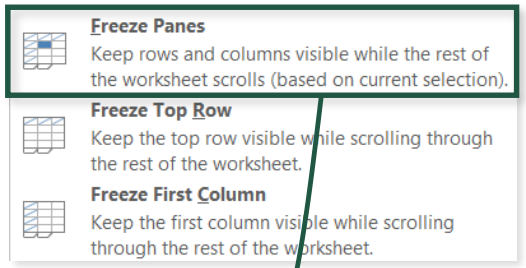
- **Freeze Panes** is commonly used to "*freeze*" or fix the first row or column in place while allowing the rest of the sheet to scroll:
  - **Freeze Top Row:** Freezes the first row in place (*usually the header row*)
  - **Freeze First Column:** Freezes the first column in place
  - **Freeze Panes:** Freezes all cells ***above and to the left of*** the selected cell
- **NOTE:** Use the **Freeze Panes** option when you want to fix both rows and columns

FORMATTING

2 STARS (BASIC)

	A	B	C	D
1	**Title**	**Release Date**	**Release Year**	**Release Month**
2	Over the Hill to the Poorhouse	9/15/1920	1920	9
3	Metropolis	1/26/1927	1927	1
4	The Broadway Melody	11/11/1929	1929	11
5	42nd Street	8/29/1933	1933	8
6	Top Hat	4/15/1935	1935	4
7	Modern Times	10/7/1936	1936	10
8	Snow White and the Seven Dwarfs	2/2/1937	1937	2
9	Gone with the Wind	11/12/1939	1939	11
10	The Wizard of Oz	4/9/1939	1939	4
11	Fantasia	5/18/1940	1940	5
12	Pinocchio	10/12/1940	1940	10
13	Duel in the Sun	6/7/1946	1946	6

(View > Freeze Panes)



	A	P	Q	R	S
1	**Title**	**Lead Actor FB Likes**	**Cast FB Likes**	**Director FB Likes**	**Movie FB Likes**
32	A Fistful of Dollars	16000	16534	0	0
33	A Hard Day's Night	785	2538	44	0
34	Goldfinger	387	1198	82	0
35	Mary Poppins	382	2045	55	0
36	My Fair Lady	453	1164	165	0
37	Nothing But a Man	581	835	0	363
38	Doctor Zhivago	597	1966	767	7000
39	Major Dundee	773	2888	541	251
40	The Greatest Story Ever Told	940	1934	126	1000
41	The Sound of Music	354	1495	338	15000
42	Thunderball	244	1164	92	0
43	The Good, the Bad and the Ugly	16000	16089	0	20000
44	You Only Live Twice	742	2127	43	0
45	2001: A Space Odyssey	273	727	0	24000
46	Oliver!	695	1593	82	0
47	Butch Cassidy and the Sundance Kid	640	2169	131	0
48	Mississippi Mermaid	963	1694	0	278

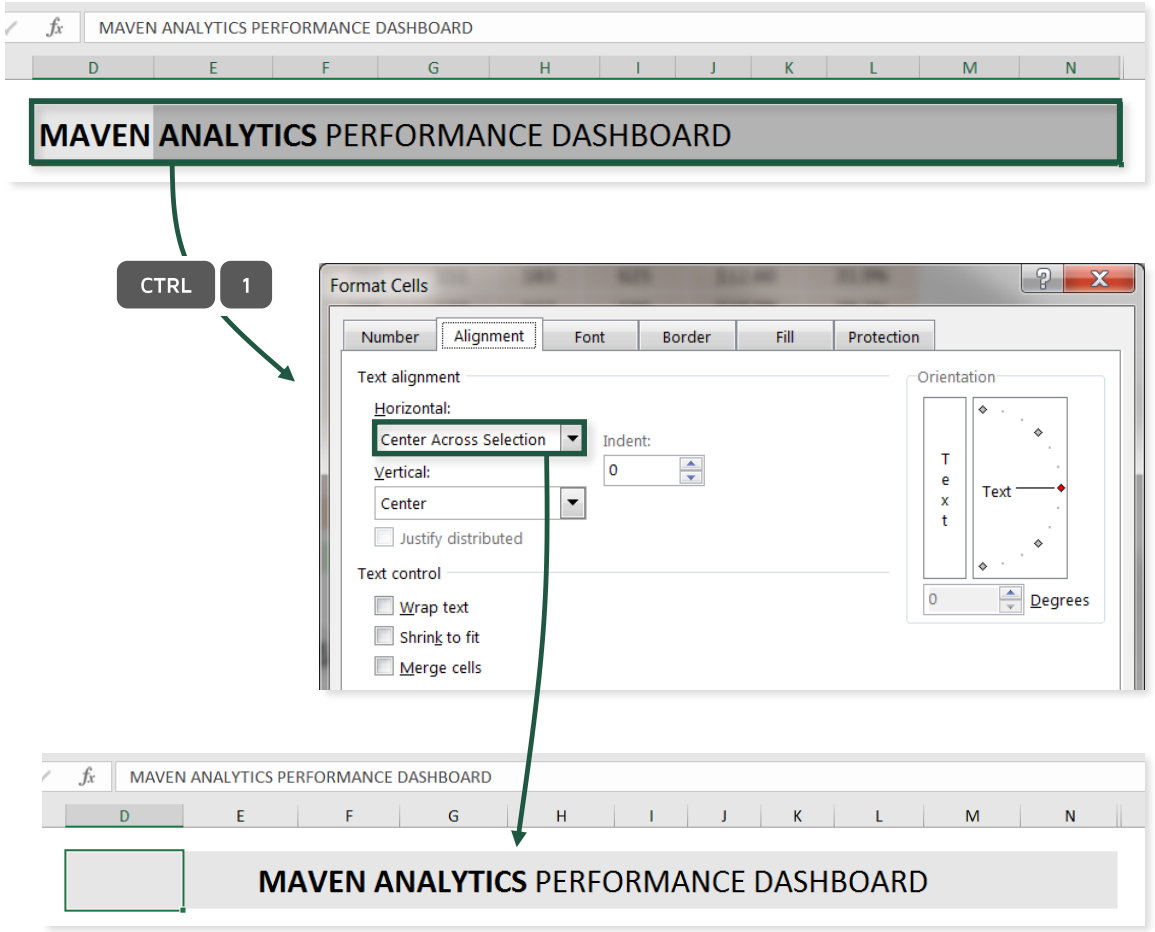
COMMON USE CASES:

- *Keeping the header row visible as you scroll through rows (**Note:** this happens by default if you format your range as a **table**)*
- *Freezing "key" columns when tables contain a large number of fields*

PRO TIP

CENTER TEXT ACROSS A SELECTION OF CELLS

- Most users center text across cells by **merging** and then **aligning to center**; however, merged cells can cause issues if you need to modify other cells in the sheet
- Instead of merging, select a cell range, launch the **Format Cells** dialog box (*CTRL-1*), and choose **Center Across Selection** in the Alignment tab (*Horizontal options*)



FORMATTING

2 STARS (BASIC)

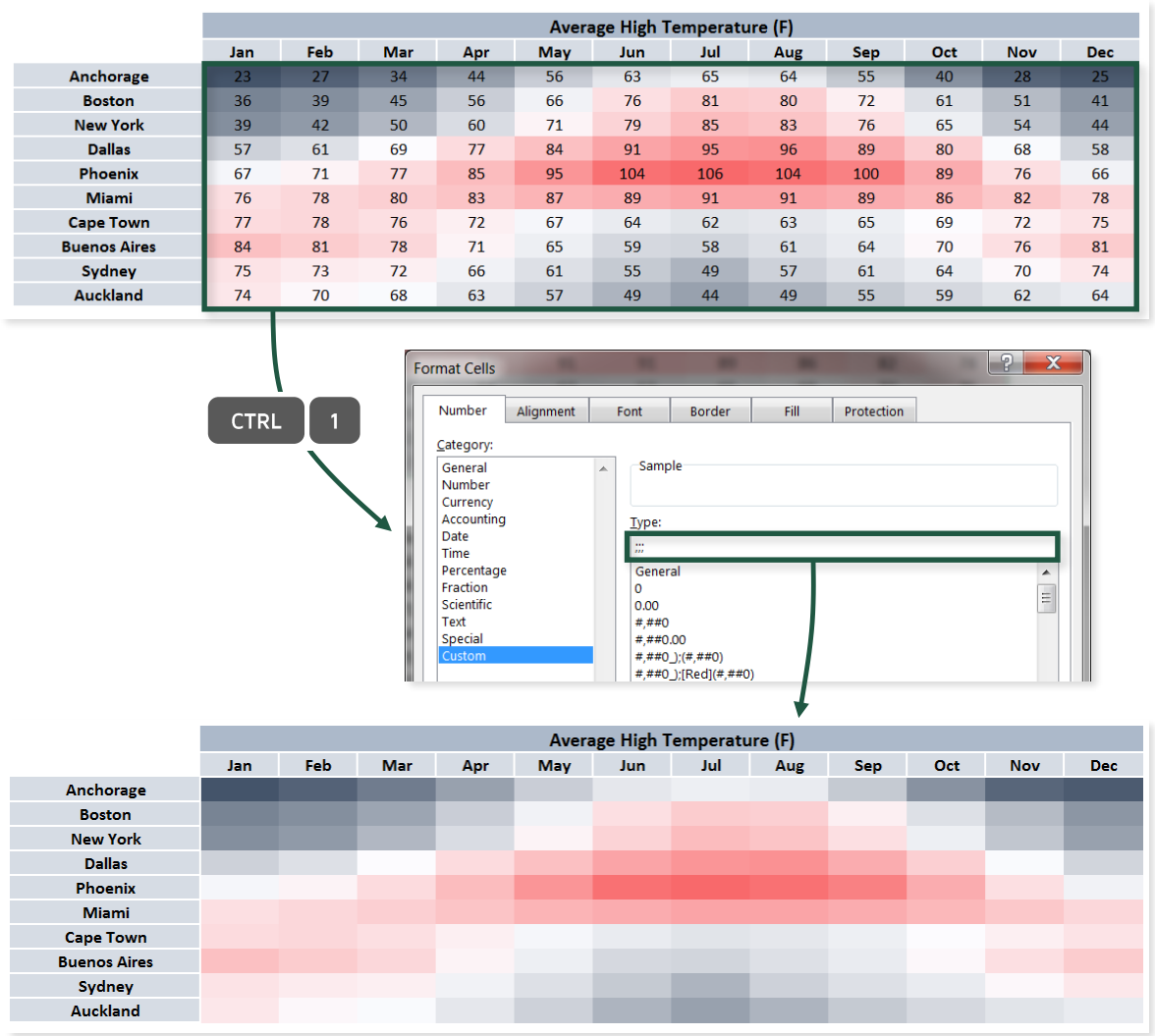
COMMON USE CASES:

- *Formatting text headers within reports or dashboards*
- *Customizing alignment in cases where merged cells cannot be used*

PRO TIP

USE CUSTOM FORMATTING TO MAKE TEXT INVISIBLE

- Use a **custom number format** with type ";;;" (three consecutive semi-colons) to make text invisible
- **NOTE:** This does *not* delete the underlying values, it simply prevents them from displaying within cells



COMMON USE CASES:

- *Creating heatmaps to visualize data while obscuring underlying values*
- *Transforming a column of values into **data bars** or **icon sets** to visualize high-level patterns or trends*

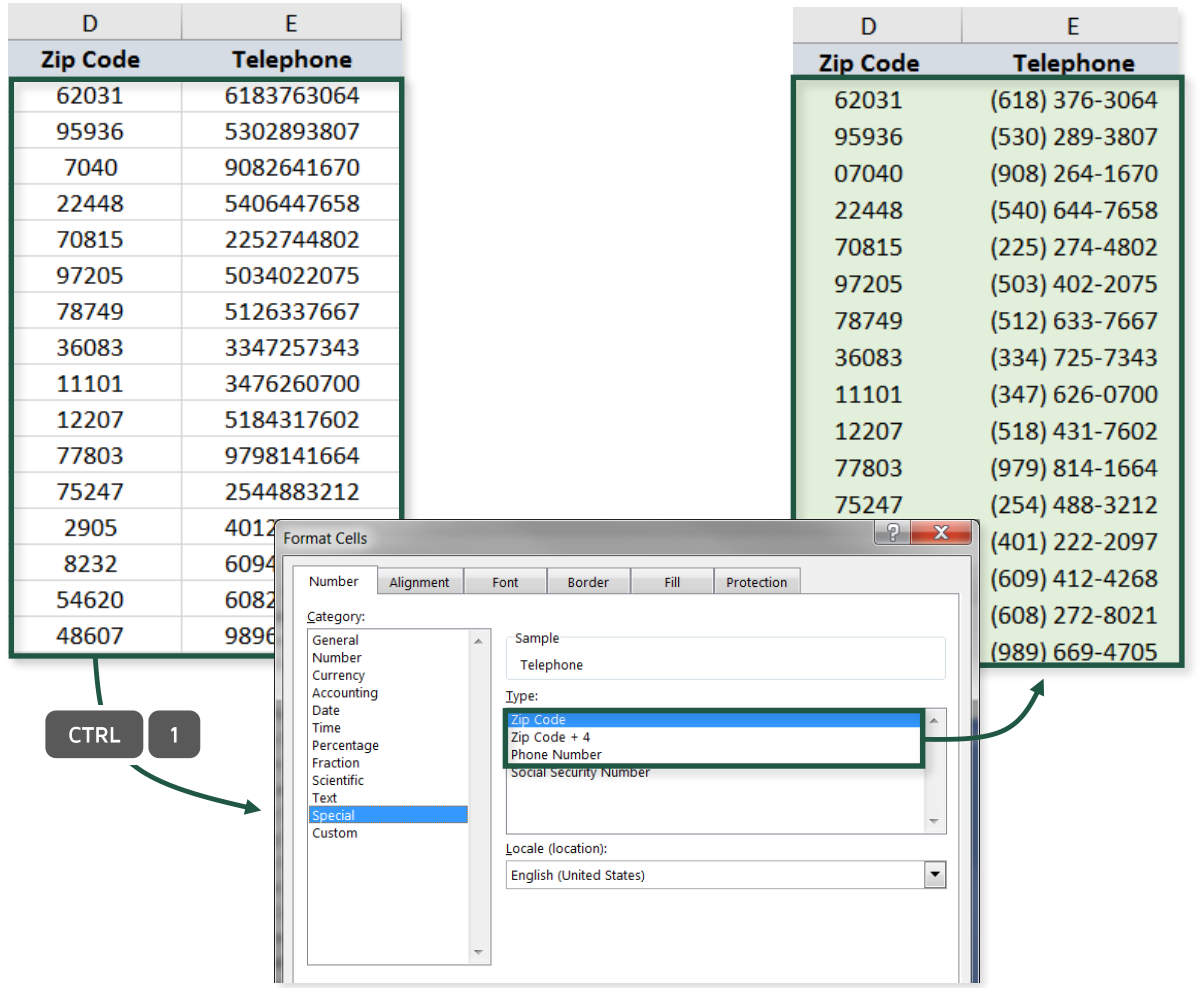
FORMATTING

★★☆☆☆ **2 STARS** (BASIC)

PRO TIP

PROPERLY FORMAT ZIP CODES & PHONE NUMBERS

- **Zip codes** and **phone numbers** can be difficult to work with, since Excel often converts them to numerical fields by default
- Most versions of Excel include a **Special** formatting category for dealing with these specific types of values (*as well as SS#*)
  - ***NOTE**: You can also use text functions like **TEXT(A1,"0####")** to format zip codes or **LEFT/MID/RIGHT** to format phone numbers*



COMMON USE CASES:

- *Reformatting zip codes or phone numbers that have automatically been converted into numerical values (you'll never add two zip codes!)*
- *Adding leading zeros to force zip codes into a consistent 5-digit format*

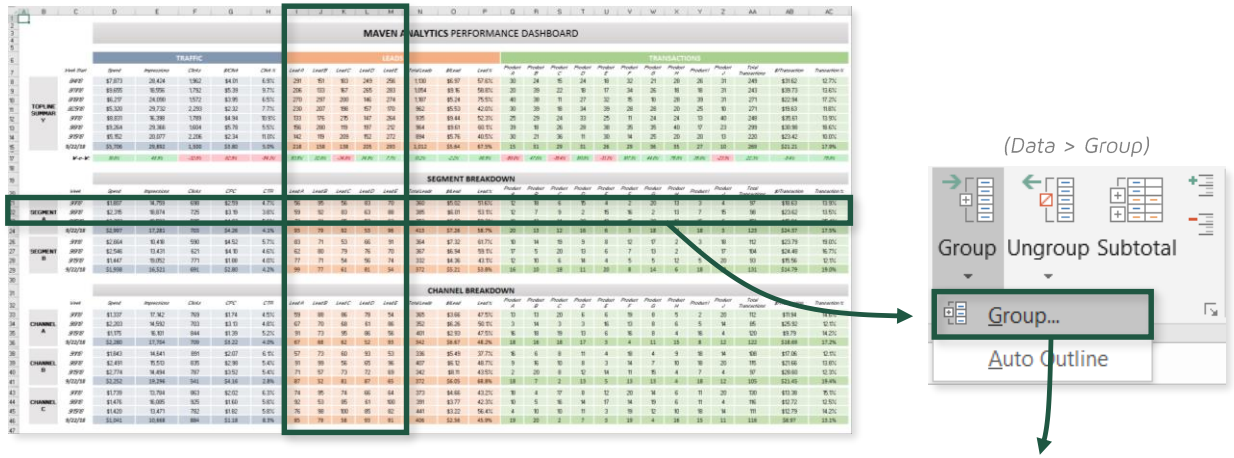
FORMATTING

2 STARS (BASIC)

PRO TIP

GROUP COLUMNS OR ROWS TO SIMPLIFY REPORTS

- Use the **Group** option (***Data*** > ***Group***) to hide selected rows or columns from view
- Excel automatically generates toggles (**+/-**) to show or hide individual groups, as well as buttons to show or hide *all* rows or columns with one click (*labeled "**1**" and "**2**"*)



MAVEN ANALYTICS PERFORMANCE DASHBOARD

		TRAFFIC					LEADS			TRANSACTIONS		
	Week Start	Spend	Impressions	Clicks	$/Click	Click %	Total Leads	$/Lead	Lead %	Total Transactions	$/Transaction	Transaction %
TOPLINE SUMMARY	9/1/18	$8,831	16,398	1,789	$4.94	10.9%	935	$9.44	52.3%	248	$35.61	13.9%
	9/8/18	$9,264	29,366	1,604	$5.78	5.5%	964	$9.61	60.1%	299	$30.98	18.6%
	9/15/18	$5,152	20,077	2,206	$2.34	11.0%	894	$5.76	40.5%	220	$23.42	10.0%
	9/22/18	$5,706	29,892	1,500	$3.80	5.0%	1,012	$5.64	67.5%	269	$21.21	17.9%
	W-o-W:	10.8%	48.9%	-32.0%	62.9%	-54.3%	13.2%	-2.2%	66.5%	22.3%	-9.4%	79.8%

SEGMENT BREAKDOWN

	Week	Spend	Impressions	Clicks	CPC	CTR	Total Leads	$/Lead	Lead %	Total Transactions	$/Transaction	Transaction %
SEGMENT A	**9/22/18**	$2,997	17,281	703	$4.26	4.1%	413	$7.26	58.7%	123	$24.37	17.5%
SEGMENT B	**9/22/18**	$1,938	16,521	691	$2.80	4.2%	372	$5.21	53.8%	131	$14.79	19.0%

CHANNEL BREAKDOWN

	Week	Spend	Impressions	Clicks	CPC	CTR	Total Leads	$/Lead	Lead %	Total Transactions	$/Transaction	Transaction %
CHANNEL A	**9/22/18**	$2,280	17,704	709	$3.22	4.0%	342	$6.67	48.2%	122	$18.69	17.2%
CHANNEL B	**9/22/18**	$2,252	19,296	541	$4.16	2.8%	372	$6.05	68.8%	105	$21.45	19.4%
CHANNEL C	**9/22/18**	$1,041	10,668	884	$1.18	8.3%	406	$2.56	45.9%	116	$8.97	13.1%

FORMATTING

★★☆☆☆ **2 STARS** (BASIC)

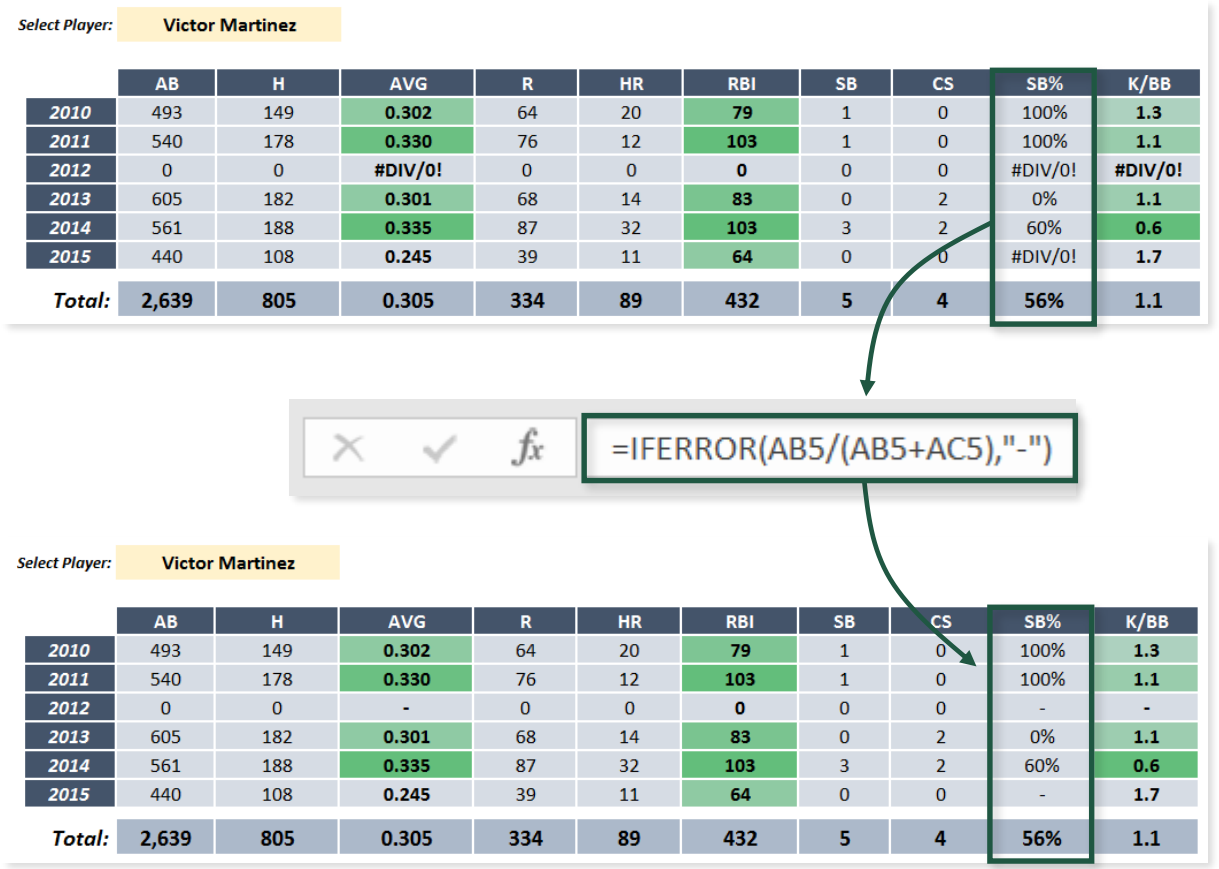
COMMON USE CASES:

- *Designing reports that contain a large amount of data, but only display the most important information by default*
- *Enabling users to quickly toggle views without manually unhiding rows and columns*

PRO TIP

REFORMAT ERRORS WITH IFERROR FUNCTIONS

- Wrap formulas with an **IFERROR** statement to customize how errors will be displayed (*i.e. "-" instead of* ***#DIV/0!***)
- **NOTE:** Use this tip carefully, and only when you understand the source of the error; **IFERROR** should be used for cosmetics, not to "fix" broken formulas or mislead users



FORMATTING

3 STARS (MODERATE)

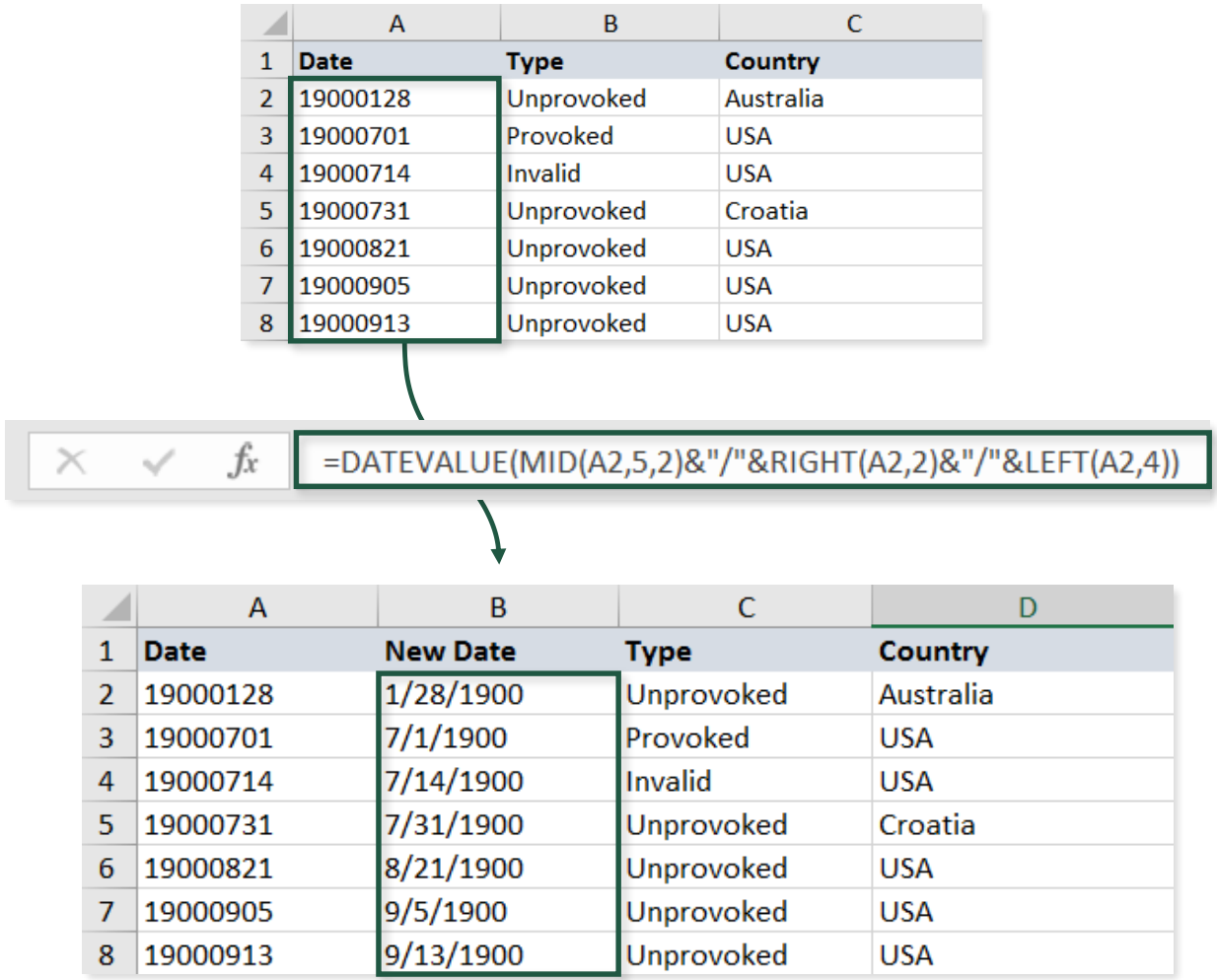
COMMON USE CASES:

- *Cleaning up a user-facing report or dashboard where occasional errors are expected (i.e. dividing by zero)*
- *Writing formulas that produce different outputs based on whether or not a conditional test yields an error*

PRO TIP

CONVERT TEXT STRINGS INTO DATE VALUES

- **Dates** are one of the most challenging data types to work with, as formats tend to vary considerably (i.e. ***MM/DD/YYYY*** *vs.* ***DD/MM/YYYY*** *vs.* ***MMDDYYYY***)
  - *Excel understands a wide range of date formats specific to your regional settings; however, unusual or non-local formats may be recognized as* ***text****, not dates*
- Use text functions (**LEFT/MID/RIGHT/&**) to rearrange date components from a string, and **DATEVALUE** to convert the result



	A	B	C
1	Date	Type	Country
2	19000128	Unprovoked	Australia
3	19000701	Provoked	USA
4	19000714	Invalid	USA
5	19000731	Unprovoked	Croatia
6	19000821	Unprovoked	USA
7	19000905	Unprovoked	USA
8	19000913	Unprovoked	USA

=DATEVALUE(MID(A2,5,2)&"/"&RIGHT(A2,2)&"/"&LEFT(A2,4))

	A	B	C	D
1	Date	New Date	Type	Country
2	19000128	1/28/1900	Unprovoked	Australia
3	19000701	7/1/1900	Provoked	USA
4	19000714	7/14/1900	Invalid	USA
5	19000731	7/31/1900	Unprovoked	Croatia
6	19000821	8/21/1900	Unprovoked	USA
7	19000905	9/5/1900	Unprovoked	USA
8	19000913	9/13/1900	Unprovoked	USA

COMMON USE CASES:

- *Reformatting fields that Excel doesn't recognize as common date types*
- *Converting dates from text to date values, in order to use them for date/time or time-series analysis*

FORMATTING

4 STARS (ADVANCED)

PRO TIP

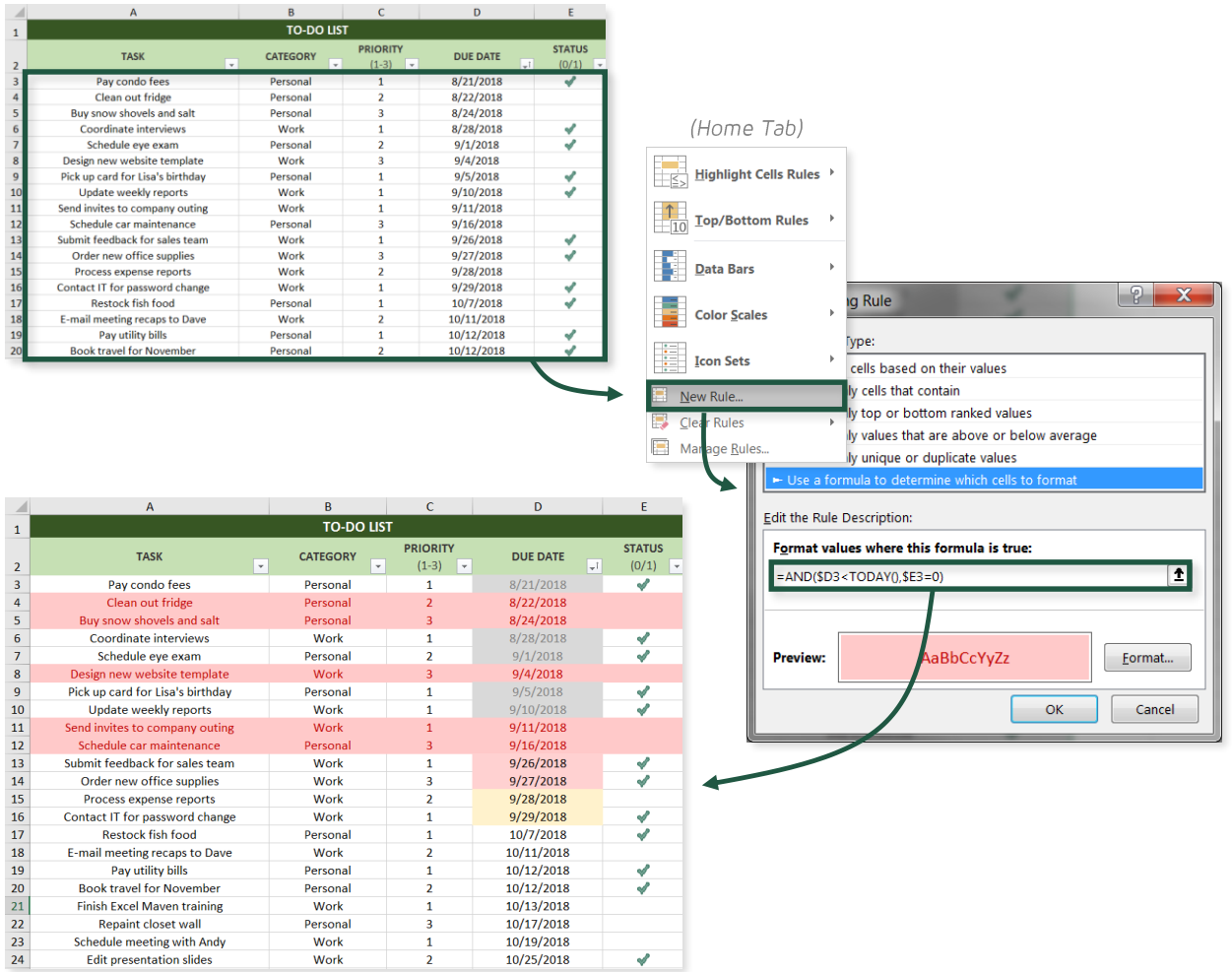
DEFINE YOUR OWN FORMULA-DRIVEN FORMATS

- Use **formula rules** to format cells using custom or complex logic that standard conditional formatting tools can't handle
  - ***For example:*** *formatting an entire row in a table or range based on the values in one column, or applying formats to cells that meet multiple, formula-based criteria*
- **NOTE:** Intellisense is not available within the conditional formatting dialog box, so make sure you know your formula syntax!

FORMATTING

★★★★★

5 STARS (EXPERT)



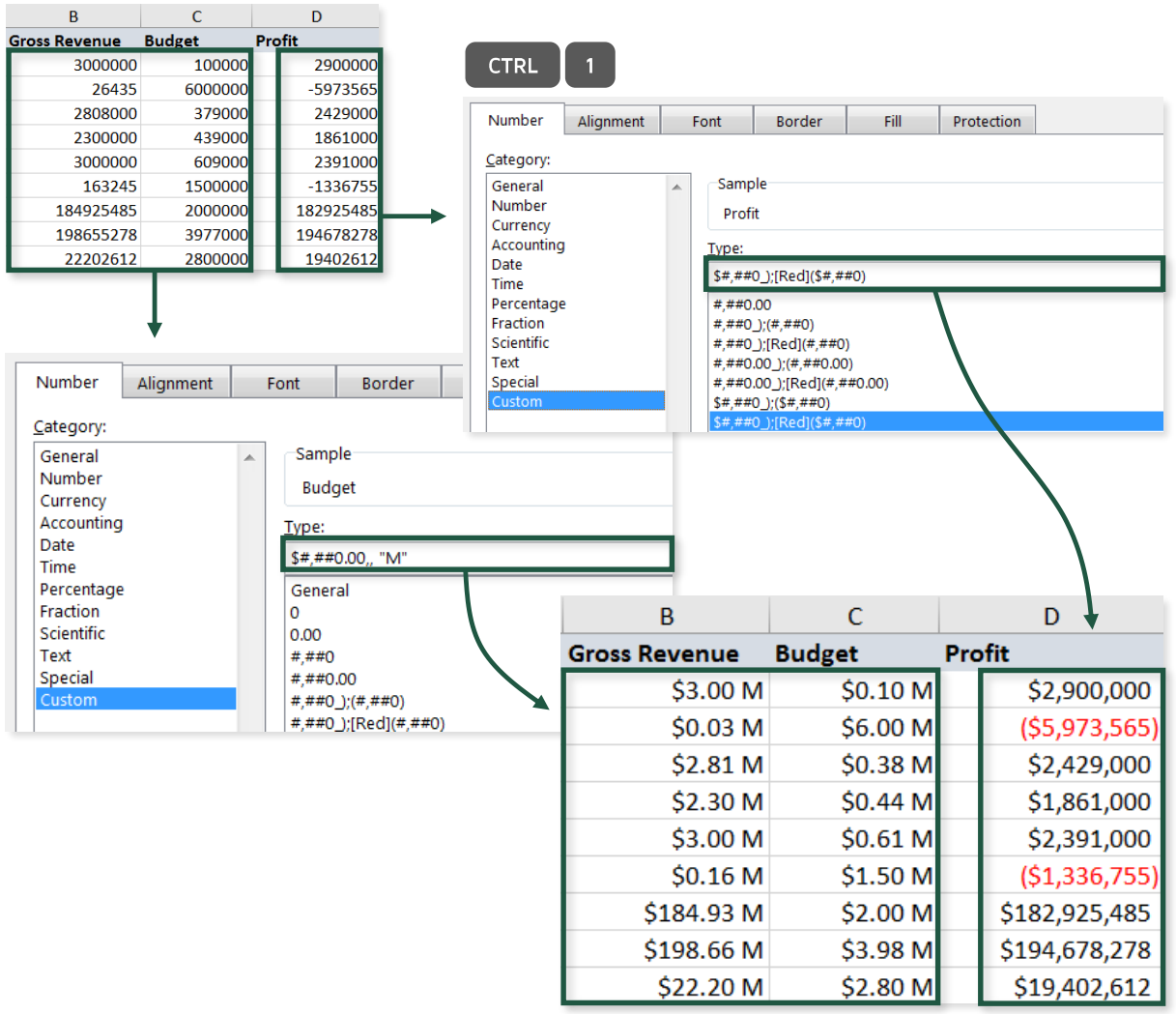
COMMON USE CASES:

- *Applying custom or complex formats beyond the standard options*
- *Highlighting cells which meet a specific set of criteria (i.e. rows where **Due Date** has passed, **Category** = "Work" and **Priority** = 1)*

PRO TIP

APPLY ADVANCED CUSTOM NUMBER FORMATS

- Use **Custom** formatting options to select more complex formats, or define your own using Excel's custom formatting syntax
- Custom formats contain up to 4 conditions, separated by semicolons (**A;B;C;D**)
  - **A**: Format for **positive** numbers
  - **B**: Format for **negative** numbers
  - **C**: Format for **zeros**
  - **D**: Format for **text**
- **LEARN MORE**: http://bit.ly/2A1JFUD



COMMON USE CASES:

- *Applying custom formats to positive vs. negative values to draw attention to important patterns or trends*
- *Abbreviating large values with "**K**" or "**M**" labels (no formulas required)*

FORMATTING

5 STARS (EXPERT)

FORMULA TIPS

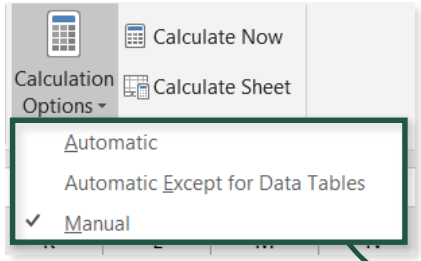
EXCEL MAVEN

PRO TIP

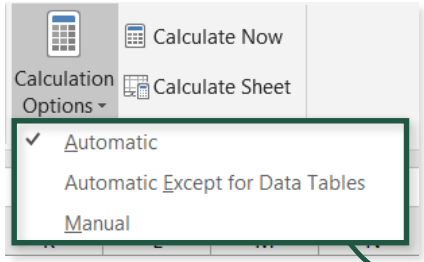
CHANGE FORMULA CALCULATION MODES

- Excel has two primary **calculation modes**: **Automatic** (*default*) and **Manual**
  - **Automatic** allows formulas to calculate automatically with any workbook change
  - **Manual** freezes all formula calculations until the **Calculate Now** (*F9*) option is selected
- **NOTE:** If you see a worksheet formula producing duplicate values, check that you are not in **Manual** calculation mode

(Formulas > Calculation Options)



B	C	D	E
Quantity	**Retail Price**	**Unit Cost**	**Sales Profit**
53,847	$0.32	$0.10	$12,058.25
47,363	$2.47	$0.74	$12,058.25
36,381	$1.72	$0.86	$12,058.25
36,334	$1.01	$0.10	$12,058.25
36,039	$0.76	$0.38	$12,058.25
35,317	$3.20	$0.32	$12,058.25
30,680	$2.38	$0.95	$12,058.25
26,437	$0.78	$0.31	$12,058.25
26,315	$0.45	$0.09	$12,058.25
24,753	$0.74	$0.22	$12,058.25
23,854	$1.65	$0.66	$12,058.25



B	C	D	E
Quantity	**Retail Price**	**Unit Cost**	**Sales Profit**
53,847	$0.32	$0.10	$12,058.25
47,363	$2.47	$0.74	$81,899.38
36,381	$1.72	$0.86	$31,324.62
36,334	$1.01	$0.10	$33,059.74
36,039	$0.76	$0.38	$13,622.48
35,317	$3.20	$0.32	$101,848.07
30,680	$2.38	$0.95	$43,726.09
26,437	$0.78	$0.31	$12,397.33
26,315	$0.45	$0.09	$9,390.94
24,753	$0.74	$0.22	$12,843.13
23,854	$1.65	$0.66	$23,634.29

COMMON USE CASES:

- *Switching large, formula-heavy files to manual mode while editing to prevent automatic recalculation*
- *Allowing formulas to calculate automatically with the exception of data tables, which typically iterate a large number of times*

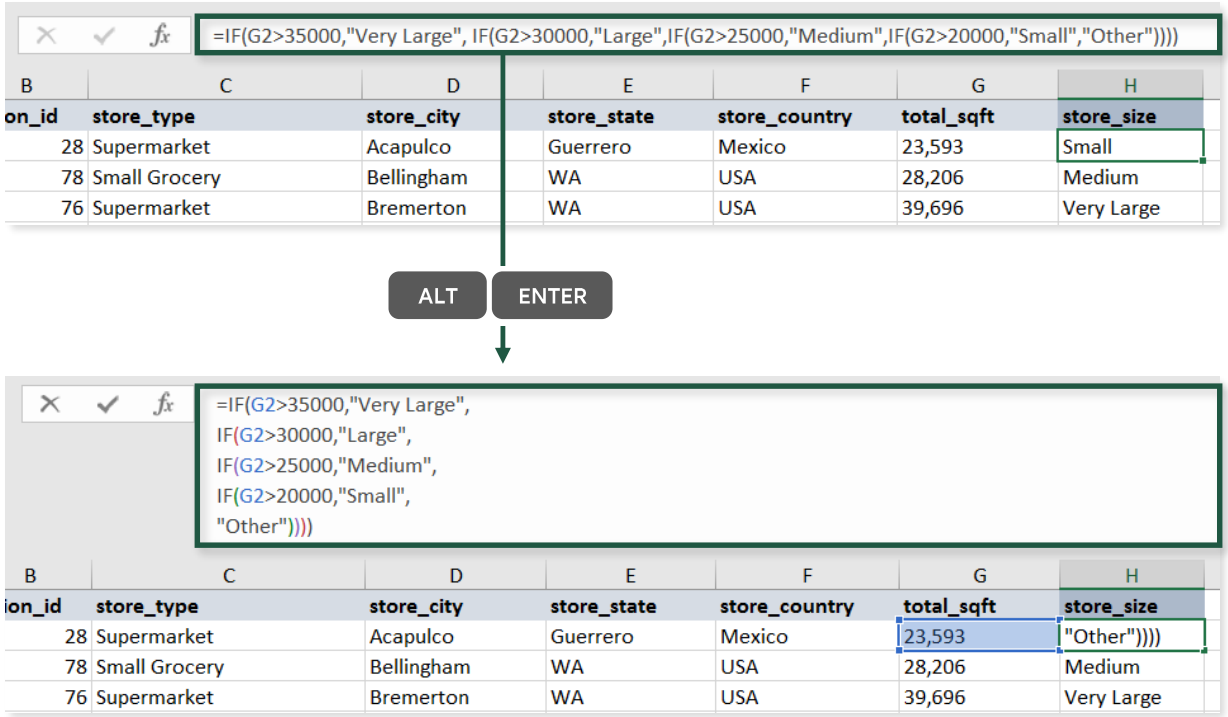
FORMULAS

1 STAR (VERY BASIC)

PRO TIP

USE FORMULA LINE BREAKS TO IMPROVE READABILITY

- While typing or editing in the formula bar, press **ALT-ENTER** to create a line break
- **Line breaks** are a great way to improve readability and provide transparency into formula logic, particularly for complex nested or conditional (IF) functions



B	C	D	E	F	G	H
on_id	store_type	store_city	store_state	store_country	total_sqft	store_size
28	Supermarket	Acapulco	Guerrero	Mexico	23,593	Small
78	Small Grocery	Bellingham	WA	USA	28,206	Medium
76	Supermarket	Bremerton	WA	USA	39,696	Very Large

B	C	D	E	F	G	H
ion_id	store_type	store_city	store_state	store_country	total_sqft	store_size
28	Supermarket	Acapulco	Guerrero	Mexico	23,593	"Other"))))
78	Small Grocery	Bellingham	WA	USA	28,206	Medium
76	Supermarket	Bremerton	WA	USA	39,696	Very Large

COMMON USE CASES:

- *Breaking conditional functions into multiple lines to isolate each criteria*
- *Making complex nested formulas easier to interpret by introducing each component function on a new line*

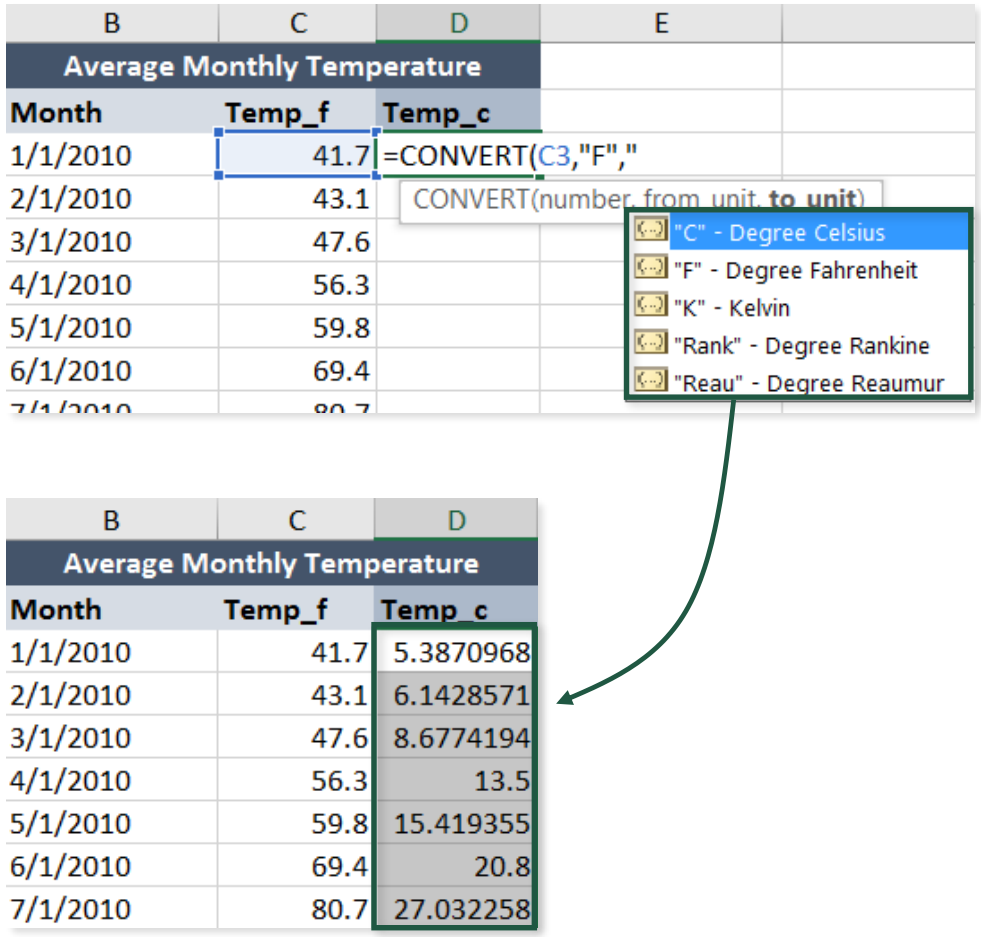
FORMULAS

1 STAR (VERY BASIC)

PRO TIP

CHANGE MEASUREMENT UNITS WITH CONVERT

- Use the **CONVERT** function to change values from one unit of measurement to another (***lbs*** *to* ***kg***, ***celcius*** *to* ***fahrenheit***, ***minutes*** *to* ***seconds***, *etc.)*
- **NOTE:** Some unit types may not be listed in the formula dialog box, but can be accessed by typing them directly (*i.e.* *"**cm**" for centimeters or "**kg**" for kilograms*)



FORMULAS

1 STAR (VERY BASIC)

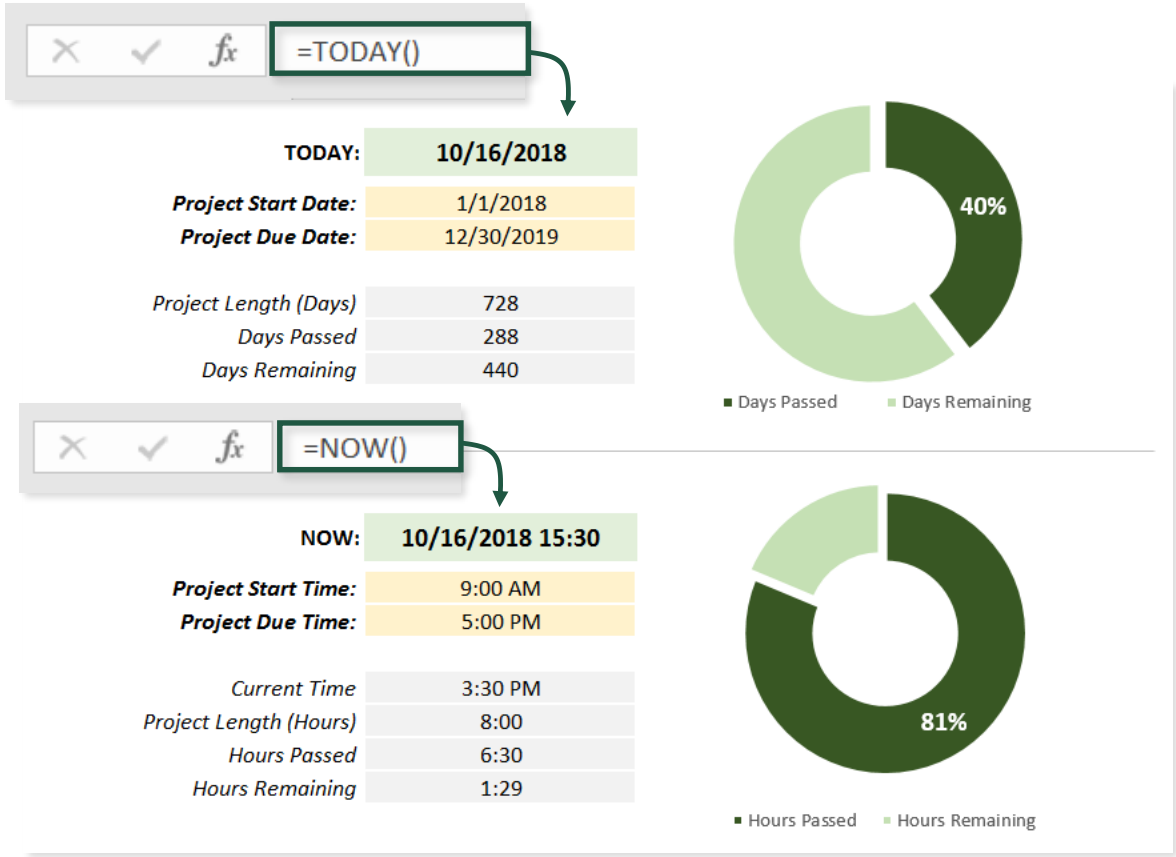
COMMON USE CASES:

- *Quickly converting measurements without needing conversion rates or calculators*
- *Converting date or time values without the use of formula constants*

PRO TIP

ADD REAL-TIME FORMULAS WITH TODAY & NOW

- Use **TODAY()** and **NOW()** functions to return the current date or time
- **NOTE**: These are **volatile** functions, meaning that they automatically recalculate with *any* workbook change
  - Use the **CTRL-;** shortcut to return the current day as a hard-coded value, or **CTRL-SHIFT-;** to return the current time



FORMULAS

2 STARS (BASIC)

COMMON USE CASES:

- *Displaying the current date or time in a worksheet cell*
- *Creating scheduling or timeline tools that update with any workbook change*

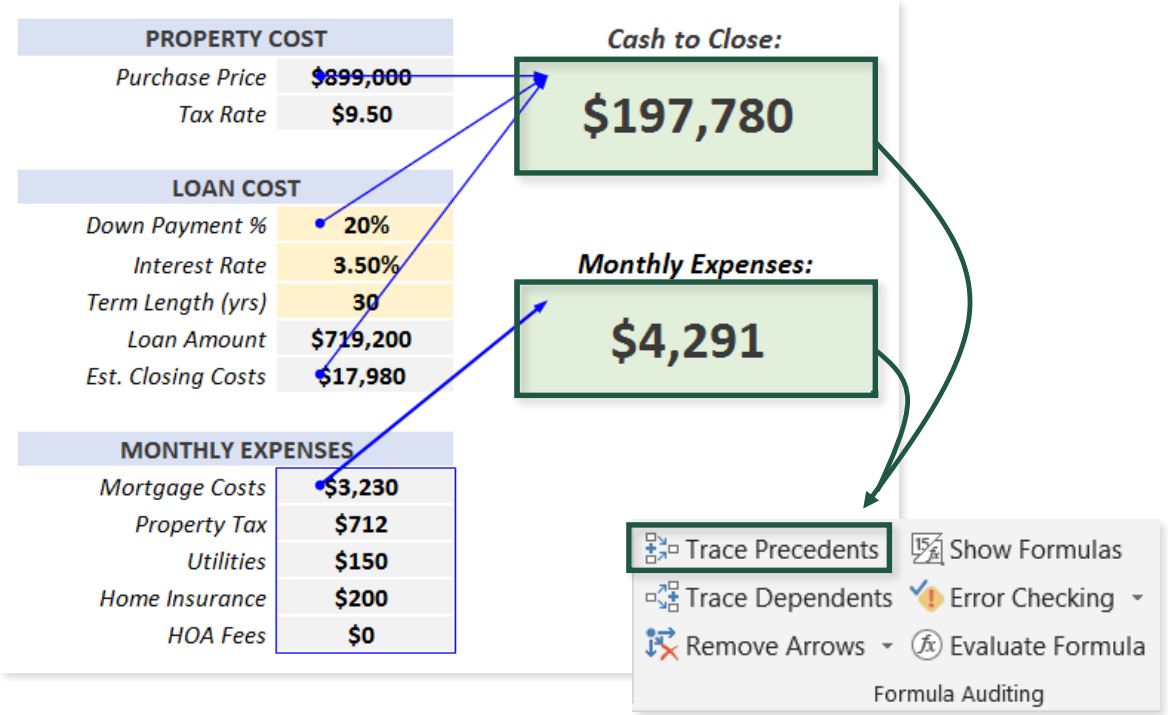
PRO TIP

FIND & FIX ERRORS USING FORMULA AUDITING TOOLS

- Use **Formula Auditing** to trace references, evaluate calculations, and diagnose errors:
  - **Trace Precedents**: Draws arrows to any cells that *impact* the selected value
  - **Trace Dependents**: Draws arrows to any cells *impacted by* the selected value
  - **Show Formulas**: Temporarily displays all formulas within the worksheet
  - **Error Checking**: Scans the sheet for errors and traces the source to precedent cells
  - **Evaluate Formula**: Evaluates each component of a formula step-by-step

FORMULAS

3 STARS (MODERATE)



(Formulas > Formula Auditing)

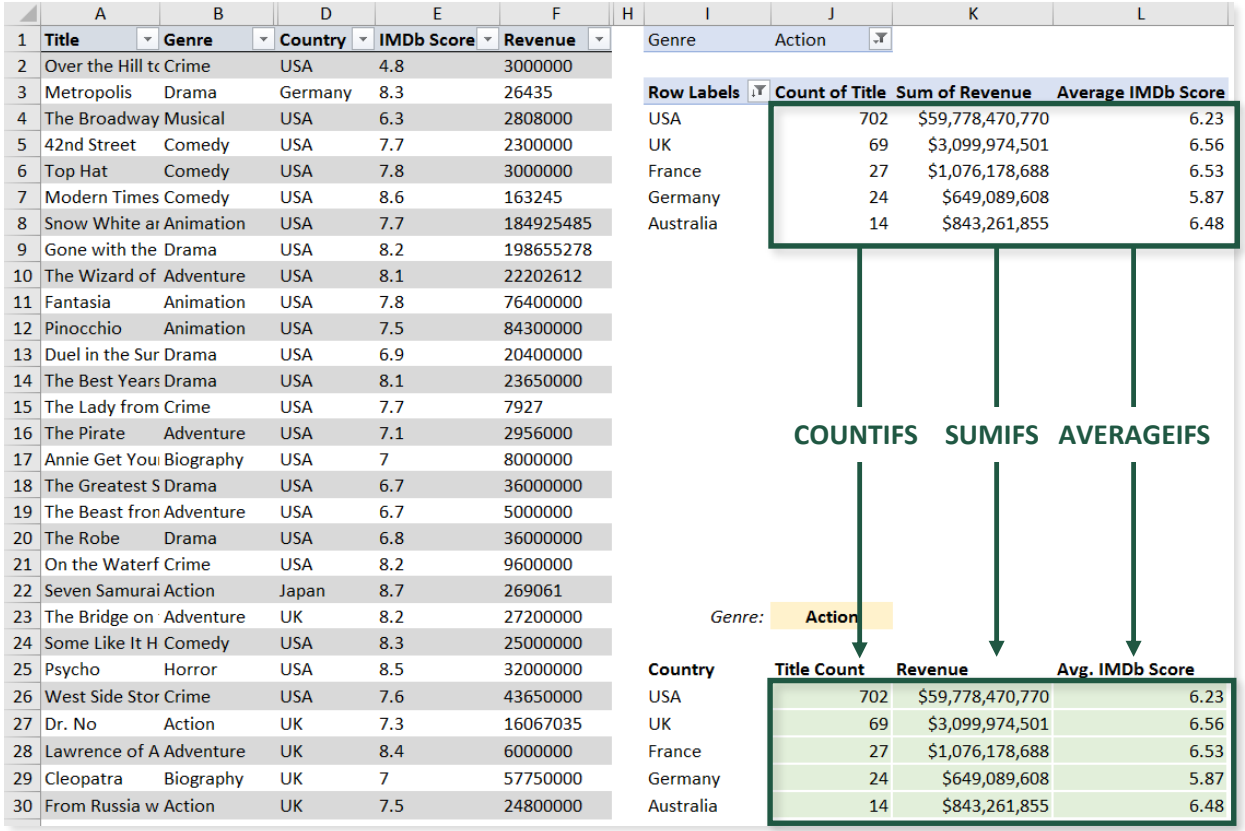
COMMON USE CASES:

- *Understanding how complex formulas and functions operate*
- *Visualizing which cells factor into a certain formula output*
- *Tracing and diagnosing the source of an error*

PRO TIP

USE STATS FUNCTIONS FOR PIVOT-STYLE REPORTS

- **Conditional stats functions** (**SUMIFS**, **COUNTIFS**, **AVERAGEIFS**) can be used to generate filtered values just like PivotTables
- While pivots are ideal for unstructured analysis, **stats functions** are a great tool for building custom reports and dashboards



	A	B	D	E	F
1	Title	Genre	Country	IMDb Score	Revenue
2	Over the Hill to	Crime	USA	4.8	3000000
3	Metropolis	Drama	Germany	8.3	26435
4	The Broadway	Musical	USA	6.3	2808000
5	42nd Street	Comedy	USA	7.7	2300000
6	Top Hat	Comedy	USA	7.8	3000000
7	Modern Times	Comedy	USA	8.6	163245
8	Snow White ar	Animation	USA	7.7	184925485
9	Gone with the	Drama	USA	8.2	198655278
10	The Wizard of	Adventure	USA	8.1	22202612
11	Fantasia	Animation	USA	7.8	76400000
12	Pinocchio	Animation	USA	7.5	84300000
13	Duel in the Sur	Drama	USA	6.9	20400000
14	The Best Years	Drama	USA	8.1	23650000
15	The Lady from	Crime	USA	7.7	7927
16	The Pirate	Adventure	USA	7.1	2956000
17	Annie Get You	Biography	USA	7	8000000
18	The Greatest S	Drama	USA	6.7	36000000
19	The Beast fron	Adventure	USA	6.7	5000000
20	The Robe	Drama	USA	6.8	36000000
21	On the Waterf	Crime	USA	8.2	9600000
22	Seven Samurai	Action	Japan	8.7	269061
23	The Bridge on	Adventure	UK	8.2	27200000
24	Some Like It H	Comedy	USA	8.3	25000000
25	Psycho	Horror	USA	8.5	32000000
26	West Side Stor	Crime	USA	7.6	43650000
27	Dr. No	Action	UK	7.3	16067035
28	Lawrence of A	Adventure	UK	8.4	6000000
29	Cleopatra	Biography	UK	7	57750000
30	From Russia w	Action	UK	7.5	24800000

	I	J	K	L
1	Genre	Action		
2				
3	Row Labels	Count of Title	Sum of Revenue	Average IMDb Score
4	USA	702	$59,778,470,770	6.23
5	UK	69	$3,099,974,501	6.56
6	France	27	$1,076,178,688	6.53
7	Germany	24	$649,089,608	5.87
8	Australia	14	$843,261,855	6.48

COUNTIFS SUMIFS AVERAGEIFS

	I	J	K	L
10	*Genre:*	Action		
11	Country	Title Count	Revenue	Avg. IMDb Score
12	USA	702	$59,778,470,770	6.23
13	UK	69	$3,099,974,501	6.56
14	France	27	$1,076,178,688	6.53
15	Germany	24	$649,089,608	5.87
16	Australia	14	$843,261,855	6.48

FORMULAS

3 STARS (MODERATE)

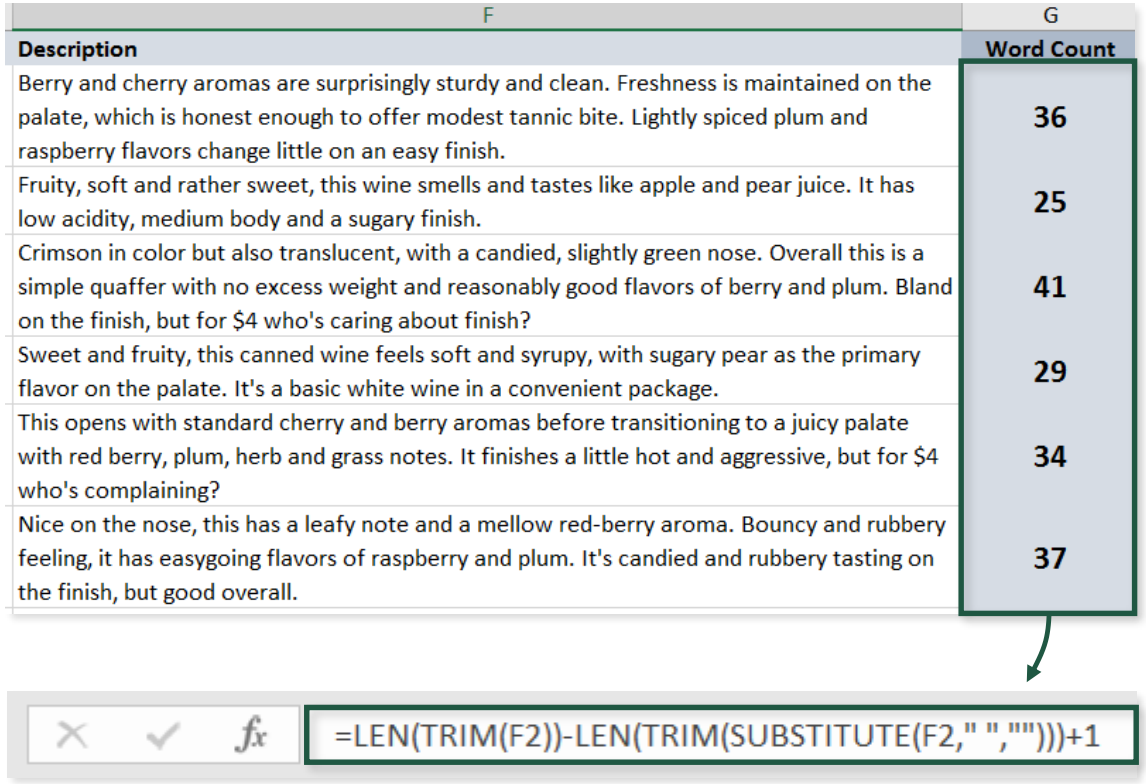
COMMON USE CASES:

- *Designing custom-formatted, dynamic reports without using PivotTables*
- *Filtering or segmenting raw data based on a given set of criteria*

PRO TIP

COUNT WORDS IN A CELL USING TEXT FORMULAS

- **LEN** and **SUBSTITUTE** can be used to count the number of spaces (and therefore determine the number of words) in a string
  - *In this example, we're counting spaces by replacing them with blanks, and comparing the original string length to the new one*
- **NOTE**: Add a **TRIM** function to avoid counting extra (leading or trailing) spaces



	F	G
	Description	**Word Count**
	Berry and cherry aromas are surprisingly sturdy and clean. Freshness is maintained on the palate, which is honest enough to offer modest tannic bite. Lightly spiced plum and raspberry flavors change little on an easy finish.	36
	Fruity, soft and rather sweet, this wine smells and tastes like apple and pear juice. It has low acidity, medium body and a sugary finish.	25
	Crimson in color but also translucent, with a candied, slightly green nose. Overall this is a simple quaffer with no excess weight and reasonably good flavors of berry and plum. Bland on the finish, but for $4 who's caring about finish?	41
	Sweet and fruity, this canned wine feels soft and syrupy, with sugary pear as the primary flavor on the palate. It's a basic white wine in a convenient package.	29
	This opens with standard cherry and berry aromas before transitioning to a juicy palate with red berry, plum, herb and grass notes. It finishes a little hot and aggressive, but for $4 who's complaining?	34
	Nice on the nose, this has a leafy note and a mellow red-berry aroma. Bouncy and rubbery feeling, it has easygoing flavors of raspberry and plum. It's candied and rubbery tasting on the finish, but good overall.	37

```
=LEN(TRIM(F2))-LEN(TRIM(SUBSTITUTE(F2," ","")))+1
```

COMMON USE CASES:

- *Adding counters in cases where word counts are important*
- *Analyzing text fields (Tweets, Blogs, Ad Copy, etc.) based on length*

FORMULAS

3 STARS (MODERATE)

PRO TIP

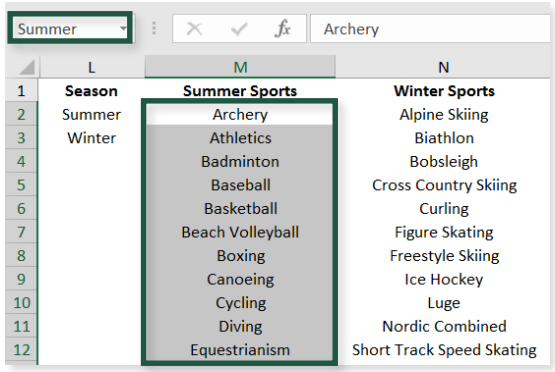
CREATE DEPENDENT DROP-DOWNS WITH INDIRECT

- Use **INDIRECT** and **Data Validation** to create drop-down lists that update based on user selections:
  - **STEP 1**: Create individual lists containing all sets of possible selections
  - **STEP 2**: Turn each list into a named range using the Name Manager or Name Box (*be consistent with your names!*)
  - **STEP 3**: Create a second drop-down, and configure the list source to reference the first drop-down cell, wrapped in **INDIRECT**

FORMULAS

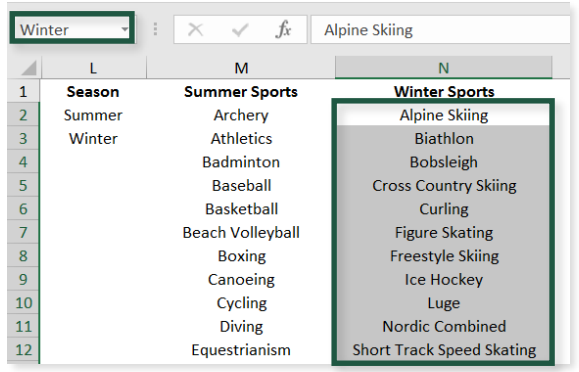
4 STARS (ADVANCED)

*(**Summer** Named Range)*

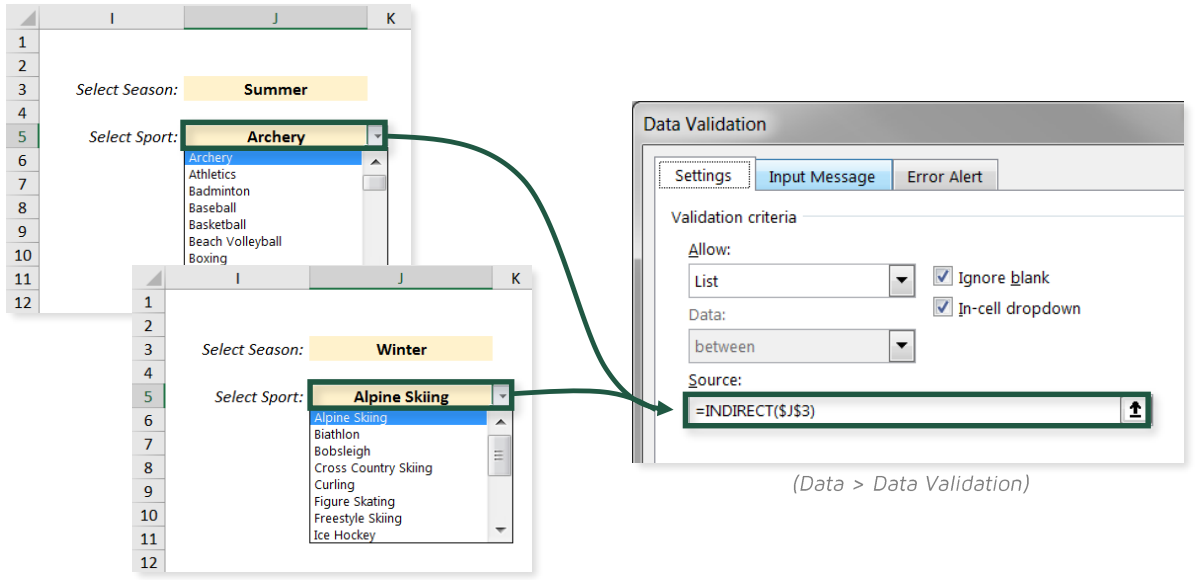


	L	M	N
1	Season	Summer Sports	Winter Sports
2	Summer	Archery	Alpine Skiing
3	Winter	Athletics	Biathlon
4		Badminton	Bobsleigh
5		Baseball	Cross Country Skiing
6		Basketball	Curling
7		Beach Volleyball	Figure Skating
8		Boxing	Freestyle Skiing
9		Canoeing	Ice Hockey
10		Cycling	Luge
11		Diving	Nordic Combined
12		Equestrianism	Short Track Speed Skating

*(**Winter** Named Range)*



	L	M	N
1	Season	Summer Sports	Winter Sports
2	Summer	Archery	Alpine Skiing
3	Winter	Athletics	Biathlon
4		Badminton	Bobsleigh
5		Baseball	Cross Country Skiing
6		Basketball	Curling
7		Beach Volleyball	Figure Skating
8		Boxing	Freestyle Skiing
9		Canoeing	Ice Hockey
10		Cycling	Luge
11		Diving	Nordic Combined
12		Equestrianism	Short Track Speed Skating



(Data > Data Validation)

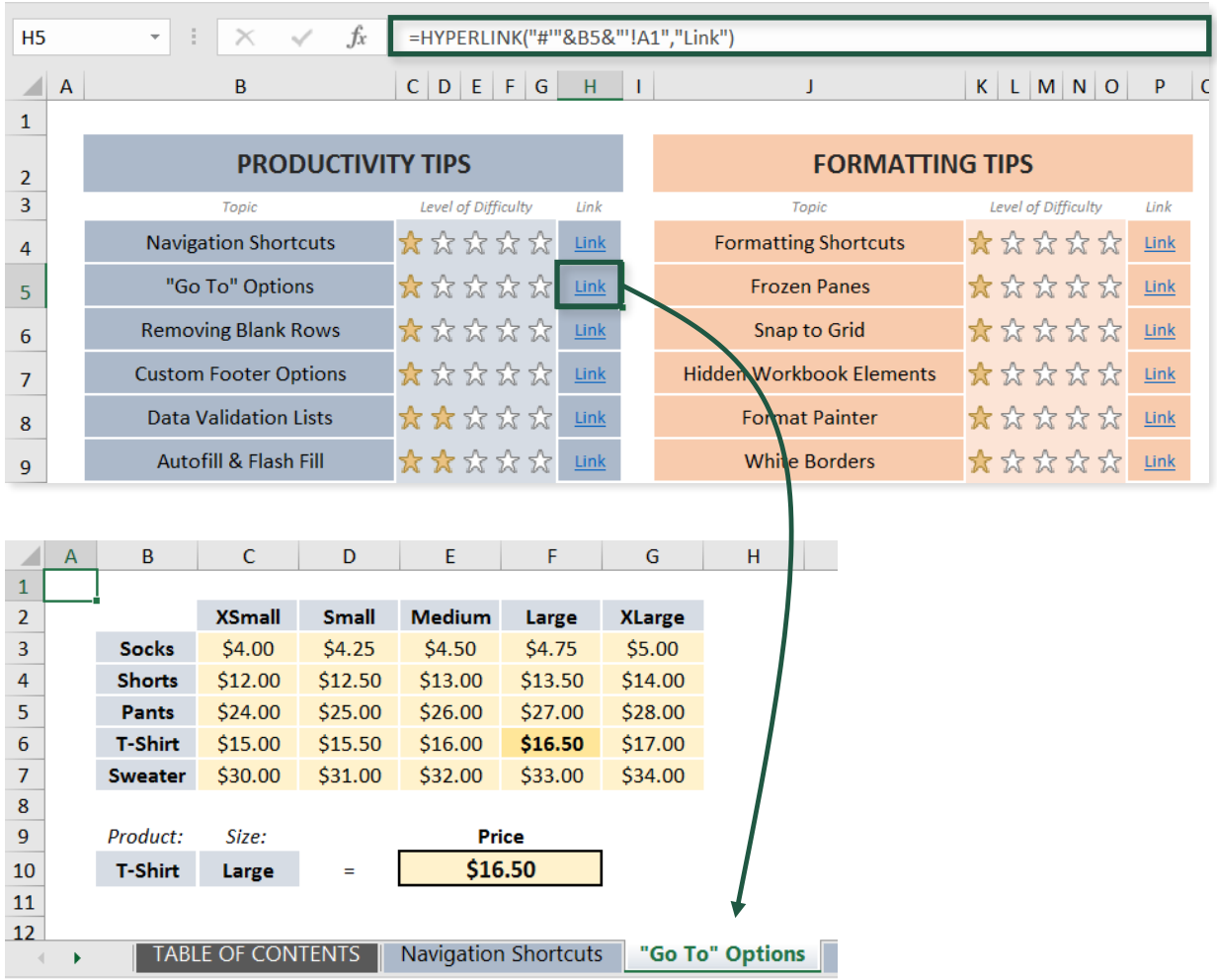
COMMON USE CASES:

- *Creating dynamic reports with multiple user inputs*
- *Configuring drop-down lists to prevent users from selecting invalid combinations of options*

PRO TIP

CREATE DYNAMIC LINKS USING HYPERLINK

- Use **HYPERLINK** to connect users straight to specific workbook locations
- Syntax is *extremely* specific; for example, the following would link to Sheet1, cell A1: =**HYPERLINK**("**#'Sheet1'!A1**","*LinkName*")
- **NOTE**: You can replace a hard-coded sheet name with a cell reference, but you need to concatenate it with the hash, exclamation point, and quotation marks as shown



COMMON USE CASES:

- *Adding tools to help users navigate large or complex workbooks*
- *Creating reports with links to additional details or data sources*

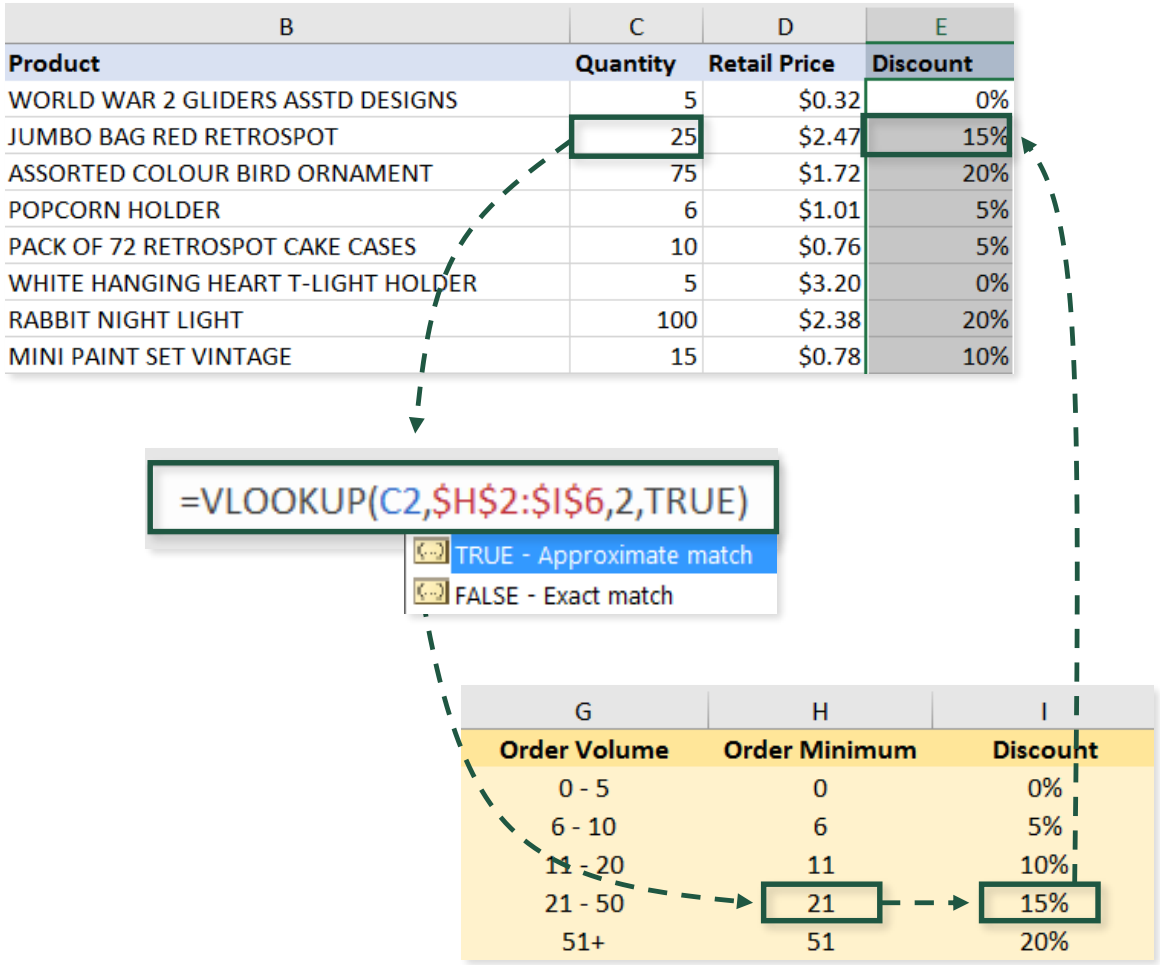
FORMULAS

4 STARS (ADVANCED)

PRO TIP

FIND VALUE RANGES USING "FUZZY MATCH" VLOOKUPS

- **Approximate** (aka "Fuzzy") **Match** can be used to determine where a specific lookup value falls within a given set of ranges
  - Instead of finding an *exact* match, this finds the closest match that is **equal to or less than** the lookup value
  - In the *[range_lookup]* argument of a VLOOKUP, use **TRUE** (or **1**) for fuzzy match
- **NOTE**: Groups must be defined by the **minimum value**, and in **ascending order**



FORMULAS

4 STARS (ADVANCED)

COMMON USE CASES:

- *Defining variable commission rates based on agent sales volume*
- *Calculating tiered discounts based on purchase quantity*

PRO TIP

RANDOMIZE ITEMS WITH OFFSET & RANDBETWEEN

- Combine **OFFSET** with **RANDBETWEEN** to jump to random rows within a cell range
  - Since **RANDBETWEEN** is a **volatile** function, the result will automatically recalculate with *any* workbook change
- **TIP**: Rather than hard-coding the max random value, use a **COUNTA** function to automatically count the length of the list

FORMULAS

4 STARS (ADVANCED)

	A	B
1	**LIST A**	**LIST B**
2	AQUATIC	AARDVAARKS
3	BALD	AIRBAGS
4	BELLIGERENT	ANGELS
5	CHUBBY	BANANAS
6	CLEAN	BEASTS
7	DAZZLING	BUBBLES
8	ELASTIC	CRICKETS
9	ELECTRIC	CROWS
10	ELEGANT	DAUGHTERS
11	FANCY	FISH
12	FLABBY	GHOSTS
13	FRIGID	GOGGLES
14	GLAMOROUS	KITTENS
15	GLISTENING	KNIVES
16	HANDSOME	MILKMEN
17	MAGNIFICENT	MONKEYS
18	MUSCULAR	MUSHROOMS
19	MYSTERIOUS	NECKTIES
20	PLUMP	PENGUINS
21	SALTY	
22	SCREECHING	
23	SCRUFFY	
24	SHAPELY	SPOONS
25	SHORT	SQUIRRELS
26	SKINNY	
27	SLIPPERY	
28	UGLY	
29	UNDERCOVER	TODDLERS
30	WHIMSICLE	WEASELS
31	WHITE HOT	WHIPS



=OFFSET(A$1,RANDBETWEEN(1,30),0)

=OFFSET(B$1,RANDBETWEEN(1,30),0)

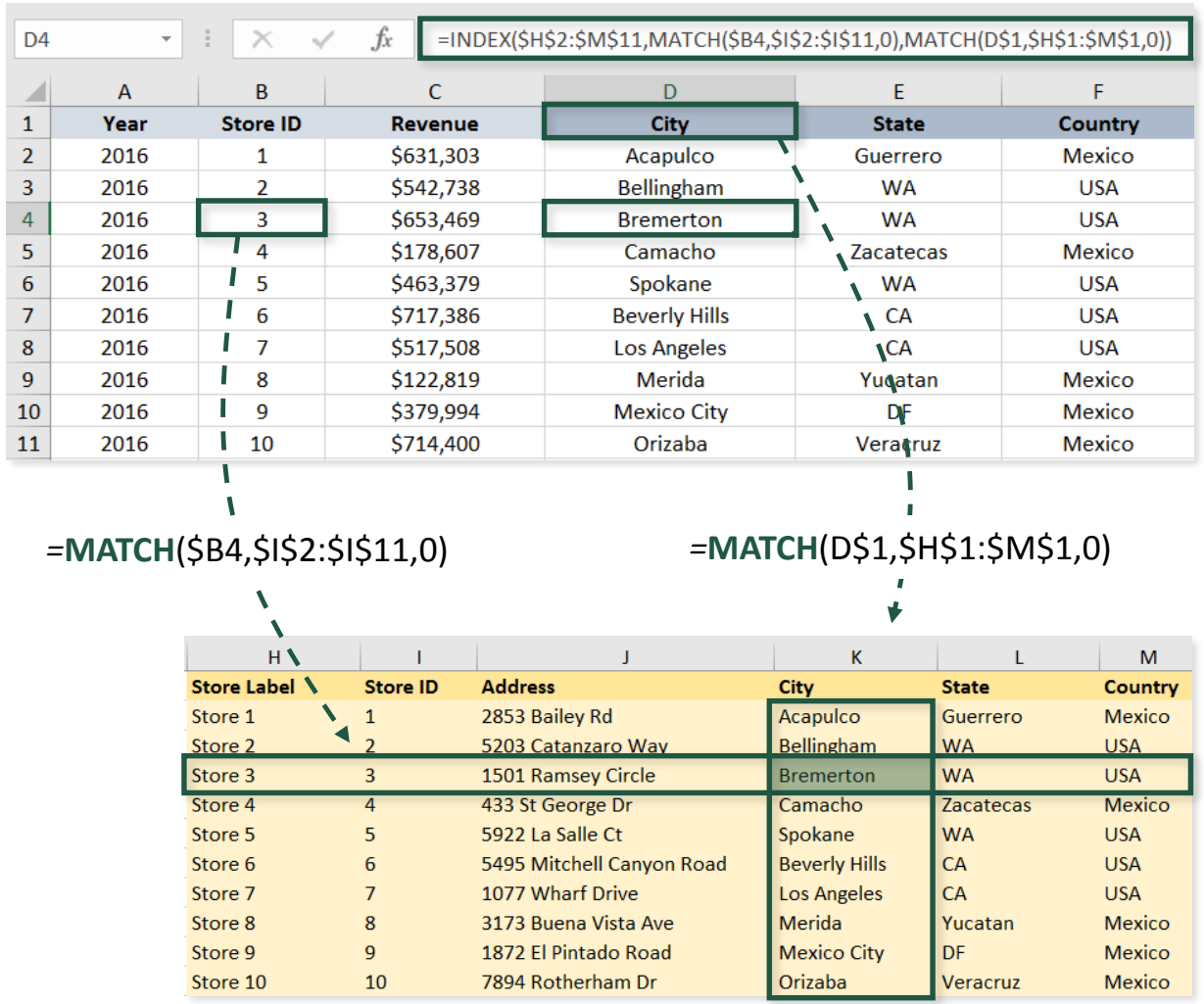
COMMON USE CASES:

- *Creating models or scenarios with randomized inputs*
- *Building your own hilarious band name generator (because why not!)*

PRO TIP

COMBINE INDEX & MATCH FOR A FLEXIBLE LOOKUP

- **INDEX** & **MATCH** offer several benefits over traditional VLOOKUP functions:
  - **No hard-coded column index**; MATCH can identify headers and automatically pull values from the correct columns
  - Lookup values **don't need to live in the first column** of the table array
  - **More flexibility** for custom or complex cases (*like finding the 2nd or 3rd matches*)



	A	B	C	D	E	F
1	Year	Store ID	Revenue	City	State	Country
2	2016	1	$631,303	Acapulco	Guerrero	Mexico
3	2016	2	$542,738	Bellingham	WA	USA
4	2016	3	$653,469	Bremerton	WA	USA
5	2016	4	$178,607	Camacho	Zacatecas	Mexico
6	2016	5	$463,379	Spokane	WA	USA
7	2016	6	$717,386	Beverly Hills	CA	USA
8	2016	7	$517,508	Los Angeles	CA	USA
9	2016	8	$122,819	Merida	Yucatan	Mexico
10	2016	9	$379,994	Mexico City	DF	Mexico
11	2016	10	$714,400	Orizaba	Veracruz	Mexico

	H	I	J	K	L	M
1	Store Label	Store ID	Address	City	State	Country
2	Store 1	1	2853 Bailey Rd	Acapulco	Guerrero	Mexico
3	Store 2	2	5203 Catanzaro Way	Bellingham	WA	USA
4	Store 3	3	1501 Ramsey Circle	Bremerton	WA	USA
5	Store 4	4	433 St George Dr	Camacho	Zacatecas	Mexico
6	Store 5	5	5922 La Salle Ct	Spokane	WA	USA
7	Store 6	6	5495 Mitchell Canyon Road	Beverly Hills	CA	USA
8	Store 7	7	1077 Wharf Drive	Los Angeles	CA	USA
9	Store 8	8	3173 Buena Vista Ave	Merida	Yucatan	Mexico
10	Store 9	9	1872 El Pintado Road	Mexico City	DF	Mexico
11	Store 10	10	7894 Rotherham Dr	Orizaba	Veracruz	Mexico

COMMON USE CASES:

- *Populating many lookup columns without having to manually update formulas*
- *Working with more complex scenarios (like lookups with multiple matches)*

FORMULAS

4 STARS (ADVANCED)

PRO TIP

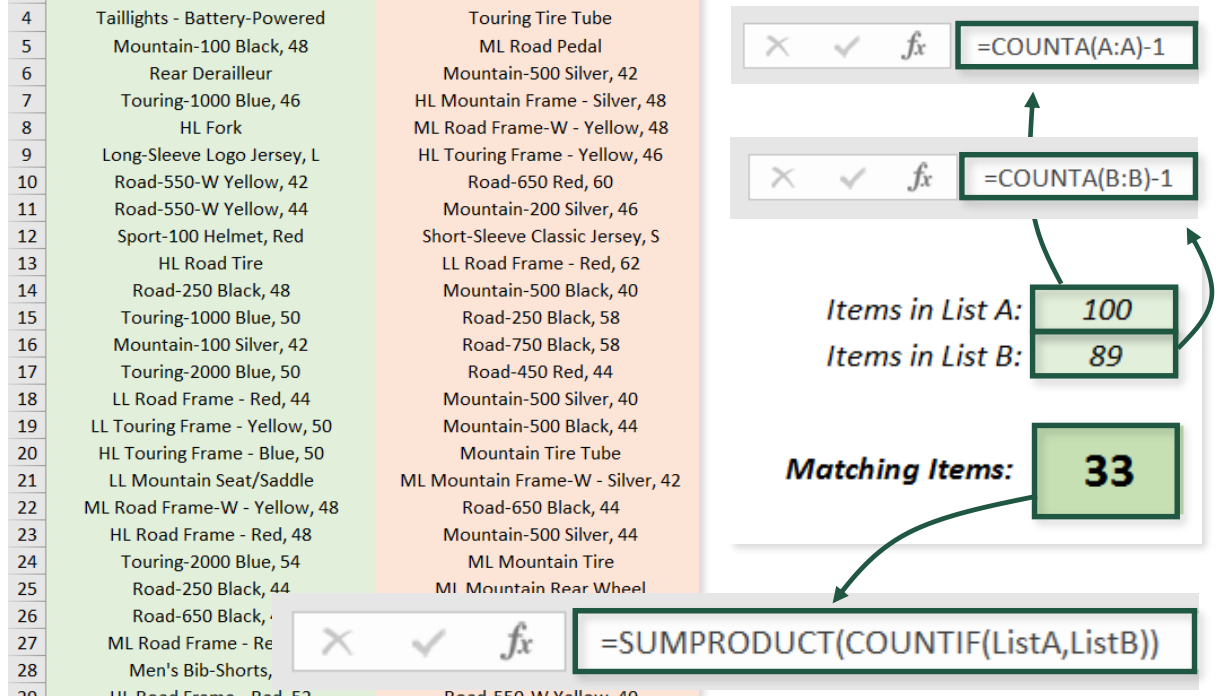
COUNT MATCHING ITEMS BETWEEN TWO LISTS

- Combine **SUMPRODUCT** with **COUNTIF** to return the number of matching values across multiple lists
- **SUMPRODUCT** acts like an array formula, iterating through each row of one list to search for matches in the other
  - **COUNTIF** assigns each row as a **1** (if a match exists) or a **0**, and **SUMPRODUCT** sums the resulting array of values
- **NOTE**: Either list can be used as the basis

FORMULAS

5 STARS (EXPERT)

	A	B
1	**Product List A**	**Product List B**
2	Racing Socks, L	Touring-1000 Blue, 50
3	HL Touring Frame - Blue, 46	Mountain-100 Black, 48
4	Taillights - Battery-Powered	Touring Tire Tube
5	Mountain-100 Black, 48	ML Road Pedal
6	Rear Derailleur	Mountain-500 Silver, 42
7	Touring-1000 Blue, 46	HL Mountain Frame - Silver, 48
8	HL Fork	ML Road Frame-W - Yellow, 48
9	Long-Sleeve Logo Jersey, L	HL Touring Frame - Yellow, 46
10	Road-550-W Yellow, 42	Road-650 Red, 60
11	Road-550-W Yellow, 44	Mountain-200 Silver, 46
12	Sport-100 Helmet, Red	Short-Sleeve Classic Jersey, S
13	HL Road Tire	LL Road Frame - Red, 62
14	Road-250 Black, 48	Mountain-500 Black, 40
15	Touring-1000 Blue, 50	Road-250 Black, 58
16	Mountain-100 Silver, 42	Road-750 Black, 58
17	Touring-2000 Blue, 50	Road-450 Red, 44
18	LL Road Frame - Red, 44	Mountain-500 Silver, 40
19	LL Touring Frame - Yellow, 50	Mountain-500 Black, 44
20	HL Touring Frame - Blue, 50	Mountain Tire Tube
21	LL Mountain Seat/Saddle	ML Mountain Frame-W - Silver, 42
22	ML Road Frame-W - Yellow, 48	Road-650 Black, 44
23	HL Road Frame - Red, 48	Mountain-500 Silver, 44
24	Touring-2000 Blue, 54	ML Mountain Tire
25	Road-250 Black, 44	ML Mountain Rear Wheel
26	Road-650 Black,	
27	ML Road Frame - Re	
28	Men's Bib-Shorts,	
29	HL Road Frame - Red, 52	Road-550-W Yellow, 40
30	Headlights - Dual-Beam	ML Touring Seat/Saddle
31	Road-250 Red, 44	Road-650 Red, 48



(*ListA* = A2:A101, *ListB* = B2:B90)

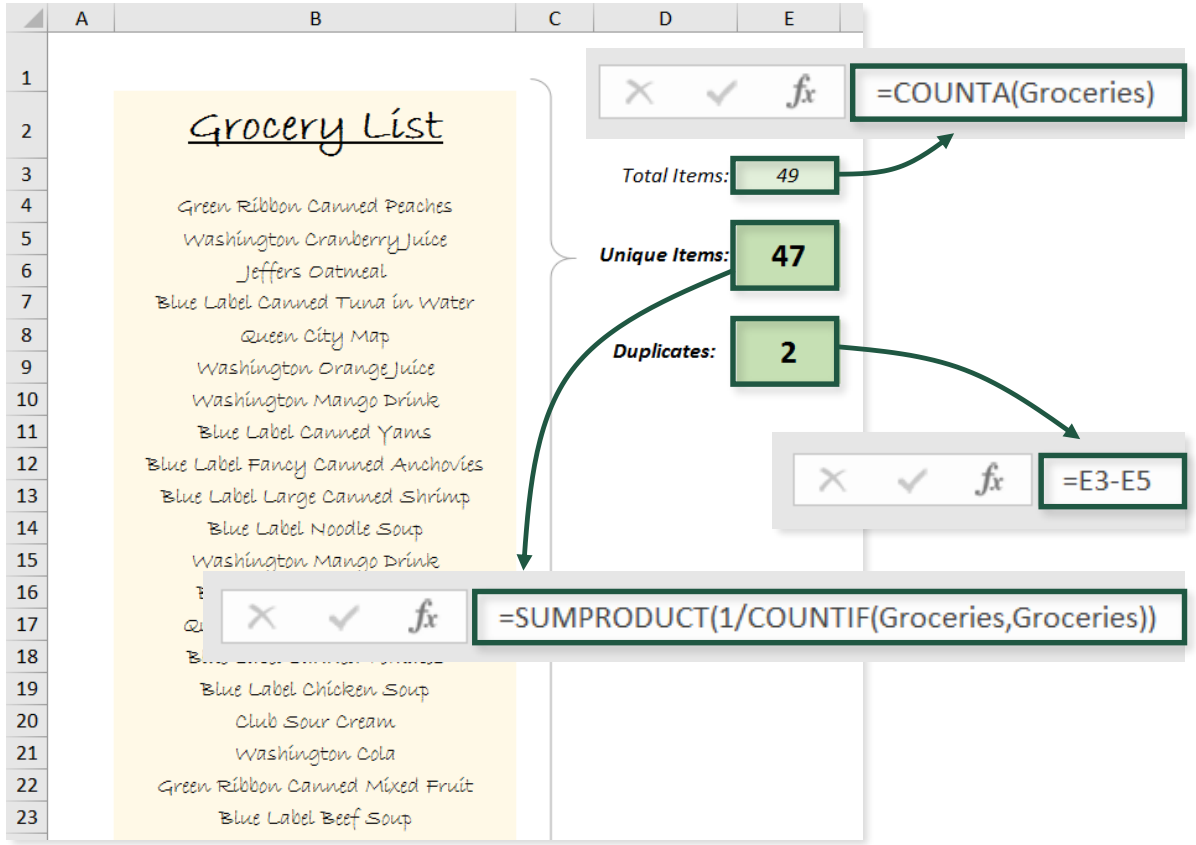
COMMON USE CASES:

- *Calculating the overlap between two lists*
- *Confirming that lists are unique to avoid double-counting errors*

PRO TIP

COUNT DUPLICATE OR UNIQUE ROWS IN A LIST

- Counting duplicates and uniques with cell formulas alone can be deceptively tricky
  - **NOTE**: Tools like Pivots or Power Query (*which includes a "Count Distinct" option*) often offer simpler alternatives
- One option is to use **SUMPRODUCT** and **COUNTIF** to count *unique* values, and then subtract from the total count to identify the number of *duplicate* values



(*Groceries* = B4:B52)

FORMULAS

5 STARS (EXPERT)

COMMON USE CASES:

- *Calculating the number of unique or distinct products ordered*
- *Error-checking to avoid inflation caused by duplicate records*

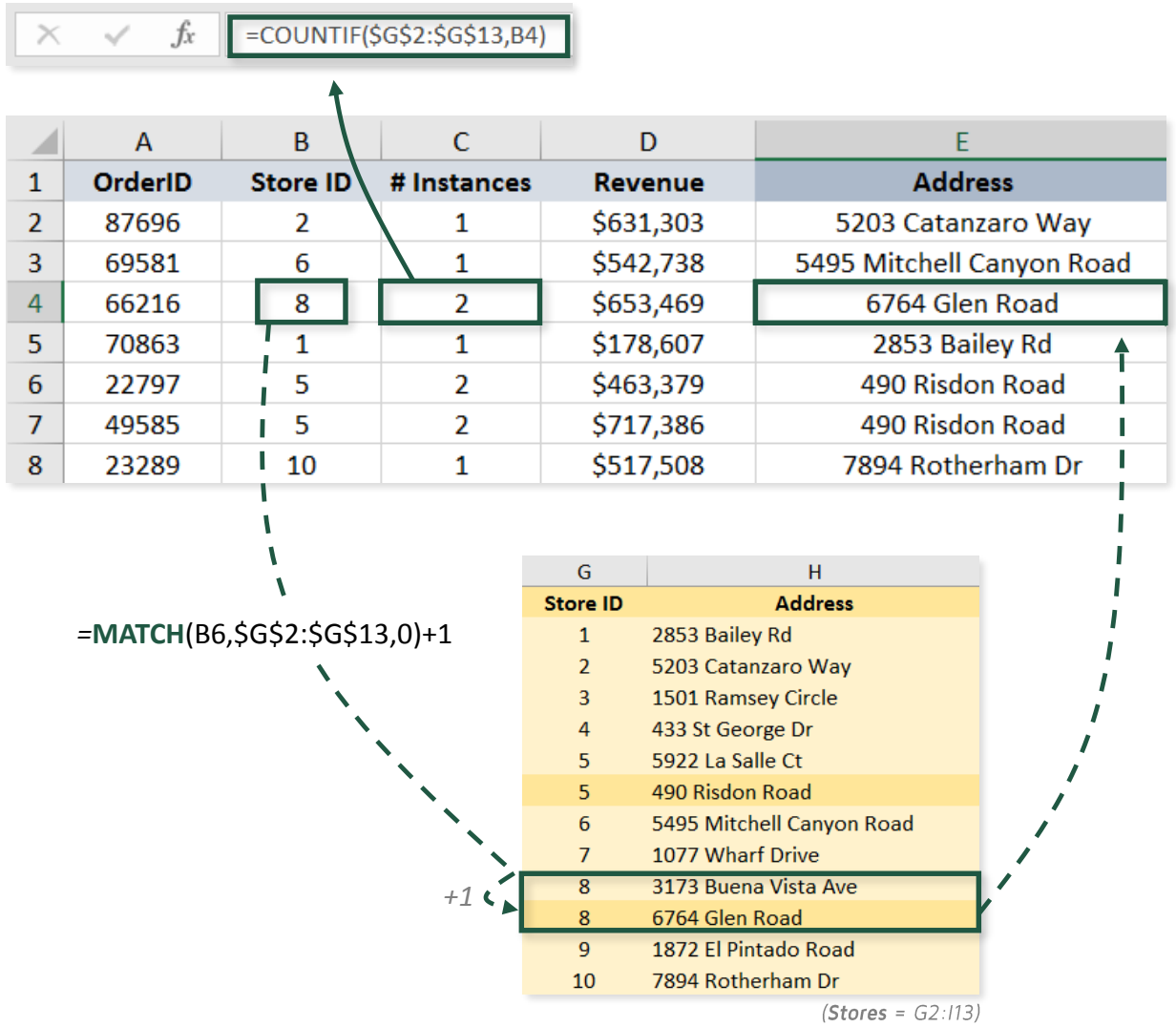
PRO TIP

USE COUNTIF TO HANDLE MANY-TO-MANY LOOKUPS

- Excel lookup formulas are designed to work with **1-to-1** or **1-to-Many** relationships (*only one instance of each primary key*)
- If you have a **Many-to-Many relationship** (multiple matches), you can use **COUNTIF**, **INDEX** and **MATCH** formulas to return values from subsequent matches
- **NOTE**: Make sure to understand *why* there are multiple instances of your key, and confirm that they are valid records

FORMULAS

5 STARS (EXPERT)



	A	B	C	D	E
1	OrderID	Store ID	# Instances	Revenue	Address
2	87696	2	1	$631,303	5203 Catanzaro Way
3	69581	6	1	$542,738	5495 Mitchell Canyon Road
4	66216	8	2	$653,469	6764 Glen Road
5	70863	1	1	$178,607	2853 Bailey Rd
6	22797	5	2	$463,379	490 Risdon Road
7	49585	5	2	$717,386	490 Risdon Road
8	23289	10	1	$517,508	7894 Rotherham Dr

G	H
Store ID	Address
1	2853 Bailey Rd
2	5203 Catanzaro Way
3	1501 Ramsey Circle
4	433 St George Dr
5	5922 La Salle Ct
5	490 Risdon Road
6	5495 Mitchell Canyon Road
7	1077 Wharf Drive
8	3173 Buena Vista Ave
8	6764 Glen Road
9	1872 El Pintado Road
10	7894 Rotherham Dr

COMMON USE CASES:

- *Tracking historical changes within an existing lookup table (i.e. product prices or store addresses that may have changed at one point in time)*

VISUALIZATION TIPS

EXCEL MAVEN

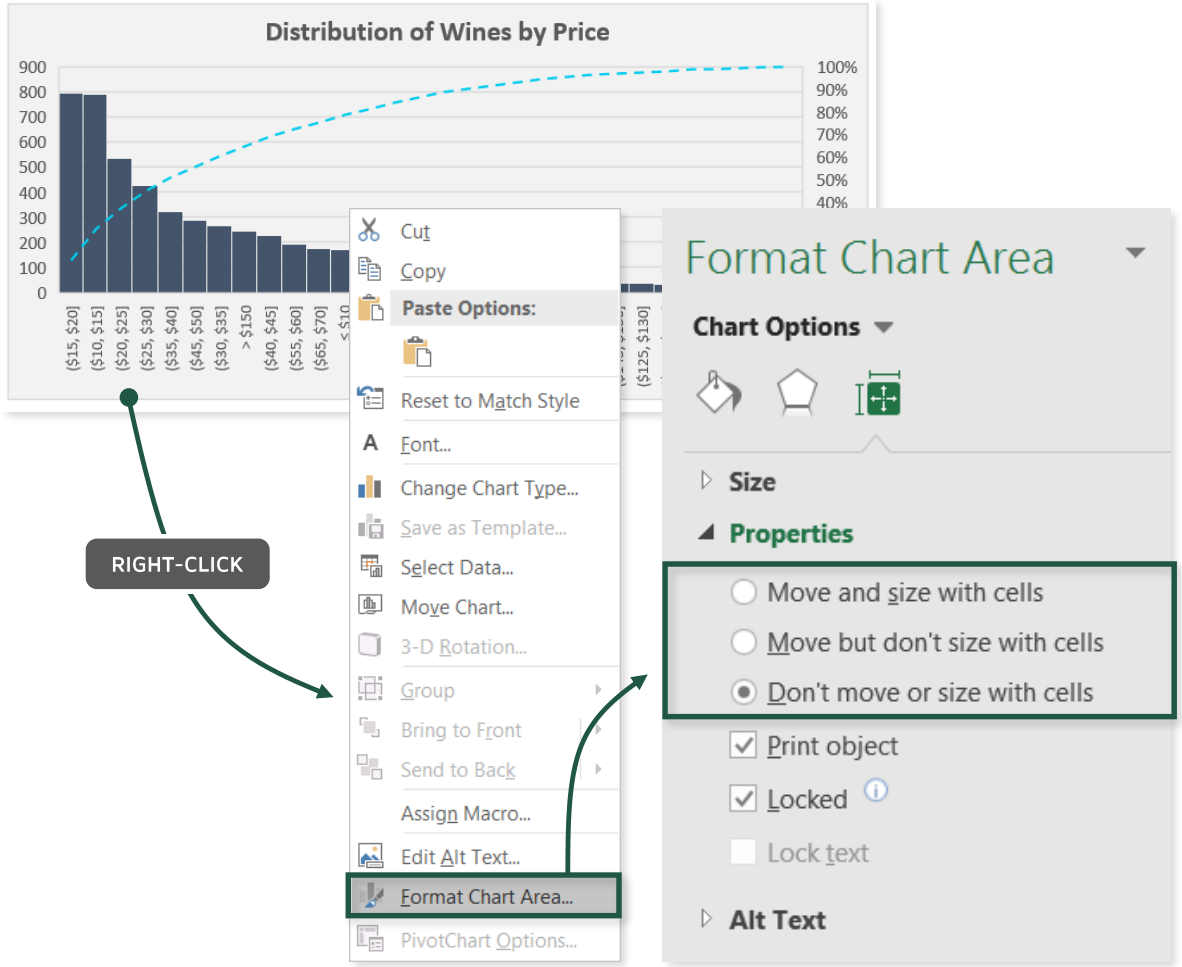
PRO TIP

PREVENT CHARTS FROM MOVING OR SIZING

- By default, charts **move** and **size** with cells, stretching and shifting as underlying rows or columns are resized
- Change chart behavior using the **Format Chart Area** options, under **Properties**:
  - **Move and Size:** Allows charts to move and stretch with columns or rows
  - **Move but don't Size:** Allows charts to move but not stretch or distort
  - **Don't move or Size:** Fixes chart position in place, regardless of how cells change

DATA VIZ

1 STAR (VERY BASIC)



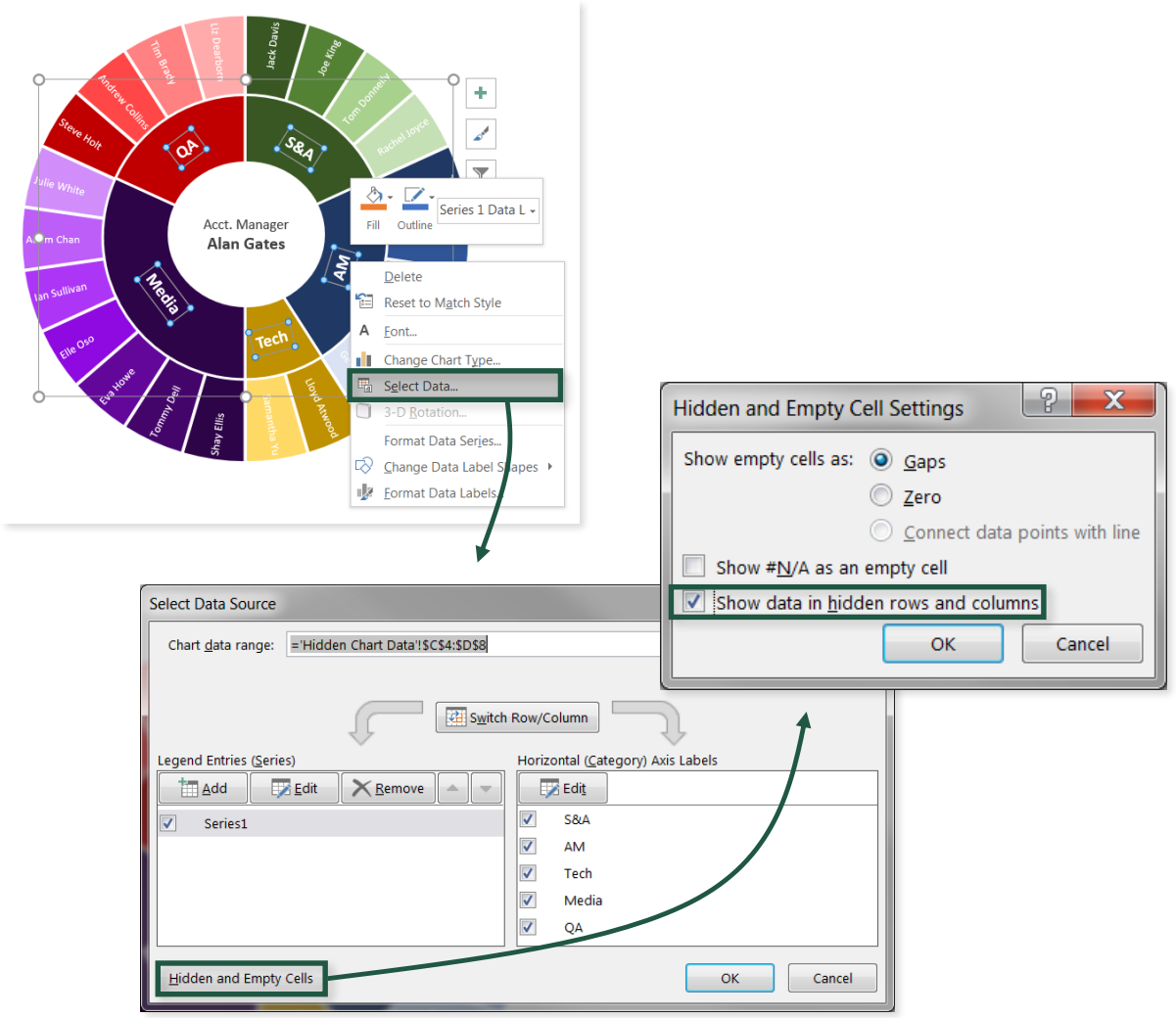
COMMON USE CASES:

- *Locking in chart positions within reports or dashboards to prevent them from distorting if underlying rows or columns are modified*
- *Allowing charts to move (but not stretch) if rows are filtered or sorted*

PRO TIP

ALLOW CHARTS TO REFERENCE HIDDEN DATA

- By default, Excel charts will not display source data in **hidden rows or columns**
- To allow charts to display hidden data, right-click, **Select Data**, click **Hidden and Empty Cells**, and activate the "***Show data in hidden rows and columns***" checkbox



DATA VIZ

1 STAR (VERY BASIC)

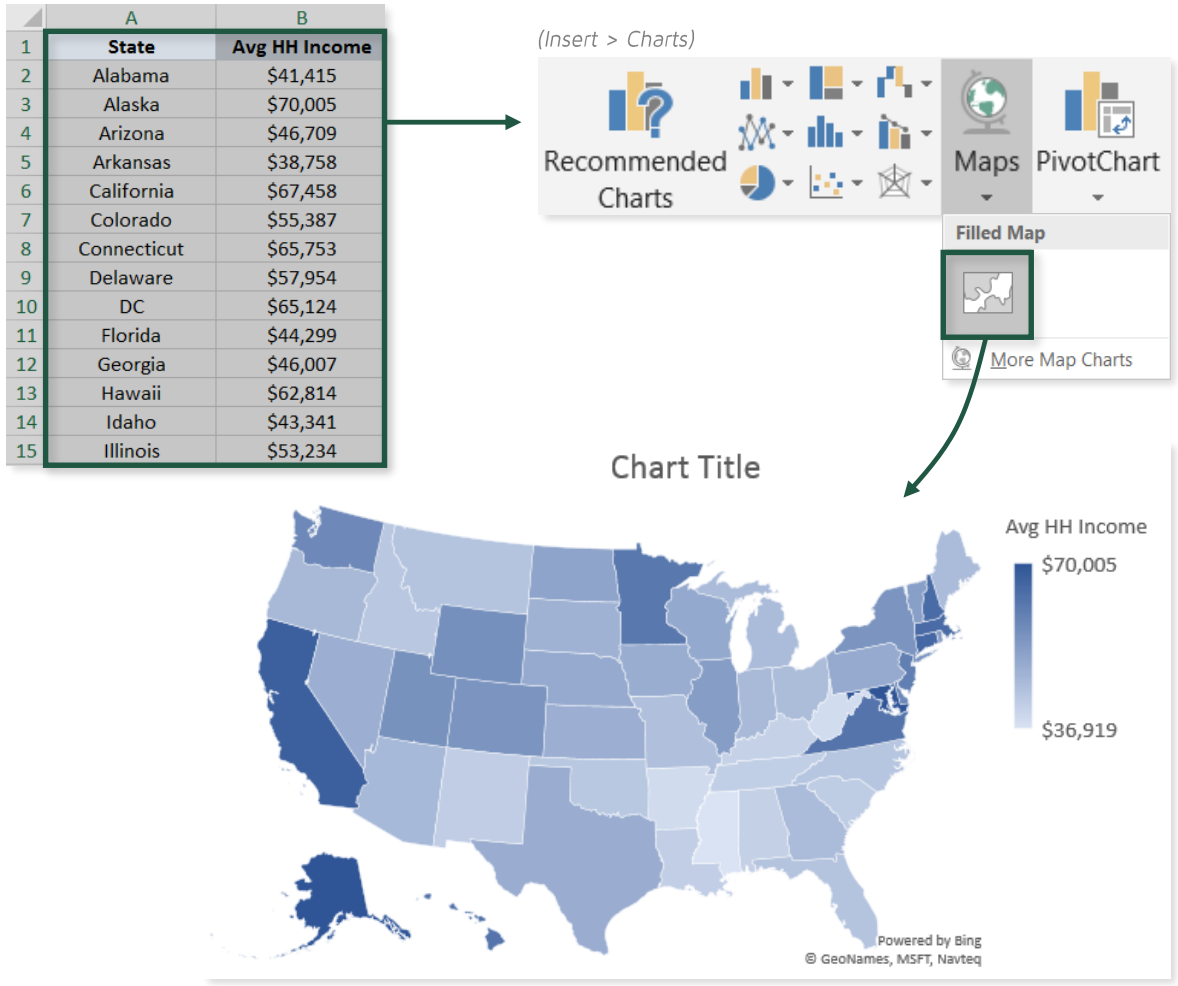
COMMON USE CASES:

- *Designing reports with the source data and visuals on the same tab*
- *Preventing users from accessing or modifying a chart's raw source data*

PRO TIP

USE FILLED MAPS FOR GEOSPATIAL ANALYSIS

- Recent versions of Excel (*2016/365*) include a **Filled Map** visual to plot geospatial data
  - Like dates, Excel automatically recognizes many types of geospatial information (*states, countries, zip codes, lat/long, etc.*)
  - If some values aren't recognized, you may see a warning icon with a statement like "*We plotted 80% of the locations from your data with high confidence*"
- **NOTE**: Filled maps are currently unavailable in older versions of Excel (pre-2016)



COMMON USE CASES:

- *Quickly visualizing regional patterns or trends*
- *Comparing census information like population, GDP, household income, or birth rates across countries*

DATA VIZ

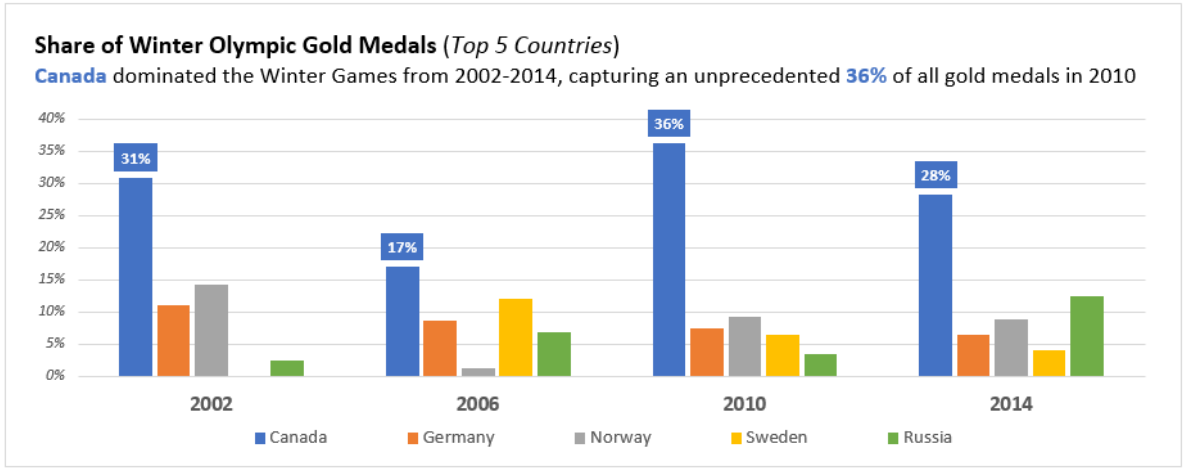
1 STAR (VERY BASIC)

PRO TIP

CUSTOMIZE YOUR CHARTS TO TELL A STORY

- Don't settle for boring, default formats; use **chart titles** and **formatting tools** to add insight and bring your data to life
  - **Tell a story** in the chart title or subtitle rather than making the user interpret it
  - Add **individual labels or markers** to draw attention to key points
  - **Use colors** consistently and deliberately
- **TIP**: Add text or shapes while the chart is selected to automatically group the objects

	% of Winter Olympic Gold Medals (Top 5 Countries)			
	2002	2006	2010	2014
Canada	31%	17%	36%	28%
Germany	11%	9%	7%	6%
Norway	14%	1%	9%	9%
Sweden	0%	12%	6%	4%
Russia	2%	7%	3%	12%



COMMON USE CASES:

- *Preparing visuals for use in presentations, where comprehension is key*
- *Reducing the chance of users misinterpreting the story that a chart or graph is designed to communicate*

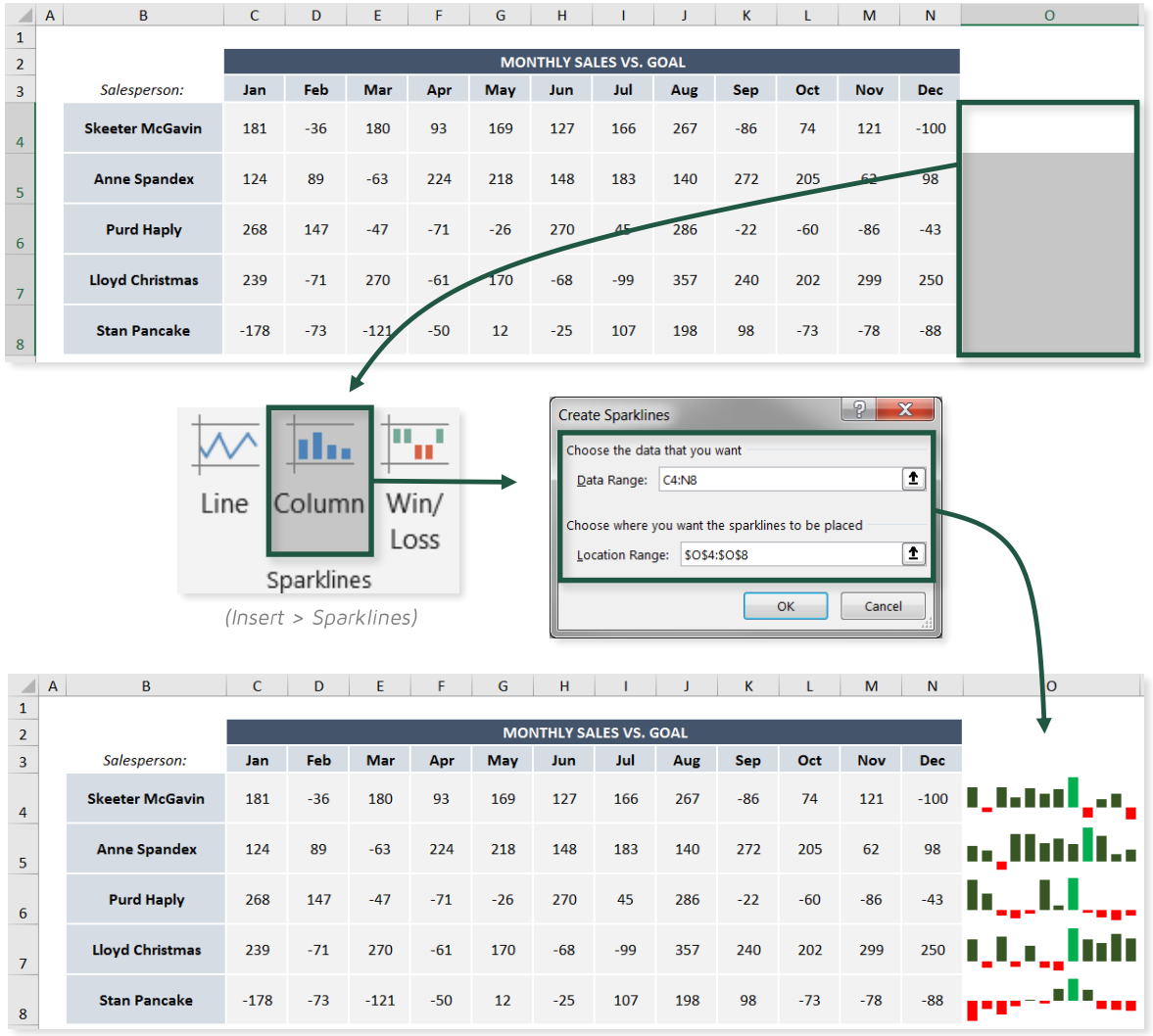
DATA VIZ

2 STARS (BASIC)

PRO TIP

QUICKLY VISUALIZE TRENDS WITH SPARKLINES

- **Sparklines** are basically tiny charts that live within the confines of a single cell
- While formatting options are somewhat limited, sparklines can be a great way to quickly (*and subtly*) visualize patterns
  - Use the **Sparkline Tools** ribbon to change chart types, highlight specific points (*high/low, first/last, +/-*), and adjust colors, markers, or axis options



COMMON USE CASES:

- *Embedding a simple line or column sparkline to show trends at a glance*
- *Creating KPI cells to show key metrics with sparklines in the background*

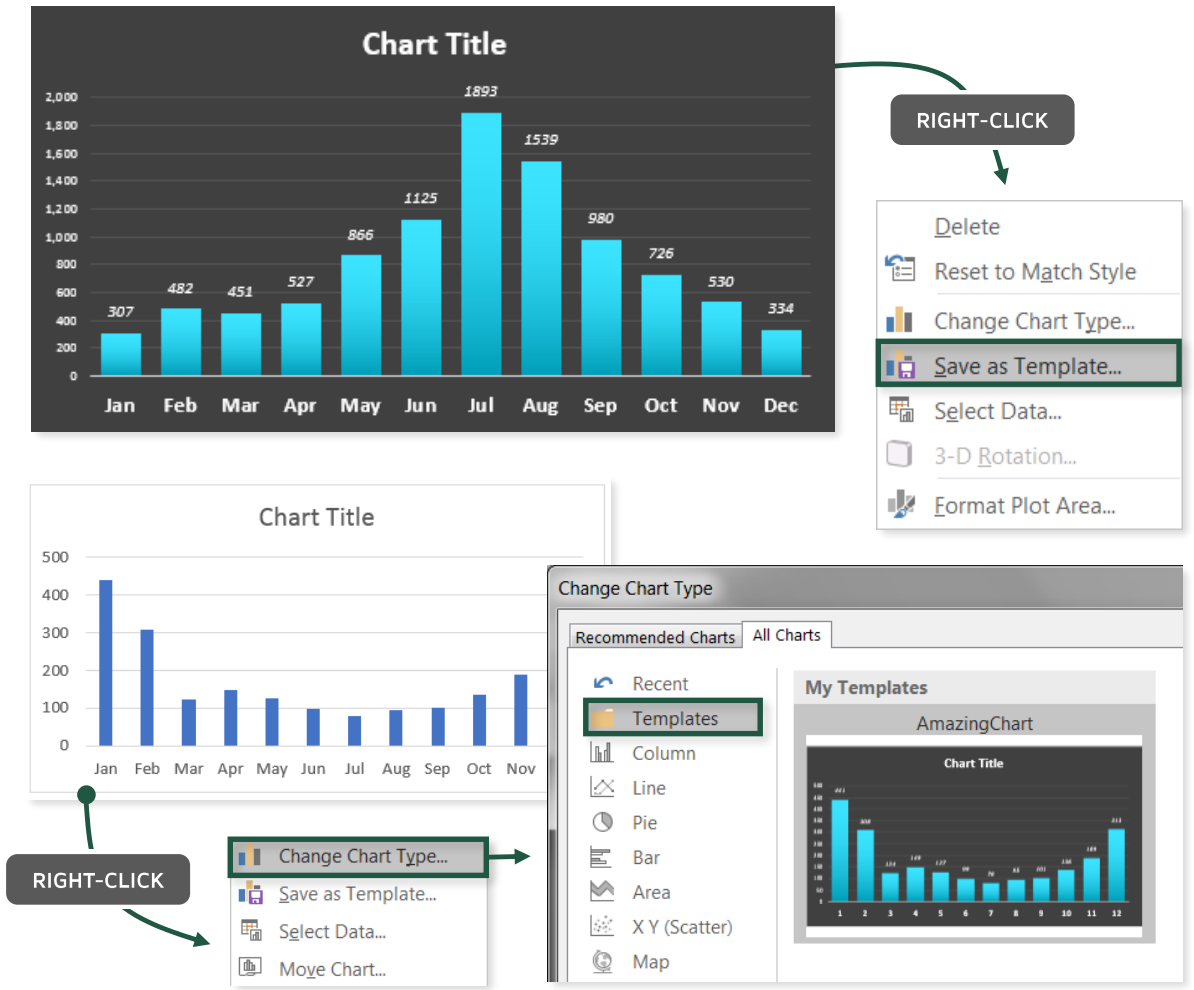
DATA VIZ

2 STARS (BASIC)

PRO TIP

DESIGN & APPLY CUSTOM CHART TEMPLATES

- To apply a specific set of formatting rules across multiple charts, you can create your own **custom chart templates**
  - Once you've applied the formatting you'd like to replicate, **right-click** the chart and select **Save As Template**
  - To apply the template to a new chart, navigate to the **Templates** folder within the **Change Chart Type** dialog box



COMMON USE CASES:

- *Saving specific formats for future use in other workbooks*
- *Ensuring consistent formatting across all visuals in a report*

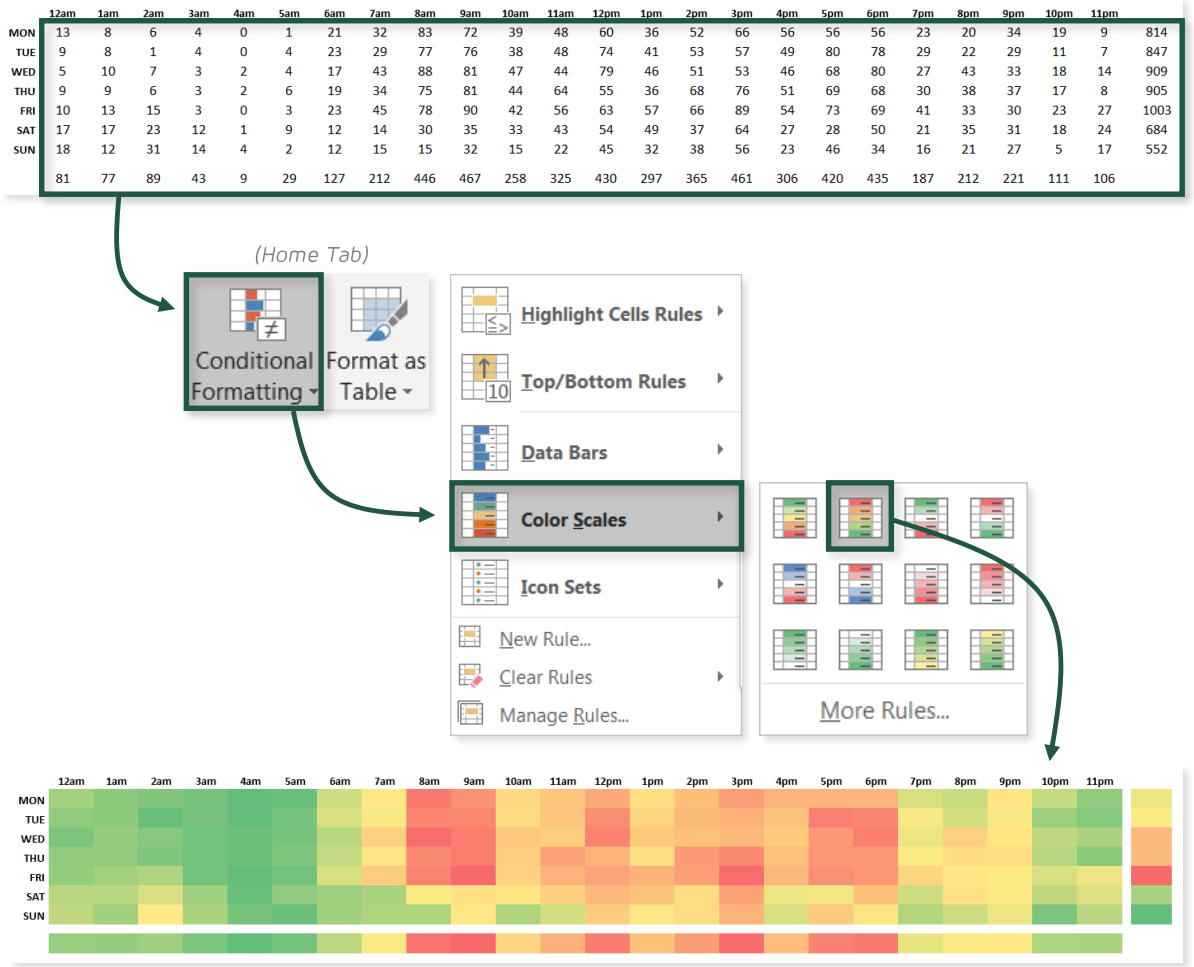
DATA VIZ

2 STARS (BASIC)

PRO TIP

DESIGN YOUR OWN CONDITIONAL HEAT MAPS

- One of the most effective visualizations in Excel isn't a chart at all, but a **heat map** created with **conditional formatting**
  - Select your source data and navigate to ***Home > Conditional Formatting > Color Scales*** to format cells based on their values
- **BONUS**: To hide text, apply a **Custom** number format and set the **Type** to ";;;"



COMMON USE CASES:

- *Quickly identifying patterns or trends in the underlying values using common color scales (red to green, white to red, etc)*
- *Highlighting "hot spots" or outliers based on the colors alone*

DATA VIZ

2 STARS (BASIC)

PRO TIP

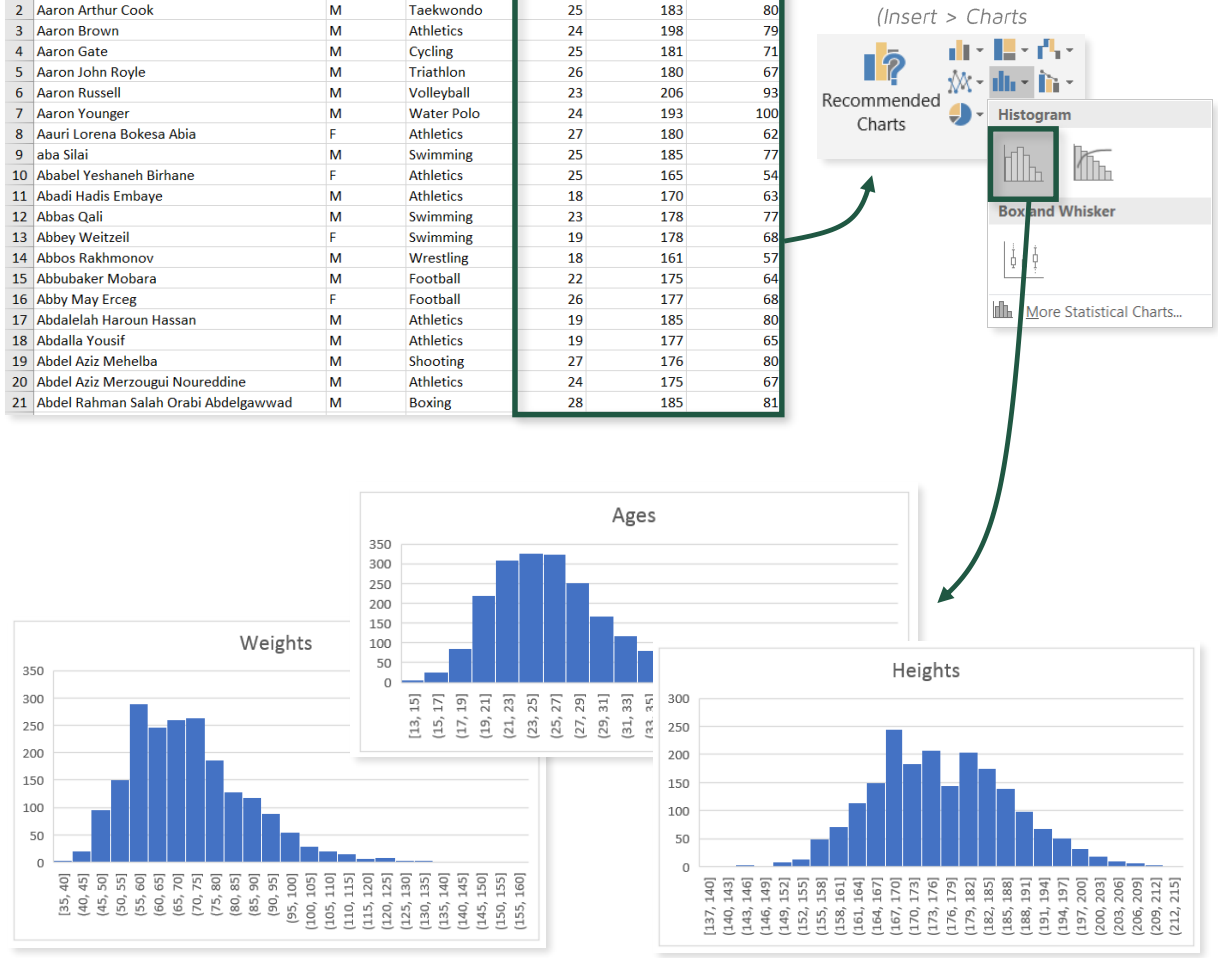
ANALYZE DISTRIBUTIONS WITH HISTOGRAMS

- **Histograms** are a great tool to understand and visualize how data is distributed
  - Unlike traditional column charts, Histograms show the **frequency** of observations that fall within given ranges of values (aka "**bins**")
- **NOTE**: Histograms are currently unavailable in older versions of Excel (pre-2016)

DATA VIZ

3 STARS (MODERATE)

	A	B	C	D	E	F
1	Name	Gender	Sport	Age	Height_cm	Weight_kg
2	Aaron Arthur Cook	M	Taekwondo	25	183	80
3	Aaron Brown	M	Athletics	24	198	79
4	Aaron Gate	M	Cycling	25	181	71
5	Aaron John Royle	M	Triathlon	26	180	67
6	Aaron Russell	M	Volleyball	23	206	93
7	Aaron Younger	M	Water Polo	24	193	100
8	Aauri Lorena Bokesa Abia	F	Athletics	27	180	62
9	aba Silai	M	Swimming	25	185	77
10	Ababel Yeshaneh Birhane	F	Athletics	25	165	54
11	Abadi Hadis Embaye	M	Athletics	18	170	63
12	Abbas Qali	M	Swimming	23	178	77
13	Abbey Weitzeil	F	Swimming	19	178	68
14	Abbos Rakhmonov	M	Wrestling	18	161	57
15	Abbubaker Mobara	M	Football	22	175	64
16	Abby May Erceg	F	Football	26	177	68
17	Abdalelah Haroun Hassan	M	Athletics	19	185	80
18	Abdalla Yousif	M	Athletics	19	177	65
19	Abdel Aziz Mehelba	M	Shooting	27	176	80
20	Abdel Aziz Merzougui Noureddine	M	Athletics	24	175	67
21	Abdel Rahman Salah Orabi Abdelgawwad	M	Boxing	28	185	81



COMMON USE CASES:

- *Visualizing population demographics (height, weight, income, age, etc.)*
- *Identifying the most and least common values in a sample*

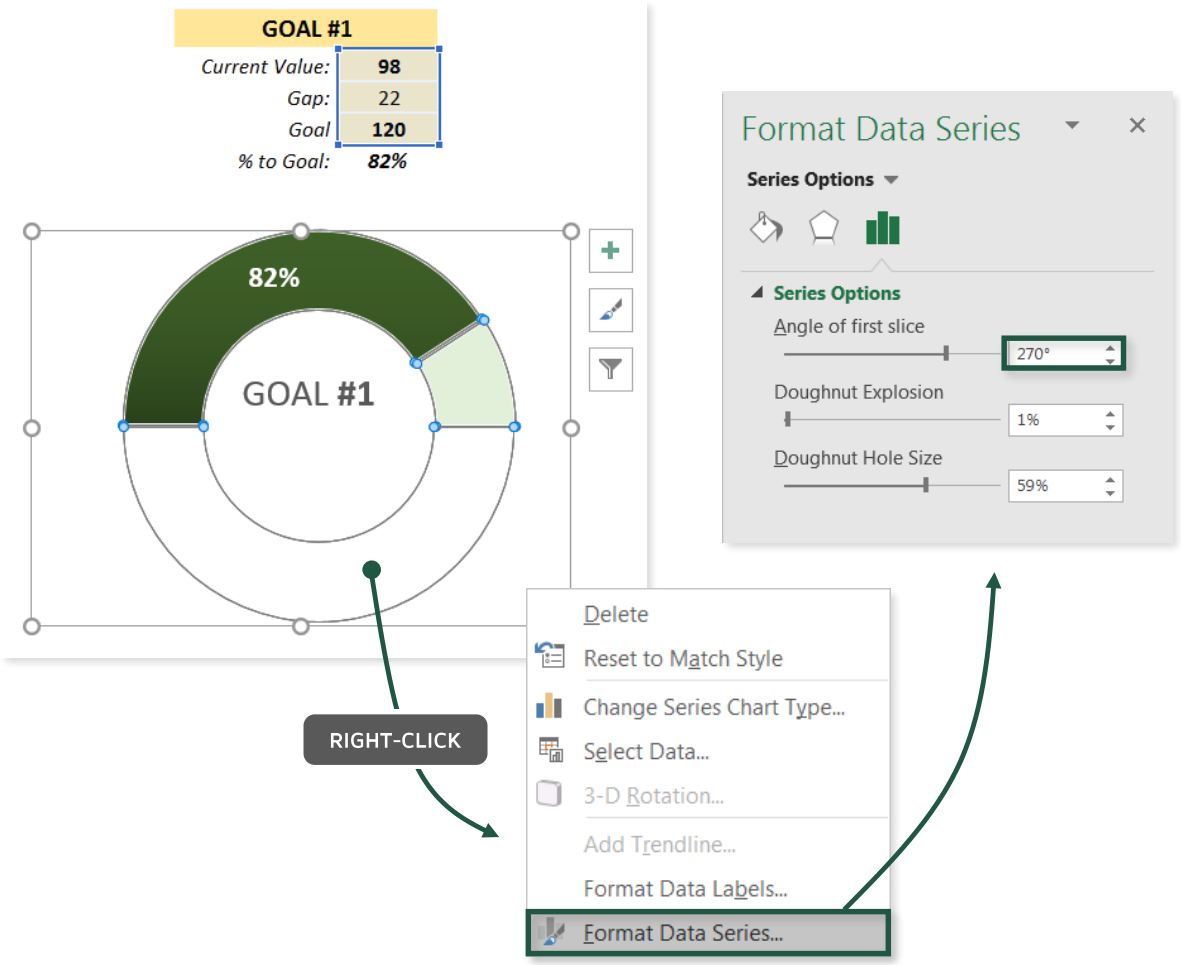
PRO TIP

TURN DONUTS INTO CUSTOM GAUGE CHARTS

- To show progress towards a goal, create a custom **Gauge Chart** by formatting a standard **Pie** or **Donut Chart**:
  - **STEP 1**: Create cells with 3 values: **Current Value**, **Gap**, and **Goal** (*in that order*)
  - **STEP 2**: Insert a **Donut or Pie Chart**, and format the **Goal** series with no fill or border
  - **STEP 3**: Format the data series, and change the *Angle of First Slice* to **270 degrees**
- **TIP**: Add a "*% to Goal*" calculation, and insert a data label linked to that cell value

DATA VIZ

3 STARS (MODERATE)



COMMON USE CASES:

- *Tracking performance against benchmarks or goals*
- *Designing executive scorecards to visualize key metrics at a glance*

PRO TIP

USE COMBO CHARTS TO HIGHLIGHT DATE RANGES

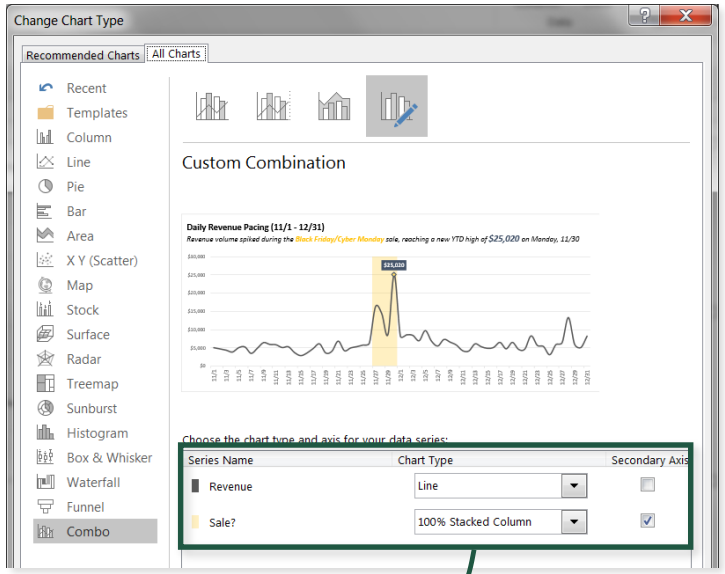
- Use **Combo Charts** to visually highlight specific time periods within a line chart:
  - **STEP 1**: Add a new column and "flag" key rows with a "**1**" (leave others **blank** or **0**)
  - **STEP 2**: Add the new series, change the type to **Combo**, and plot it as a **100% Stacked Column**, on the **Secondary Axis**
  - **STEP 3**: Right-click to format the new series and change the **Gap Width** to **0%**
- **TIP**: Apply a custom number format (;;;) to the secondary axis to hide the labels

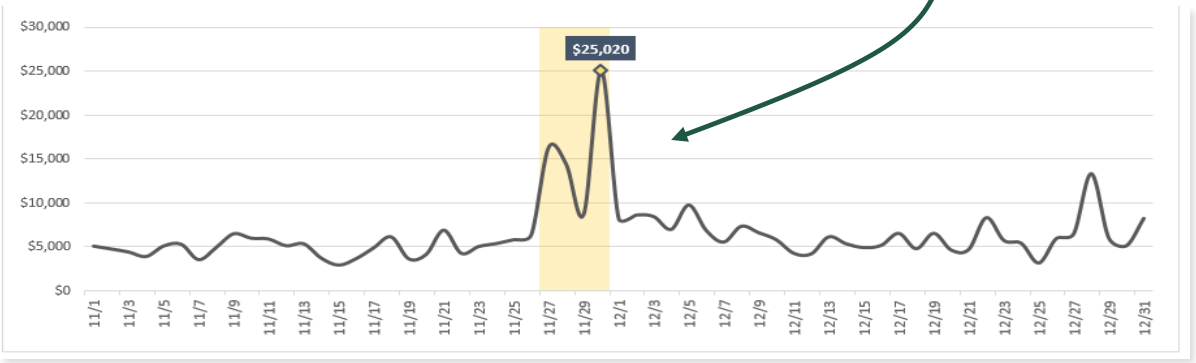
DATA VIZ

4 STARS (ADVANCED)

(Chart Tools > Type)

	A	B	C
1	Date	Revenue	Sale?
17	11/16/2015	$3,598	
18	11/17/2015	$4,830	
19	11/18/2015	$6,077	
20	11/19/2015	$3,572	
21	11/20/2015	$4,077	
22	11/21/2015	$6,842	
23	11/22/2015	$4,201	
24	11/23/2015	$4,976	
25	11/24/2015	$5,309	
26	11/25/2015	$5,749	
27	11/26/2015	$6,278	
28	11/27/2015	$16,279	1
29	11/28/2015	$14,333	1
30	11/29/2015	$8,638	1
31	11/30/2015	$25,020	1
32	12/1/2015	$8,142	
33	12/2/2015	$8,561	
34	12/3/2015	$8,390	
35	12/4/2015	$6,925	
36	12/5/2015	$9,707	





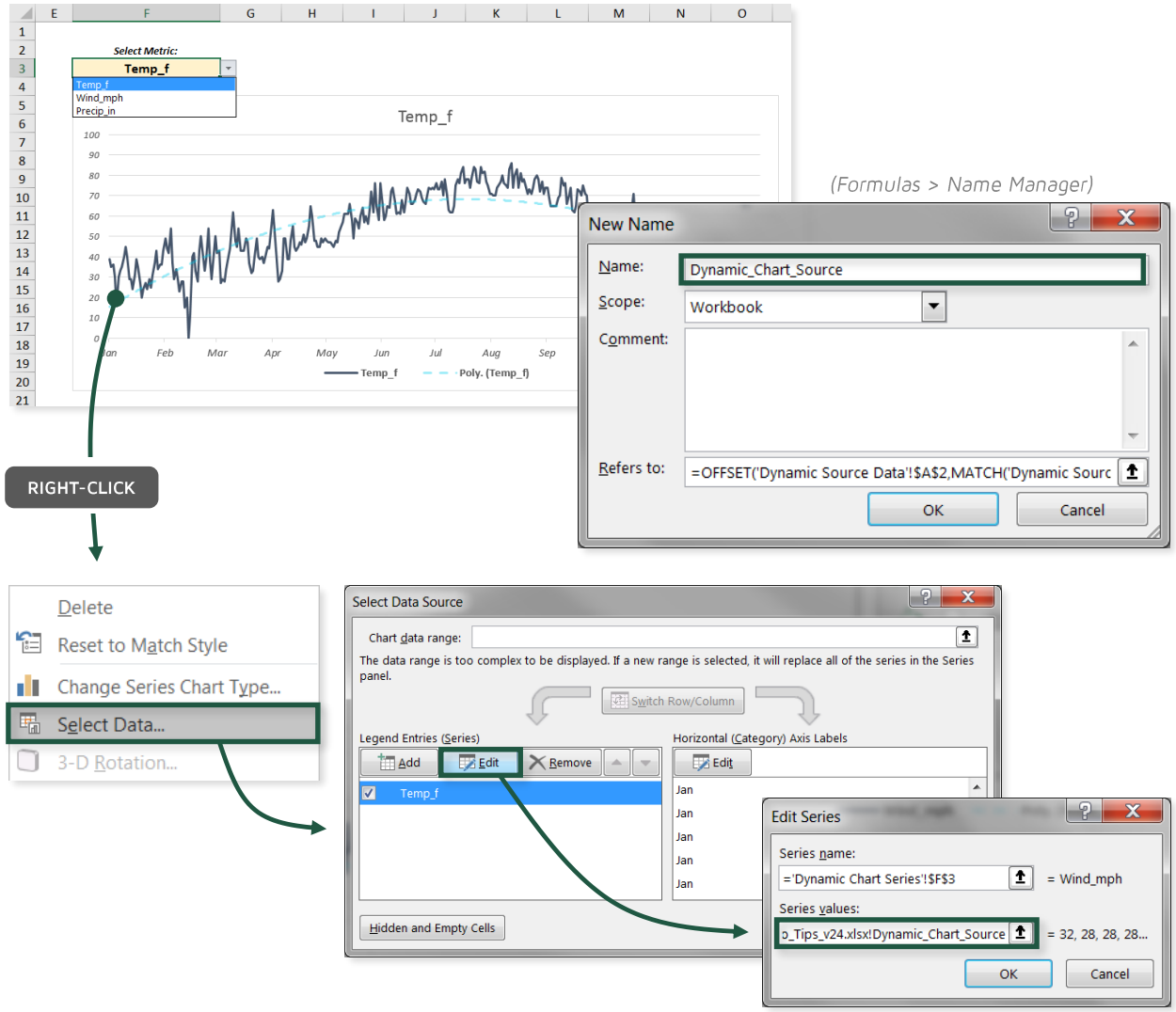
COMMON USE CASES:

- *Highlighting sales or promotional periods to add context to the chart*
- *Drawing attention to seasonal patterns or trends (peak vs. off-peak, weekends vs. weekdays, etc.)*

PRO TIP

CREATE DYNAMIC SOURCE DATA WITH NAMED RANGES

- **Named Ranges** allow you to define dynamic cell references and charts using functions like **OFFSET** and **MATCH**:
  - **STEP 1**: Create a **data validation** list containing each column header
  - **STEP 2**: Add a named range using **OFFSET**, **MATCH** and **COUNTA** to dynamically reference the column selected from the list
  - **STEP 3**: Insert a new chart, edit the data source, and reference the named range in **Series Values** (*after the workbook name*)



DATA VIZ

5 STARS (EXPERT)

COMMON USE CASES:

- *Creating interactive dashboards with user-controlled charts and graphs*
- *Consolidating multiple visuals into a single chart template*

PRO TIP

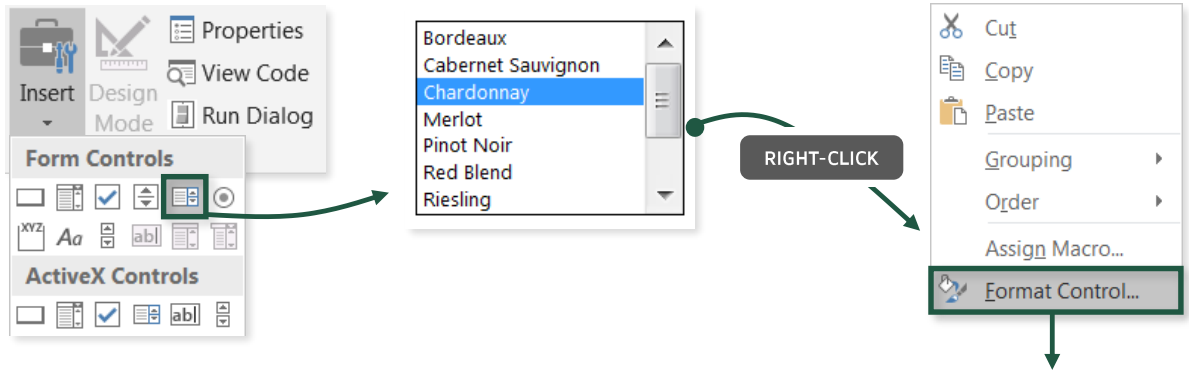
ADD FORM CONTROLS FOR INTERACTIVE VISUALS

- Use **Form Controls** to create interactive charts based on lists, buttons or scroll bars:
  - **STEP 1**: Insert a **List Box**, with an *Input Range* referencing a unique list of options and a *Cell Link* to store the selected value
  - **STEP 2**: Use cell functions (**INDEX**, **MATCH** **VLOOKUP**, etc.) to create the chart source data, based on the list box selection
- **NOTE**: If you don't see the **Developer** tab, check ***File > Options > Customize Ribbon***

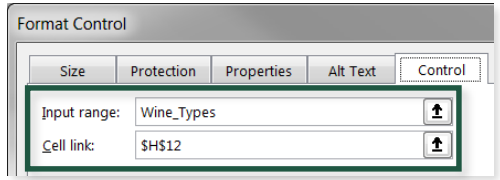
DATA VIZ

5 STARS (EXPERT)

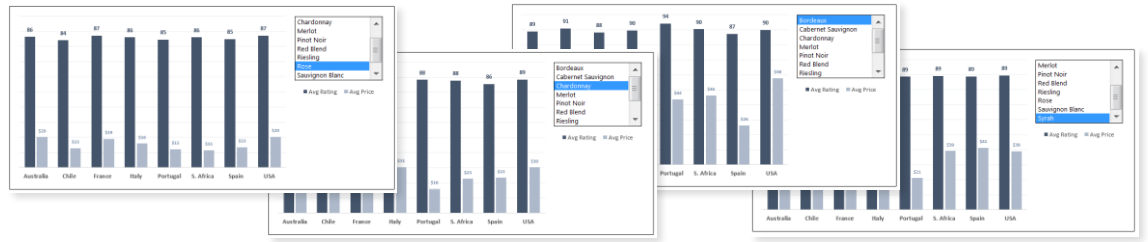
(Developer > Controls)



	Avg Rating	Avg Price
Australia	89.16	$22.02
Chile	85.63	$16.25
France	90.78	$32.24
Italy	88.25	$24.50
Portugal	86.50	$18.00
S. Africa	87.00	$18.50
Spain	82.00	$20.00
USA	87.91	$19.28



*For this demo we'll use **AVERAGEIFS** and **VLOOKUP** to aggregate ratings and prices by country, based on the wine type selected*



COMMON USE CASES:

- *Integrating user controls into a dynamic report or dashboard*
- *Consolidating multiple visuals into a single chart template*

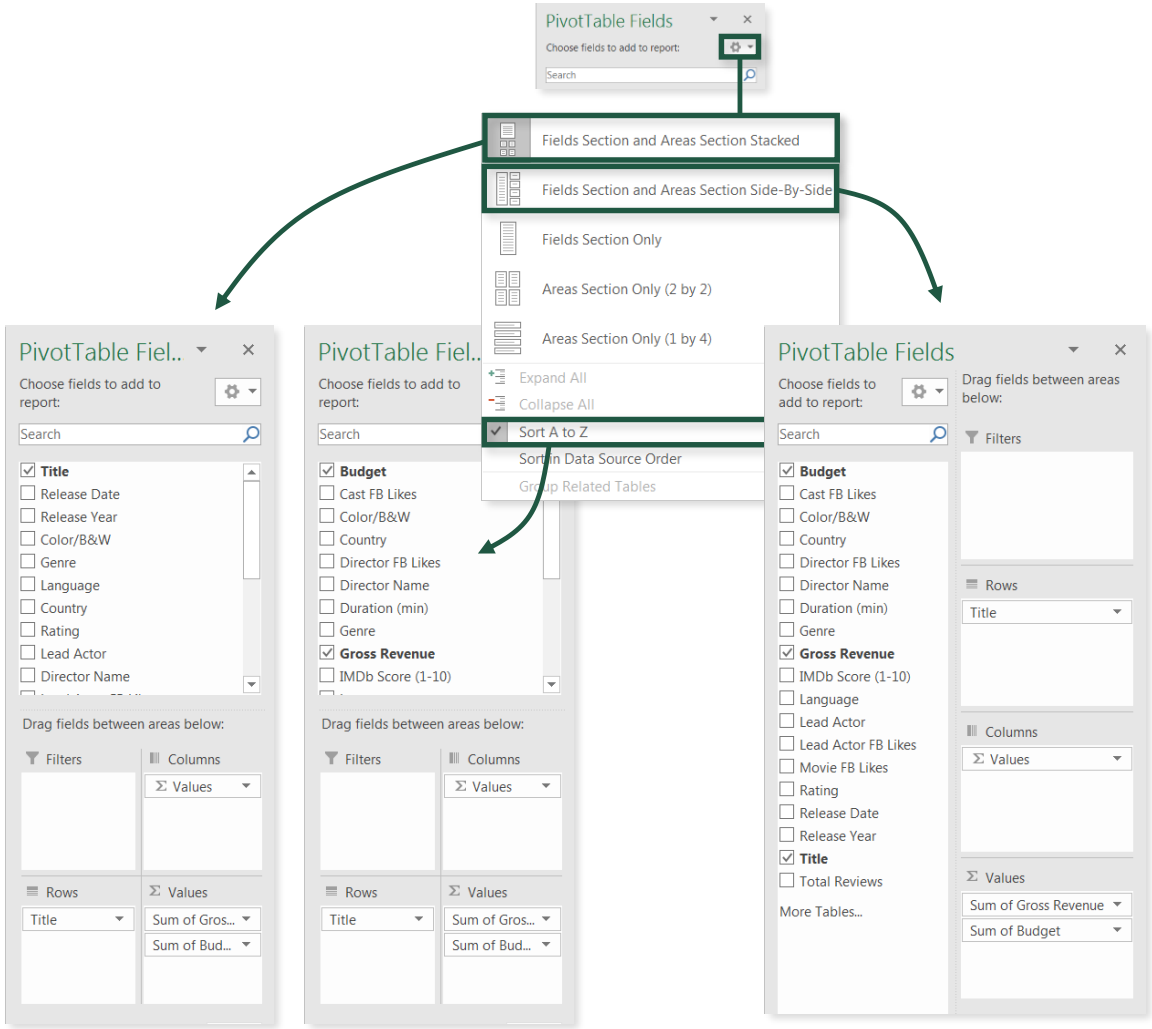
PIVOT TABLE TIPS

EXCEL MAVEN

PRO TIP

USE FIELD LIST OPTIONS TO STAY ORGANIZED

- The **Field List** is typically "*stacked*" by default, and sorted in **Data Source** order
- If you have a large number of source tables or fields, you may want to adjust settings:
  - **Fields & Areas Side-by-Side**: Displays all tables and fields in one vertical pane, next to the Filter/Rows/Columns/Values options
  - **Sort A to Z**: Sorts column names in alphabetical order, rather than source order (*note that you can **search** for fields as well*)



COMMON USE CASES:

- *Updating the layout to show all tables and fields in one vertical pane for easy access (especially for Power Pivot views with many tables)*
- *Sorting fields alphabetically when the source order isn't meaningful*

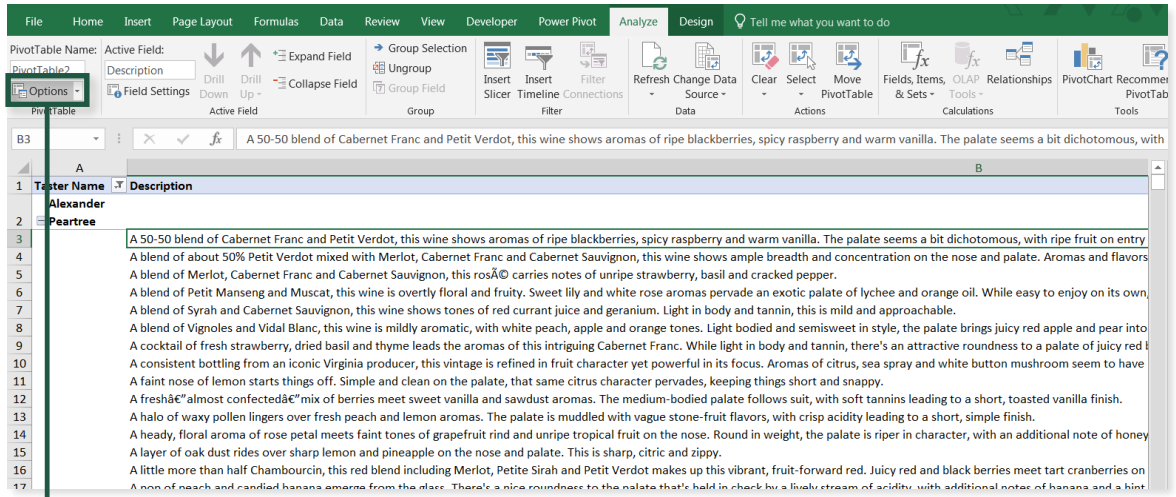
PIVOT TABLES

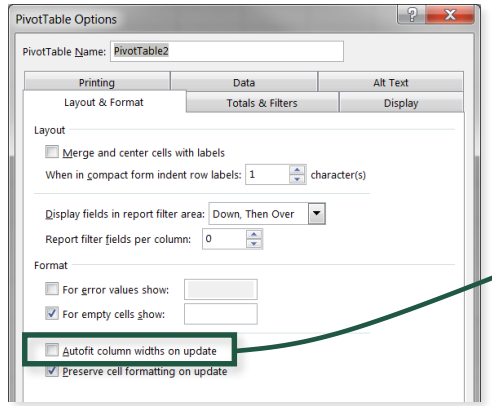
1 STAR (VERY BASIC)

PRO TIP

PREVENT COLUMNS FROM AUTOFITTING WIDTH

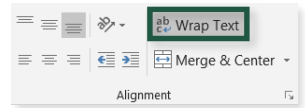
- By default, pivots will **adjust column width** to accommodate the longest value in a field
  - This can be helpful, but inconvenient when working with long, text-based fields
- To prevent automatic resizing, uncheck the "***Autofit Columns Widths***" box in the PivotTable options (**Layout & Format** tab)
- **TIP**: Use the **wrap text** option to make text fields more readable within the pivot

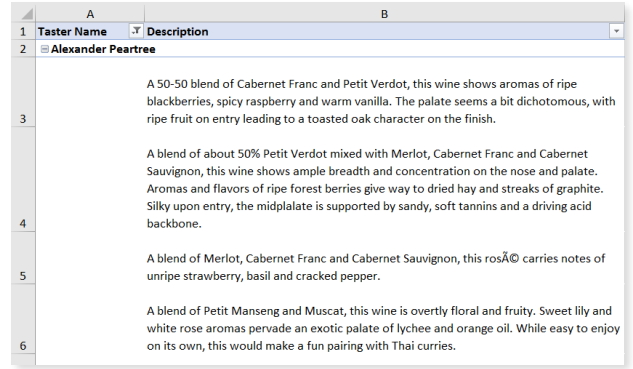




(PivotTable Tools > Options)

(Home > Wrap Text)





COMMON USE CASES:

- *Working with text fields like survey results or social media posts*
- *Forcing the PivotTable column widths to remain fixed, regardless of how the table layout changes*

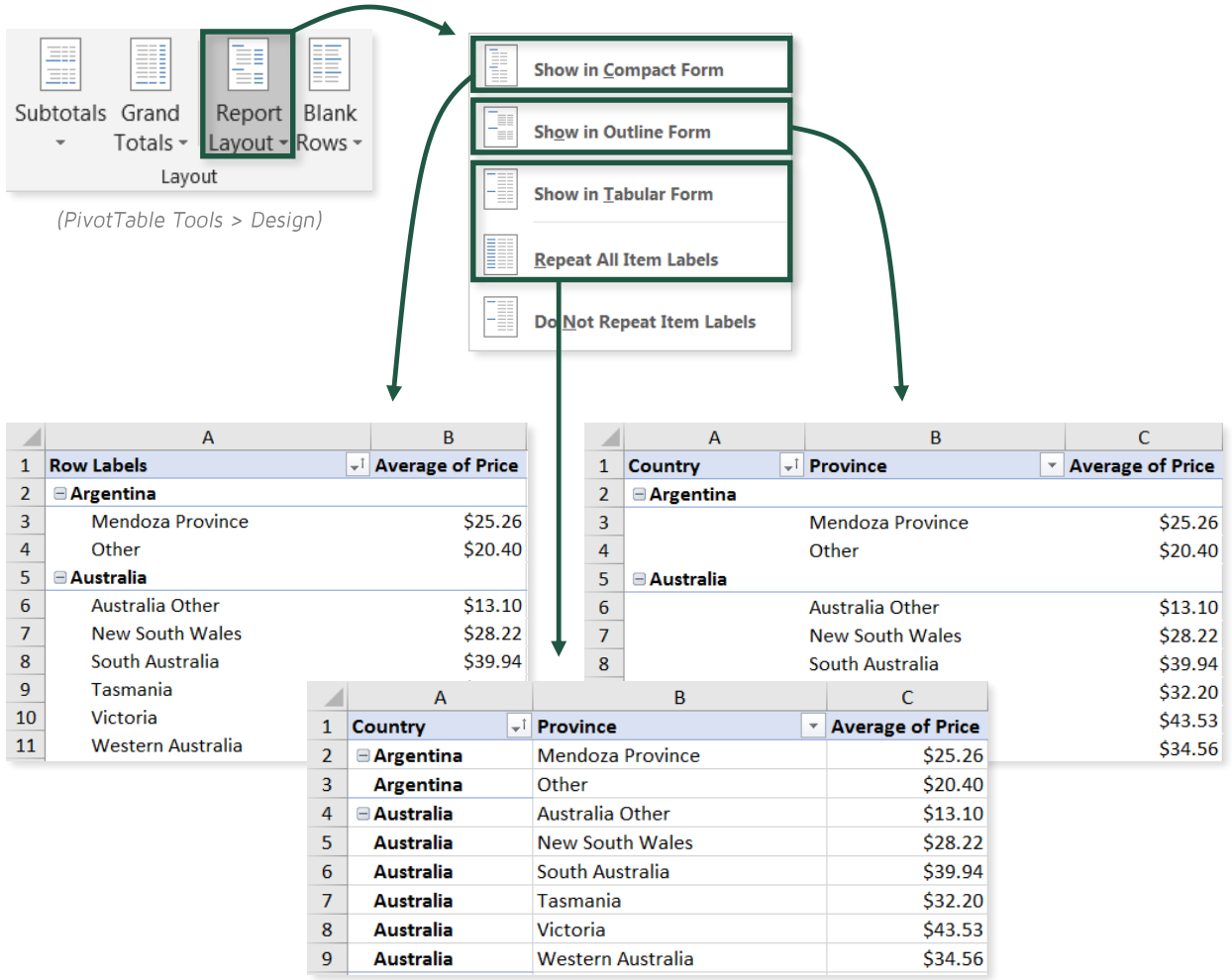
PIVOT TABLES

1 STAR (VERY BASIC)

PRO TIP

SHOW PIVOTS IN OUTLINE OR TABULAR FORM

- Pivots typically default to **Compact Form**, grouping all row labels into a single column
- You can change the layout using ***PivotTable Tools > Design > Report Layout:***
  - **Outline Form:** Splits row labels into separate columns, to sort/filter individually
  - **Tabular Form:** Formats the pivot like a table, without extra spacing rows
- **NOTE:** Head to ***File > Options > Data > Edit Default Layout*** to change your default



(PivotTable Tools > Design)

Row Labels	Average of Price
Argentina	
Mendoza Province	$25.26
Other	$20.40
Australia	
Australia Other	$13.10
New South Wales	$28.22
South Australia	$39.94
Tasmania	$32.20
Victoria	$43.53
Western Australia	$34.56

Country	Province	Average of Price
Argentina		
	Mendoza Province	$25.26
	Other	$20.40
Australia		
	Australia Other	$13.10
	New South Wales	$28.22
	South Australia	$39.94

Country	Province	Average of Price
Argentina	Mendoza Province	$25.26
Argentina	Other	$20.40
Australia	Australia Other	$13.10
Australia	New South Wales	$28.22
Australia	South Australia	$39.94
Australia	Tasmania	$32.20
Australia	Victoria	$43.53
Australia	Western Australia	$34.56

COMMON USE CASES:

- *Using **Outline Form** to sort and filter multiple fields individually*
- *Using **Tabular Form** with repeating item labels and no grand totals or subtotals to create new source data for further analysis*

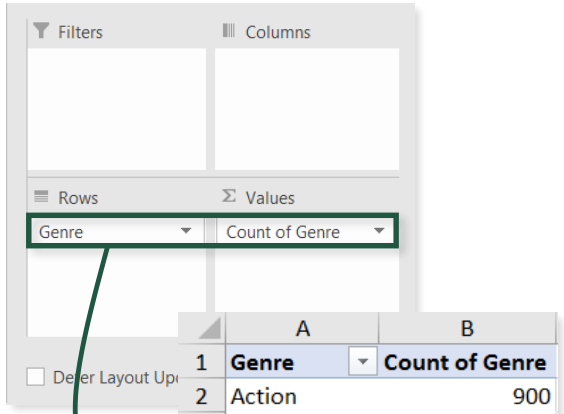
PIVOT TABLES

1 STAR (VERY BASIC)

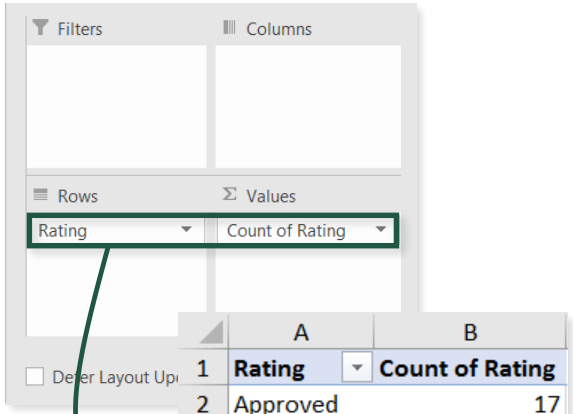
PRO TIP

USE PIVOT TABLES TO COUNT TEXT FIELDS

- While the **Values** pane is typically for numerical fields, it can be used to analyze the **count or frequency of text fields**
  - Since you can't aggregate non-numerical fields, use **Count Of** summarization to display frequencies for text-based values
- **NOTE:** If a *numerical* field defaults to a count, it typically indicates values formatted as text or blank rows in the source data



	A	B
1	**Genre**	**Count of Genre**
2	Action	900
3	Adventure	362
4	Animation	45
5	Biography	204
6	Comedy	1,016
7	Crime	252
8	Documentary	43
9	Drama	672
10	Family	3
11	Fantasy	35
12	Horror	151
13	Musical	2
14	Mystery	23
15	Romance	3
16	Sci-Fi	8
17	Thriller	3
18	Western	3
19	**Grand Total**	**3,725**



	A	B
1	**Rating**	**Count of Rating**
2	Approved	17
3	G	91
4	GP	1
5	M	2
6	NC-17	6
7	Not Rated	43
8	Passed	3
9	PG	553
10	PG-13	1,248
11	R	1,679
12	Unrated	24
13	X	9
14	(blank)	
15	**Grand Total**	**3,676**

COMMON USE CASES:

- *Analyzing the number of rows or observations that fall into specific categories (i.e. movie titles by genre, products by category, etc.)*
- *Generating source data for statistical charts like histograms/pareto charts*

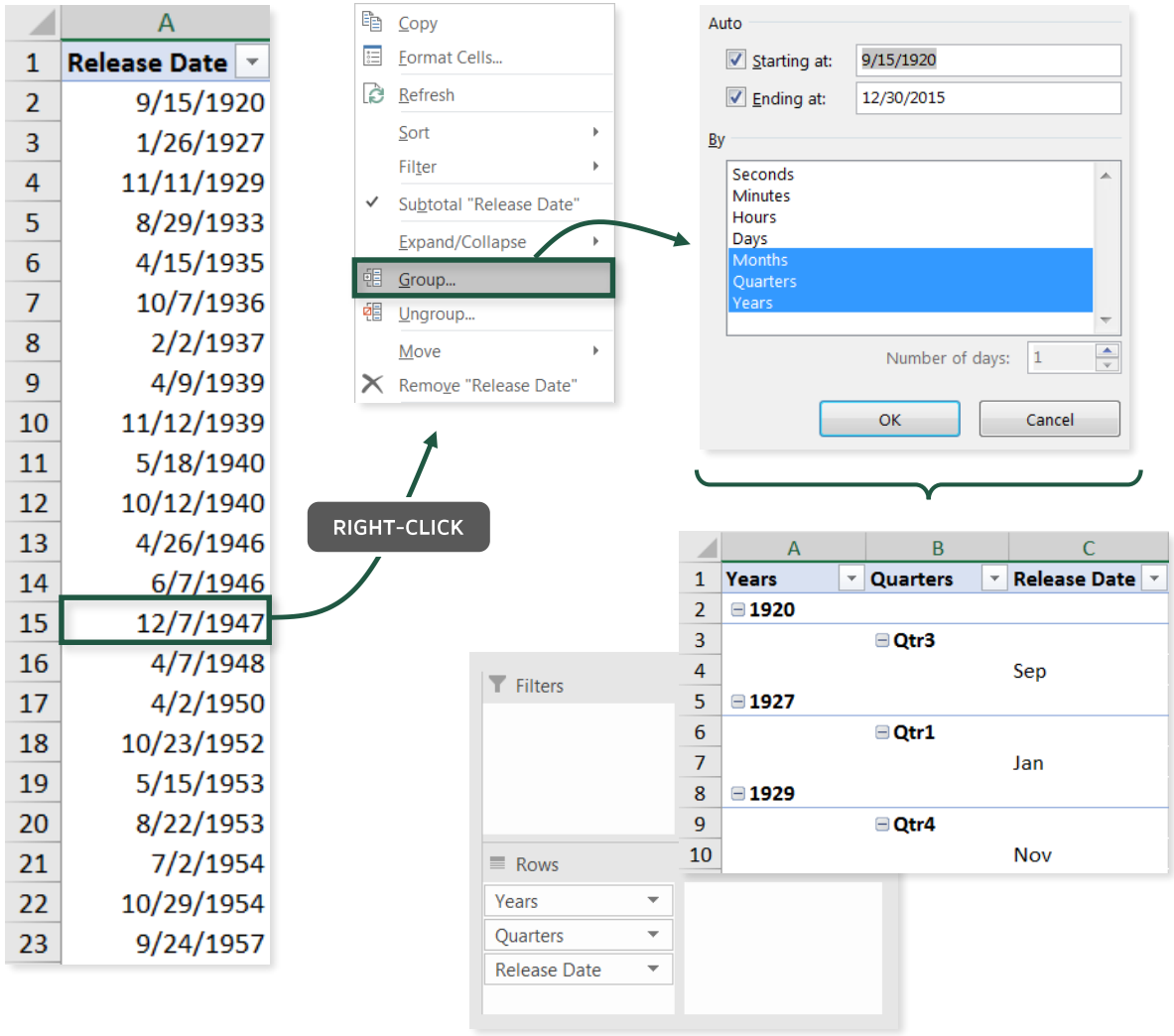
PIVOT TABLES

1 STAR (VERY BASIC)

PRO TIP

GROUP DATES USING PIVOTS OR SOURCE DATA

- PivotTables include standard tools to **group date fields** into months, years, etc.
  - **NOTE:** Head to ***Options > Data > Data Options*** to check whether or not automatic date/time grouping is enabled
- Grouping creates new fields that *only exist in the pivot*, but can be used like any other
- **NOTE:** An alternative (*and often more flexible*) approach is to create new fields in the source data using date & time functions



COMMON USE CASES:

- *Rolling up daily data to analyze trends by month or quarter*
- *Creating high-level summary tables or charts from granular source data*

PIVOT TABLES

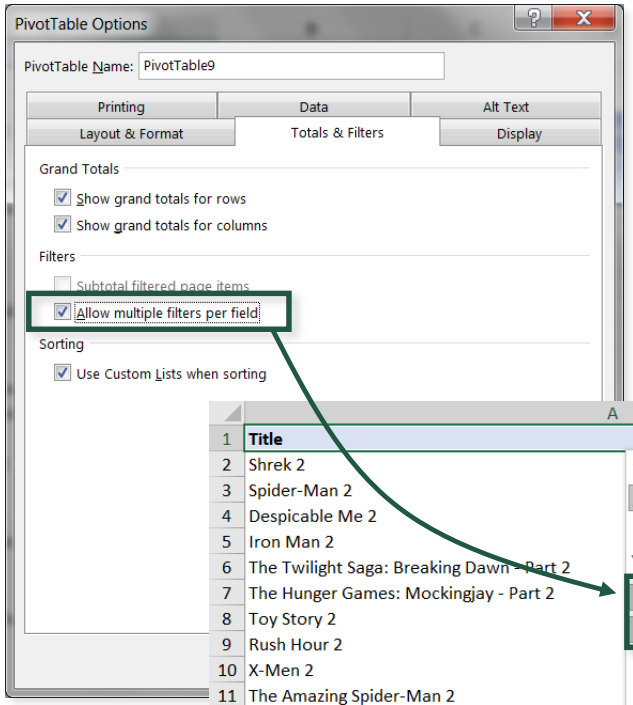
2 STARS (BASIC)

PRO TIP

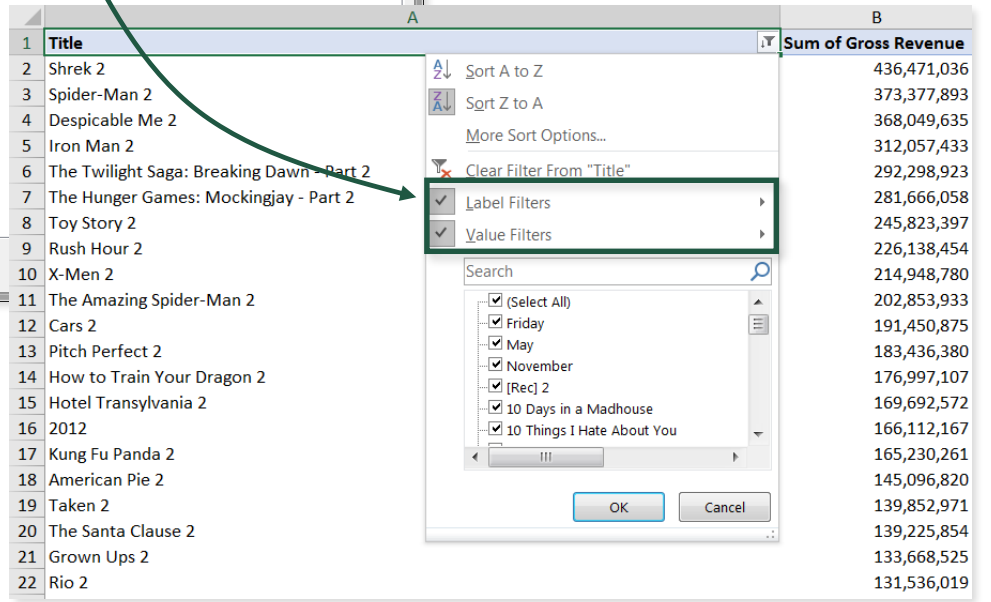
APPLY MULTIPLE FILTERS TO A SINGLE PIVOT FIELD

- By default, you can apply **label filters** or **value filters** to a single field, *but not both*
  - To change this, head to **PivotTable Options** and check the "***Allow Multiple Filters Per Field***" box in the **Totals & Filters** tab
- This will allow you to filter PivotTable fields based on both **text** and **value** attributes

(PivotTable Tools > Options)



*Titles that end with "2" **AND** drove $100,000,000+ in Gross Revenue*



COMMON USE CASES:

- *Applying more complex or custom filters that incorporate both text and value-based criteria*

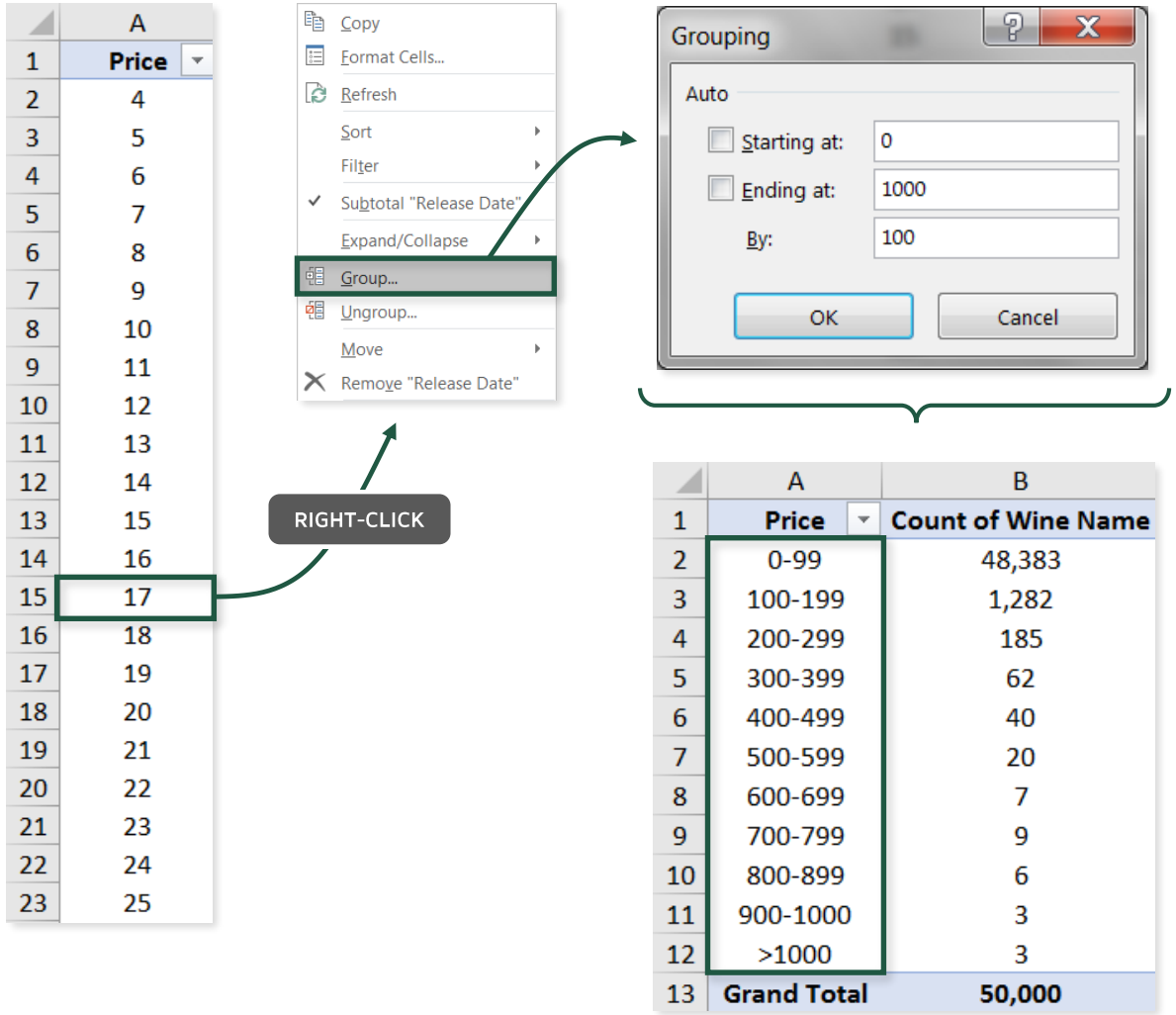
PIVOT TABLES

2 STARS (BASIC)

PRO TIP

USE PIVOT TABLES TO GROUP NUMERICAL VALUES

- Just like text, you can **group numerical fields** into custom buckets
  - Pull a numerical field into rows, right-click and select **Group**, and specify a **start value**, **end value**, and **group size**
  - To undo, right-click and select **Ungroup**
- Value grouping is a great way to analyze data distribution, or create bins as source data for a Histogram or Pareto Chart



PIVOT TABLES

2 STARS (BASIC)

COMMON USE CASES:

- *Analyzing the count or frequency of observations that fall into specific groups or "bins" of values*
- *Creating high-level summary tables or charts from granular data*

PRO TIP

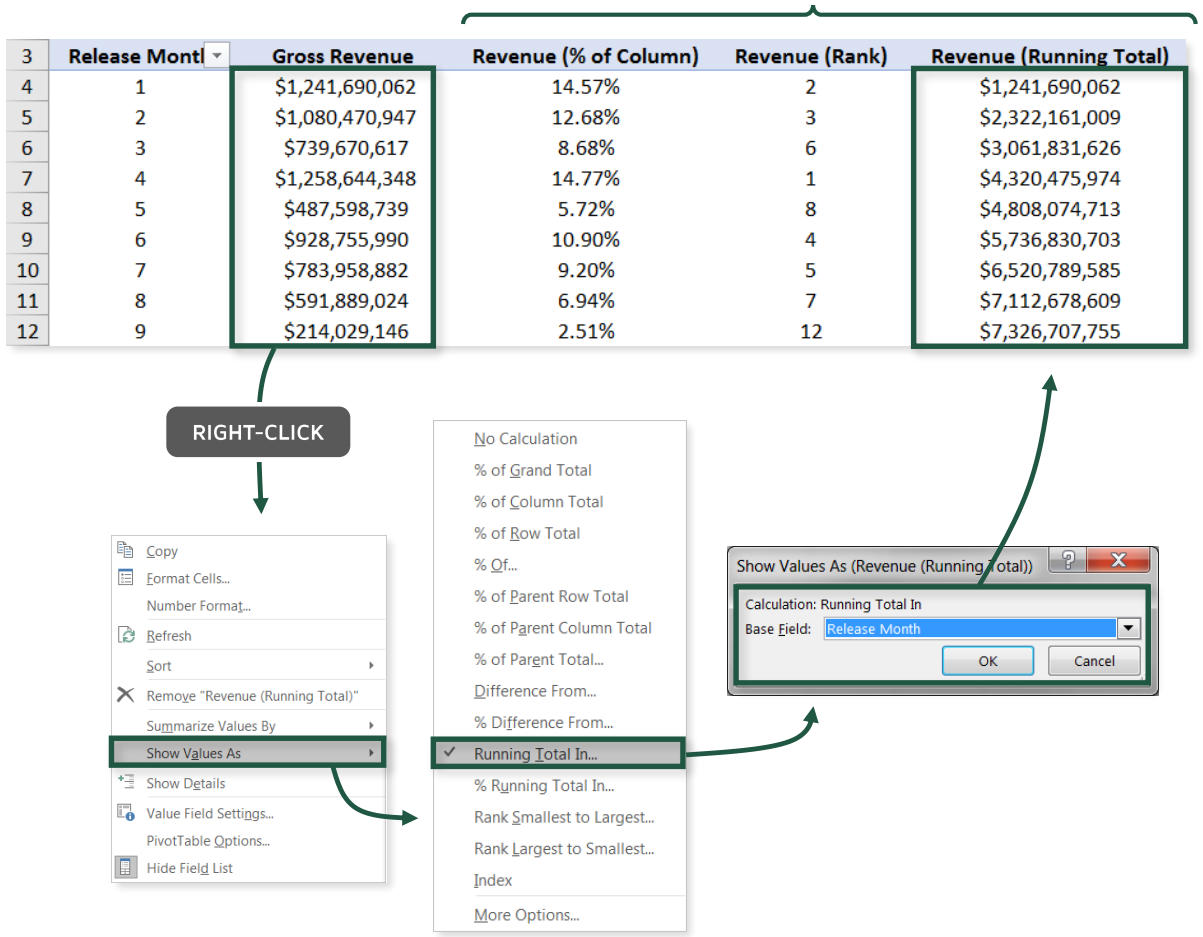
ANALYZE DATA USING VALUE CALCULATIONS

- **Value Calculations** allow you to display values from several different angles:
  - **% of Column/Row**: Displays values as a percentage of the row or column total
  - **% of Parent**: Displays values as a percentage of a given "parent" category
  - **Difference From**: Displays values in terms of the difference from a given base item
  - **Running Total**: Displays values as a cumulative total within a given base field
  - **Rank**: Displays values as a rank based on volume (*low to high or high to low*)

PIVOT TABLES

3 STARS (MODERATE)

TIP: Drag in multiple copies of the field to keep an original



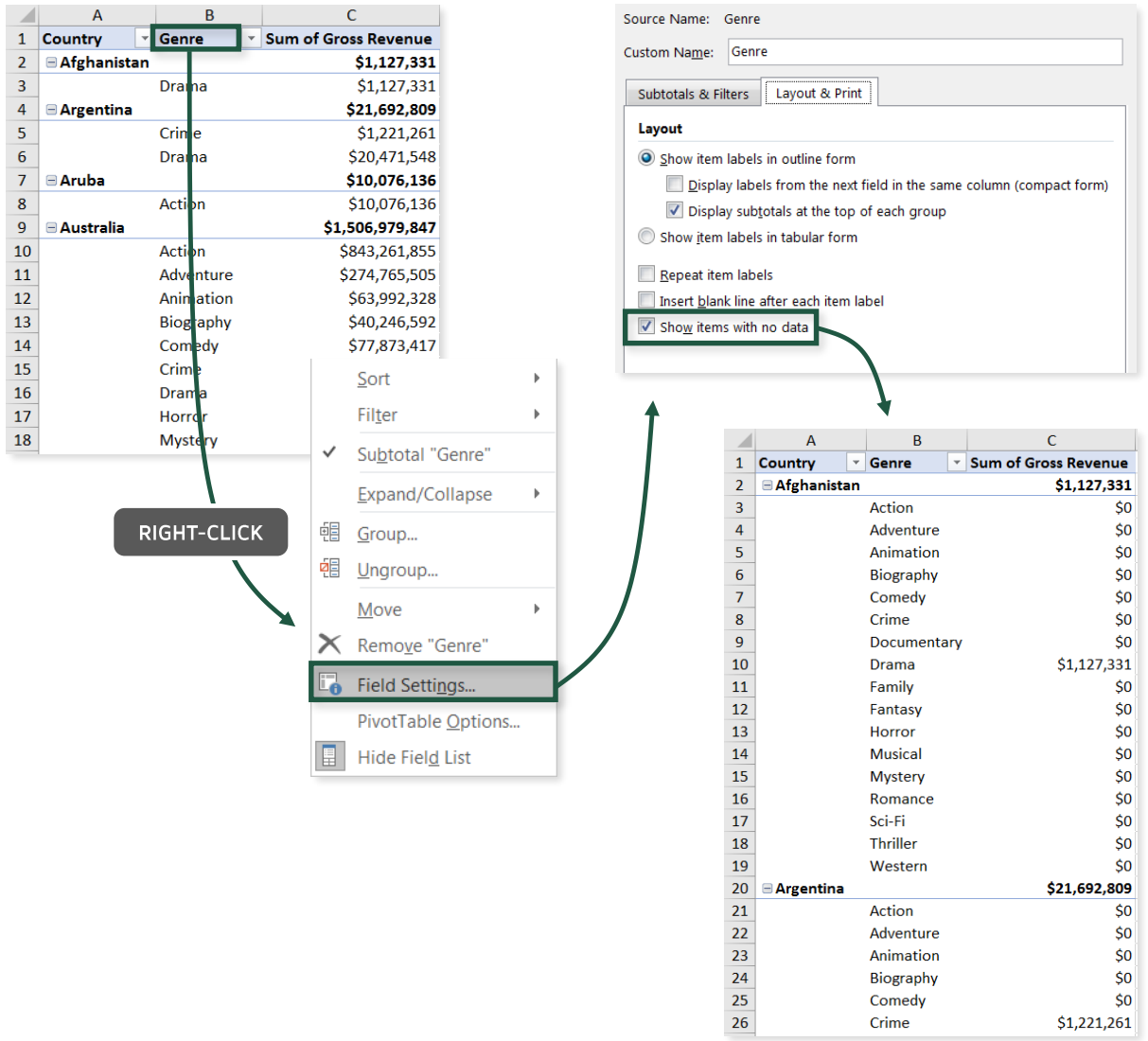
COMMON USE CASES:

- *Analyzing time-series data as a running total by day, month or year*
- *Exploring the composition of values in terms of percent share by category*

PRO TIP

CUSTOMIZE ERRORS & BLANKS WITHIN A PIVOT

- By default, pivots won't display row labels where values don't exist; however, you can choose to **show items with no data**
  - Right-click the header, select **Field Settings**, and check the "***Show items with no data***" box in the **Layout & Print** tab
- **TIP**: To customize how **empty values** and **errors** are displayed, go to the **Layout & Format** tab in the **Options** menu



COMMON USE CASES:

- *Displaying all possible items within the PivotTable (even for blanks), to create a consistent layout or template*
- *Reformatting blanks or errors to clean up user-facing reports or dashboards*

PIVOT TABLES

3 STARS (MODERATE)

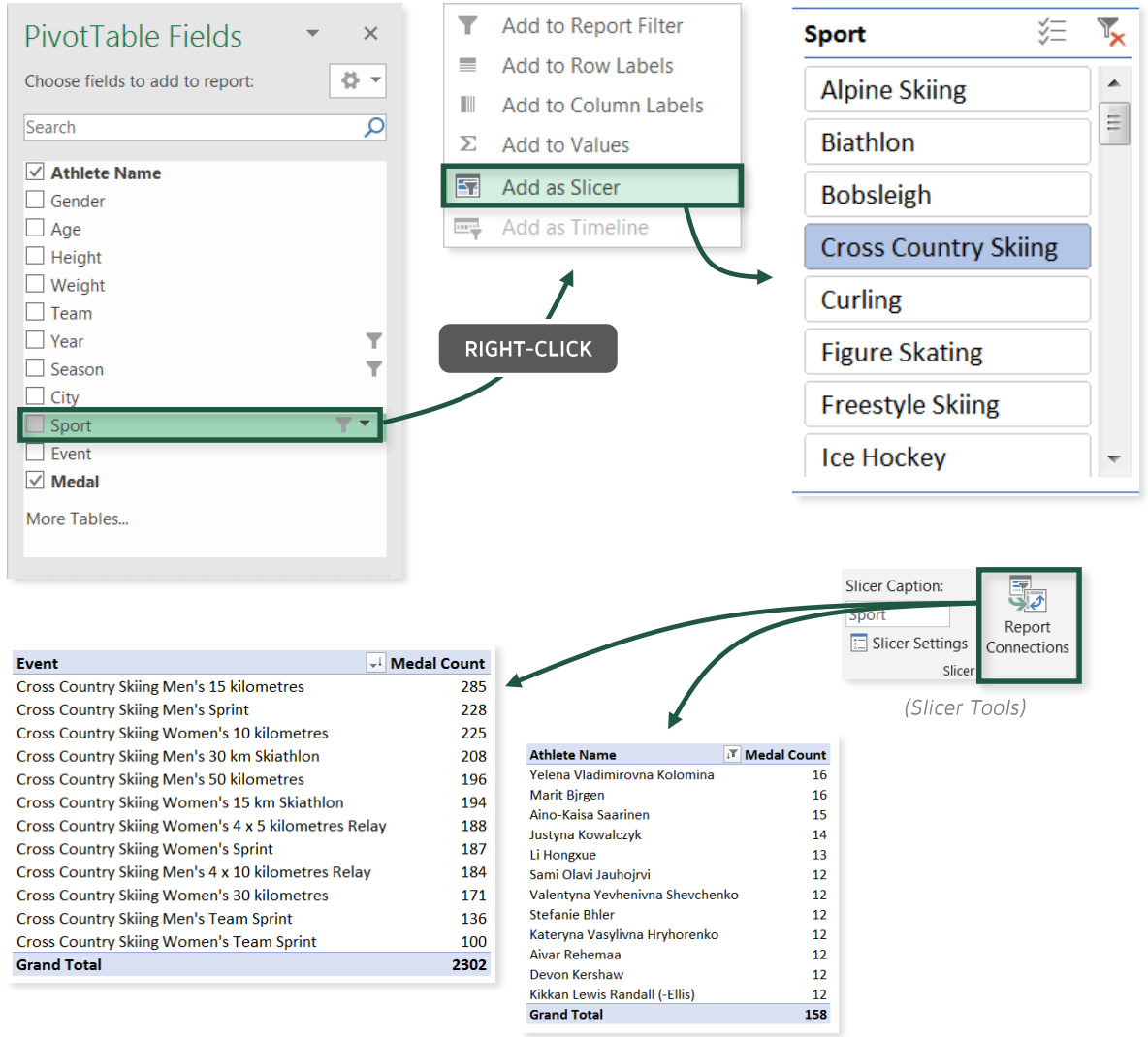
PRO TIP

ADD VISUAL FILTERS WITH SLICERS & TIMELINES

- **Slicers** are essentially interactive versions of filters, and **Timelines** are slicers designed to work specifically with dates
- Use the **Report Connections** option to link slicers to multiple PivotTables (*assuming they share the same source data*)
- **NOTE:** It typically makes sense to hide or indicate options with no data (via **Slicer Settings**), but note that this may cause slow performance for very large tables

PIVOT TABLES

3 STARS (MODERATE)



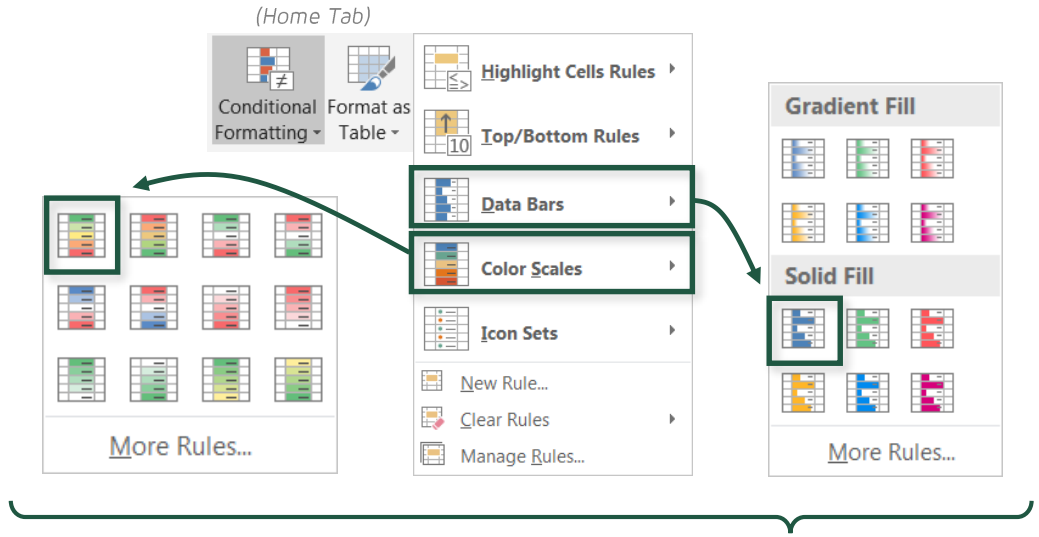
COMMON USE CASES:

- *Adding user-friendly, visual filters to reports built from PivotTables*
- *Using slicers to clearly and visually indicate how a table is being filtered*

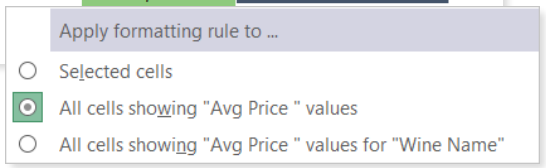
PRO TIP

BRING PIVOTS TO LIFE WITH CONDITIONAL FORMATTING

- Add **conditional formatting** rules (*data bars, color scales, icon sets, etc.*) to highlight patterns or trends in the pivot
- Unlike traditional cell formatting, you can specify how PivotTable formats react to changes in the table layout
  - **TIP:** Use "***All cells showing [X] values***" to apply the same rule to different layouts



	A	B	C
1	Country	Canada	
2	Wine Variety	(All)	
3			
4	**Row Labels**	**Avg Price**	**Avg Point Rating**
5	Cave Spring 2008 Riesling (Niagara Peninsula)	$12	
6	Cave Spring 2014 Riesling (Niagara Peninsula)	$15	
7	Laughing Stock 2015 Pinot Gris (Okanagan Valley)	$15	
8	Cave Spring 2008 Beamsville Bench Riesling (Niagara Peninsula)	$15	
9	North 42 Degrees 2016 Sweet Riesling (Lake Erie North Shore)	$16	
10	Nk'Mip Cellars 2012 Chardonnay (Okanagan Valley)	$17	
11	Cave Spring 2014 Estate Bottled Riesling (Beamsville Bench)		
12	Tinhorn Creek 2014 GewÃ¼rztraminer (Okanagan Valley)		



COMMON USE CASES:

- *Using color scales to draw attention to patterns or trends in the data*
- *Applying formatting rules that "stick" regardless of how the pivot layout changes*

PIVOT TABLES

4 STARS (ADVANCED)

PRO TIP

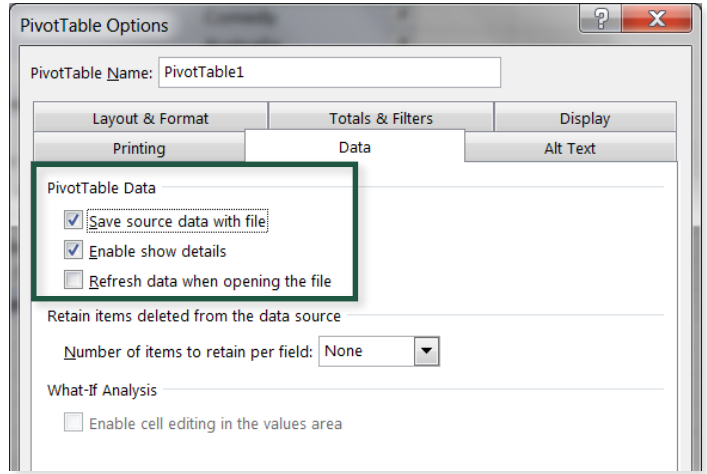
REMOVE & REVIVE PIVOT TABLE DATA FROM CACHE

- When a new PivotTable is created, a **Pivot Cache** is generated as well, which is a compressed duplicate of the source data
  - The cache allows you to **delete the original source data** without impacting the pivot, which can significantly reduce file size
- To **revive the source data** from cache, double-click an unfiltered, Grand Total cell in the pivot (***NOTE:*** *make sure "**Show Details**" is enabled from the Options menu*)

	A	B
1	**Year**	**Average of Base Pay**
2	2011	$65,045
3	2012	$65,938
4	**Grand Total**	**$65,496**

DOUBLE-CLICK

(PivotTable Tools > Options)



	A	B	C	D	E
1	**Employee Name**	**Year**	**Job Title**	**Base Pay**	**Overtime Pay**
2	A Bernard Fatooh	2011	Sheriff'S Property Keeper	19969.37	0
3	Aaric Pingree	2011	Carpenter	0	0
4	Aaron Richmond	2011	Public Svc Aide-Public Works	2175.41	0
5	Aaron A Hipolito	2011	Museum Guard	10954.9	0
6	Aaron C Ballonado	2011	Police Officer 3	123471.15	8199.27
7	Aaron D Lynch	2011	Deputy Sheriff	87296.9	3265.58
8	Aaron Del Tredici	2011	Physician Specialist	124568.92	0
9	Aaron E Vurek	2011	Roofer	79136.84	883.8
10	Aaron Hipolito	2011	Museum Guard	10233.46	70.35
11	Aaron M Del Tredici	2011	Physician Specialist	129528.39	0
12	Aaron Patterson	2011	Ps Aide Health Services	15131.57	0
13	Aaron S Duran	2011	Automotive Mechanic	73376.2	568.85
14	Aaron Sher	2011	Public Service Aide-Assistant To Professionals	4666.32	0

COMMON USE CASES:

- *Removing static source data to reduce file size or increase processing speed*
- *Limiting accessibility in order to prevent users from changing the raw data*

PIVOT TABLES

4 STARS (ADVANCED)

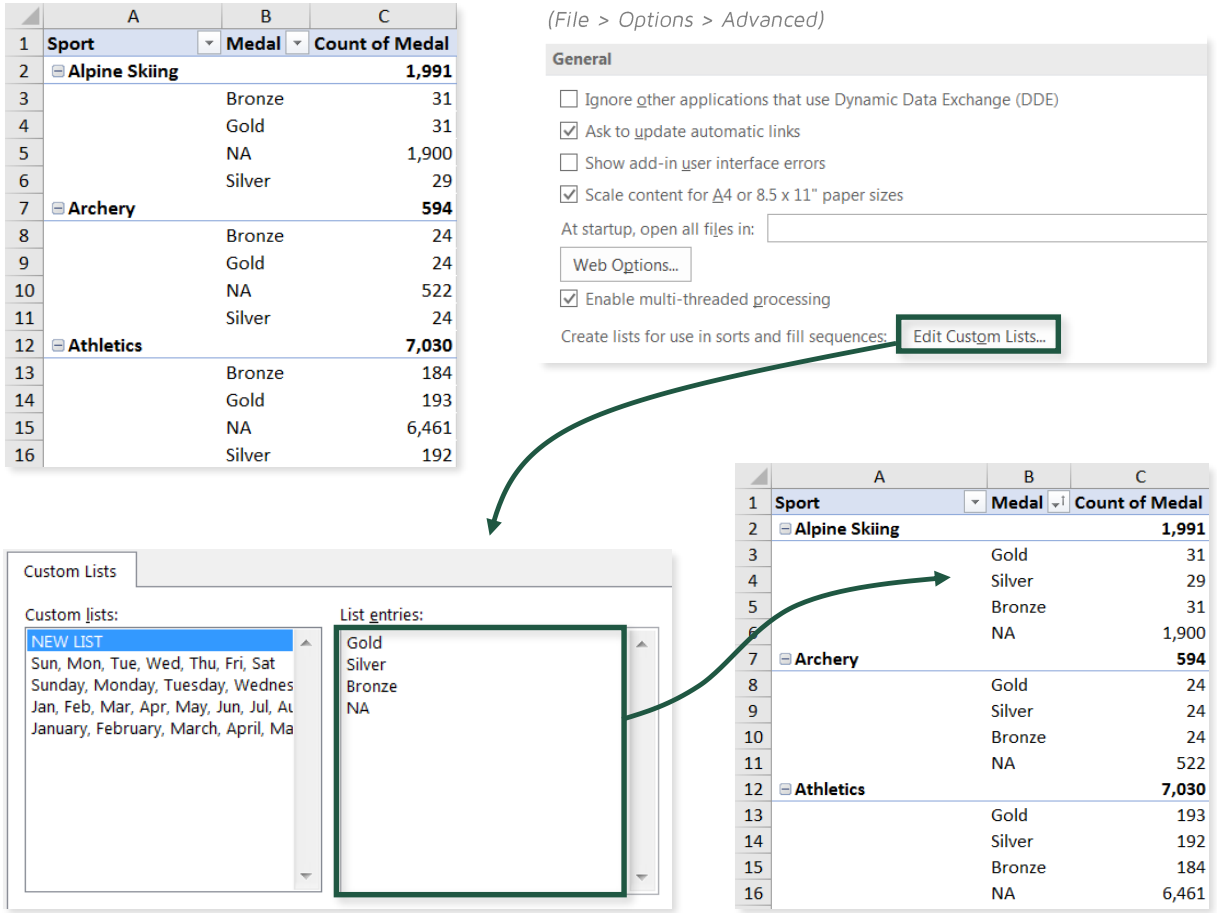
PRO TIP

ORGANIZE FIELDS USING CUSTOM SORT LISTS

- Excel has several built-in **sort lists** to help organize fields such as **weekdays** or **month names** (which use non-standard sort rules)
- To add your own **custom sort lists**, head to ***File > Options > Advanced*** and click the **Edit Custom Lists** option
  - Once a list is defined, A-Z sorting will default to the custom list, rather than standard alphabetical order

PIVOT TABLES

5 STARS (EXPERT)



	A	B	C
1	Sport	Medal	Count of Medal
2	Alpine Skiing		1,991
3		Bronze	31
4		Gold	31
5		NA	1,900
6		Silver	29
7	Archery		594
8		Bronze	24
9		Gold	24
10		NA	522
11		Silver	24
12	Athletics		7,030
13		Bronze	184
14		Gold	193
15		NA	6,461
16		Silver	192

	A	B	C
1	Sport	Medal	Count of Medal
2	Alpine Skiing		1,991
3		Gold	31
4		Silver	29
5		Bronze	31
6		NA	1,900
7	Archery		594
8		Gold	24
9		Silver	24
10		Bronze	24
11		NA	522
12	Athletics		7,030
13		Gold	193
14		Silver	192
15		Bronze	184
16		NA	6,461

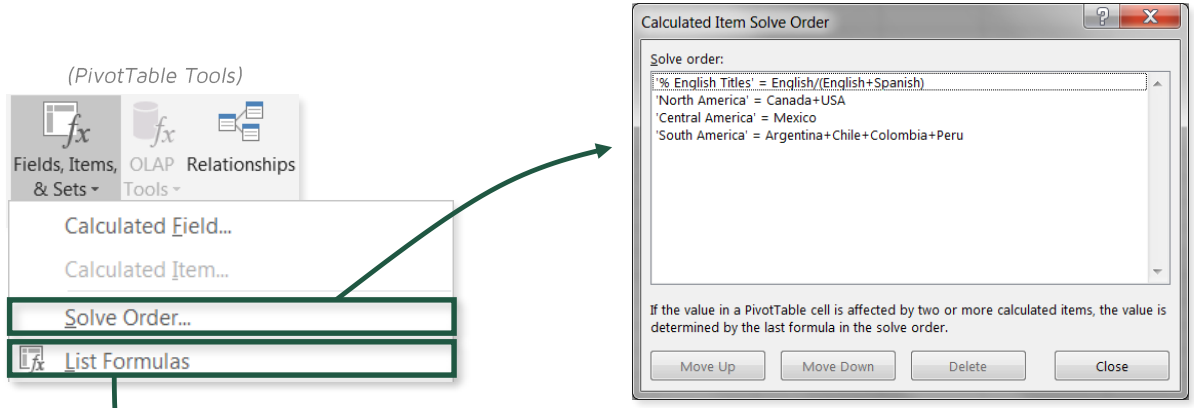
COMMON USE CASES:

- *Organizing fields which can't be sorted using standard tools, such as apparel sizes (S, M, L) or Olympic medals (Gold, Silver, Bronze)*
- *Creating custom **fiscal calendars** to override standard month sorting*

PRO TIP

CHANGE THE SOLVE ORDER FOR PIVOT CALCULATIONS

- If your pivot contains multiple calculated items and fields which could overlap, use the **Solve Order** to determine priority
  - In the **Solve Order** dialog box, the *last* formula in the list overrides those above
- **NOTE:** The **List Formulas** option generates a new tab documenting all calculated items and fields, as well as the sort order



	A	B	C
1	***Calculated Field***		
2	**Solve Order**	**Field**	**Formula**
3	1	Profit	='Gross Revenue'-Budget
4	2	ROI	='Gross Revenue'/Budget
5	3	Total FB Likes	='Lead Actor FB Likes'+'Cast FB Likes'+'Director FB Likes'+'Movie FB Likes'
6	4	Lead Actor Popularity	='Lead Actor FB Likes'/'Total FB Likes'
7			
8	***Calculated Item***		
9	**Solve Order**	**Item**	**Formula**
10	1	'% English Titles'	=English/(English+Spanish)
11	2	'North America'	=Canada+USA
12	3	'Central America'	=Mexico
13	4	'South America'	=Argentina+Chile+Colombia+Peru
14			
15			
16	***Note:***	When a cell is updated by more than one formula,	
17		the value is set by the formula with the last solve order.	
18			
19		To change the solve order for multiple calculated items or fields,	
20		on the Options tab, in the Calculations group, click Fields, Items, & Sets, and then click Solve Order.	
21			

COMMON USE CASES:

- *Determining which calculations should take priority in conflicting cases*
- *Producing documentation for complex PivotTables containing a large number of calculated items and fields*

PIVOT TABLES

5 STARS (EXPERT)

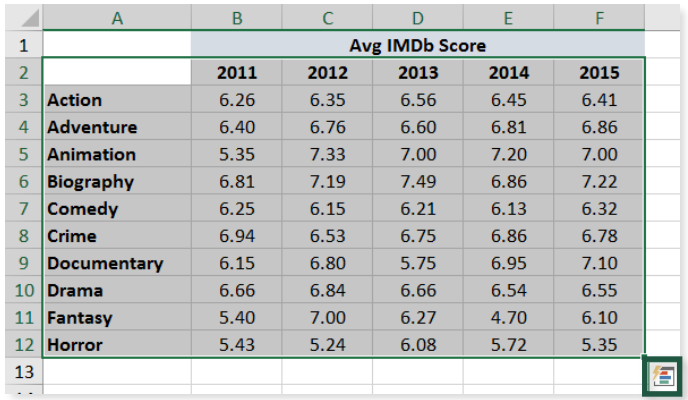
ANALYTICS TIPS

EXCEL MAVEN

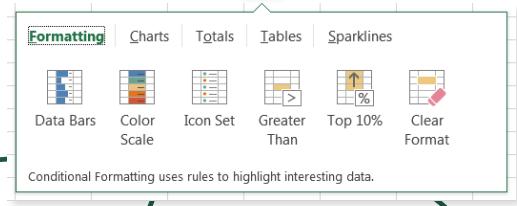
PRO TIP

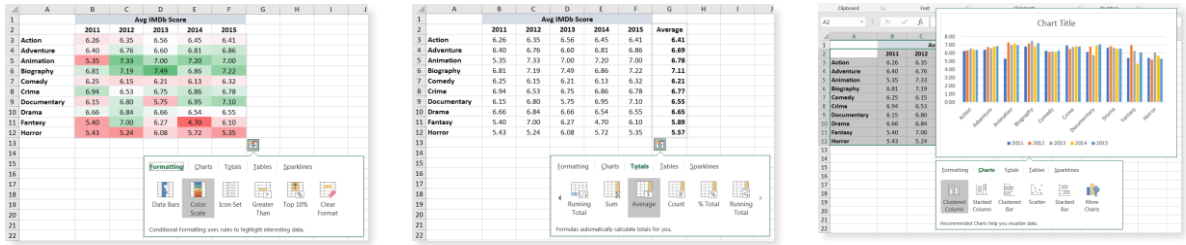
EXPLORE YOUR DATA WITH QUICK ANALYSIS TOOLS

- Select a cell range and click the pop-up (or ***CTRL-Q***) to access **Quick Analysis** tools
  - These tools allow you to quickly add **conditional formats**, **charts**, **calculated rows & columns**, **tables** & **sparklines** to help explore and analyze your data
- **NOTE:** Options may be unavailable if they aren't compatible with the selected data



	A	B	C	D	E	F
1		Avg IMDb Score				
2		2011	2012	2013	2014	2015
3	Action	6.26	6.35	6.56	6.45	6.41
4	Adventure	6.40	6.76	6.60	6.81	6.86
5	Animation	5.35	7.33	7.00	7.20	7.00
6	Biography	6.81	7.19	7.49	6.86	7.22
7	Comedy	6.25	6.15	6.21	6.13	6.32
8	Crime	6.94	6.53	6.75	6.86	6.78
9	Documentary	6.15	6.80	5.75	6.95	7.10
10	Drama	6.66	6.84	6.66	6.54	6.55
11	Fantasy	5.40	7.00	6.27	4.70	6.10
12	Horror	5.43	5.24	6.08	5.72	5.35
13						





ANALYTICS

1 STAR (VERY BASIC)

COMMON USE CASES:

- *Quickly exploring a variety of data analysis tools, without having to manually navigate through different options*
- *Adding calculated rows or columns without typing a single formula*

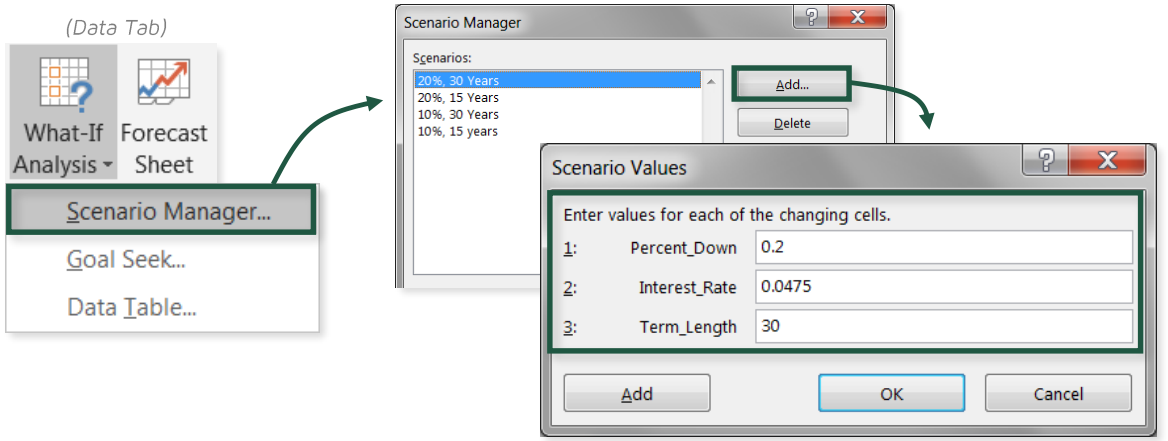
PRO TIP

COMPARE OUTPUTS WITH THE SCENARIO MANAGER

- Use the **Scenario Manager** to save and access specific combinations of cell values (***Data > What-If Analysis > Scenario Manager***)
  - This is often used for modeling exercises, allowing you to fix combinations of *inputs* in order to evaluate a calculated *output*
- **TIP**: Give your input cells meaningful names (*i.e.* "***Interest_Rate***" *vs.* ***A1***)

ANALYTICS

2 STARS (BASIC)



Scenario #1

PROPERTY COST	
Purchase Price	$899,000
Tax Rate	$9.50

LOAN COST	
Down Payment %	20%
Interest Rate	4.75%
Term Length (yrs)	30
Loan Amount	$719,200
Est. Closing Costs	$17,980

Cash to Close: **$197,780**

Monthly Expenses: **$4,813**

Scenario #2

PROPERTY COST	
Purchase Price	$899,000
Tax Rate	$9.50

LOAN COST	
Down Payment %	20%
Interest Rate	4.00%
Term Length (yrs)	15
Loan Amount	$719,200
Est. Closing Costs	$17,980

Cash to Close: **$197,780**

Monthly Expenses: **$6,382**

Scenario #3

PROPERTY COST	
Purchase Price	$899,000
Tax Rate	$9.50

LOAN COST	
Down Payment %	10%
Interest Rate	5.75%
Term Length (yrs)	30
Loan Amount	$809,100
Est. Closing Costs	$17,980

Cash to Close: **$107,880**

Monthly Expenses: **$5,783**

Scenario #4

PROPERTY COST	
Purchase Price	$899,000
Tax Rate	$9.50

LOAN COST	
Down Payment %	10%
Interest Rate	5.00%
Term Length (yrs)	15
Loan Amount	$809,100
Est. Closing Costs	$17,980

Cash to Close: **$107,880**

Monthly Expenses: **$7,460**

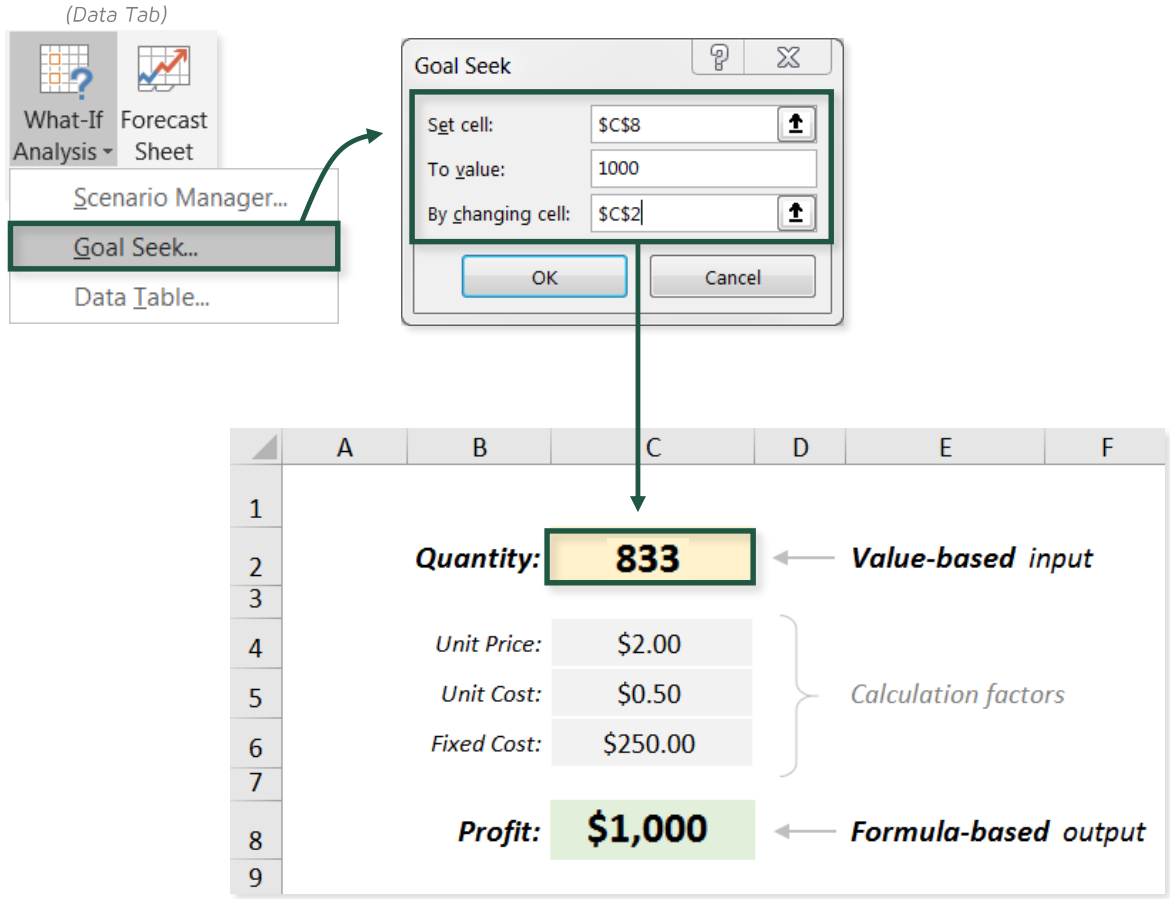
COMMON USE CASES:

- *Building forecasts based on several variables (seasonality, interest rate, etc.)*
- *Modeling several potential outcomes in cases where uncertainty is a factor (i.e. stock portfolio returns)*

PRO TIP

SOLVE FOR INDIVIDUAL OUTPUTS WITH GOAL SEEK

- Use **Goal Seek** to find the result you want by changing the value of a given input cell (***Data > What-If Analysis > Goal Seek***)
  - **Goal Seek** requires a single, hard-coded *input* cell and a single, formula-based *output* cell; you cannot test multiple inputs
- **TIP**: To evaluate multiple inputs, add constraints, or solve complex optimization problems, use Excel's **Solver** tool (*requires the Solver Add-In*)



ANALYTICS

2 STARS (BASIC)

COMMON USE CASES:

- *Determining the ideal input required to produce a specific goal or target outcome (i.e. number of sales required to yield a positive profit)*
- *Solving simple optimization problems based on a single input variable*

PRO TIP

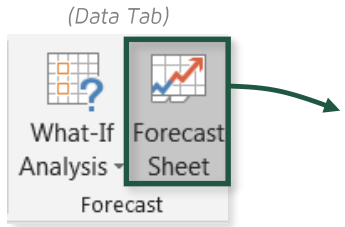
CREATE FORECAST SHEETS FROM HISTORICAL DATA

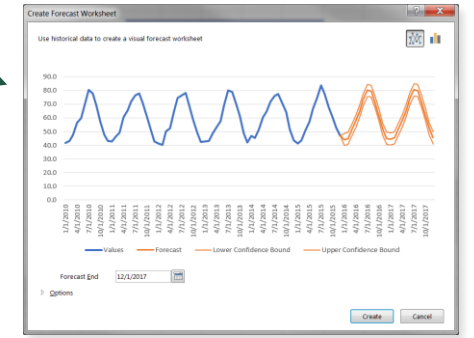
- Recent versions of Excel (*2016/365*) include a built-in **Forecast Sheet** tool to calculate forecasts based on given historical data
  - The **Forecast Sheet** dialog box allows you to set the **forecast length** and **confidence interval**, detect **seasonality**, and customize handling of **missing** or **duplicate** values
- **NOTE:** Trendlines can be used for simple forecast exercises, but do not account for confidence or seasonality

ANALYTICS

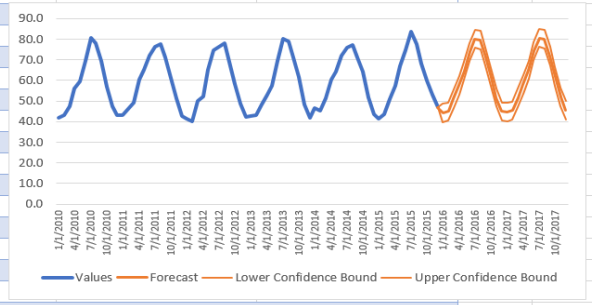
3 STARS (MODERATE)

Month	Avg Temp_F
1/1/2010	41.7
2/1/2010	43.1
3/1/2010	47.6
4/1/2010	56.3
5/1/2010	59.8
6/1/2010	69.4
7/1/2010	80.7
8/1/2010	78.0
9/1/2010	69.3
10/1/2010	57.1
11/1/2010	47.5
12/1/2010	43.0
1/1/2011	43.0
2/1/2011	46.3
3/1/2011	49.2
4/1/2011	60.4
5/1/2011	65.3
6/1/2011	72.0
7/1/2011	76.3
8/1/2011	77.8
9/1/2011	71.0
10/1/2011	61.4
11/1/2011	51.4
12/1/2011	42.8
1/1/2012	41.5
2/1/2012	40.1
3/1/2012	50.2
4/1/2012	52.4
5/1/2012	65.0
6/1/2012	74.7
7/1/2012	76.5
8/1/2012	78.2
9/1/2012	69.0
10/1/2012	58.2





	Timeline	Values	Forecast	Lower Confidence Bound	Upper Confidence Bound	F	G
56	7/1/2014	76.1					
57	8/1/2014	77.4					
58	9/1/2014	71.1					
59	10/1/2014	64.1					
60	11/1/2014	51.7					
61	12/1/2014	43.6					
62	1/1/2015	41.3					
63	2/1/2015	43.6					
64	3/1/2015	50.9					
65	4/1/2015	57.3					
66	5/1/2015	67.0					
67	6/1/2015	74.5					
68	7/1/2015	83.9					
69	8/1/2015	77.5					
70	9/1/2015	68.1					
71	10/1/2015	60.1					
72	11/1/2015	52.9					
73	12/1/2015	46.9	46.9	46.9	46.9		
74	1/1/2016		44.2	39.8	48.6		
75	2/1/2016		45.0	40.6	49.4		
76	3/1/2016		51.1	46.8	55.5		
77	4/1/2016		57.7	53.3	62.0		
78	5/1/2016		64.6	60.3	69.0		
79	6/1/2016		73.8	69.4	78.1		
80	7/1/2016		80.2	75.8	84.5		
81	8/1/2016		79.6	75.2	84.0		



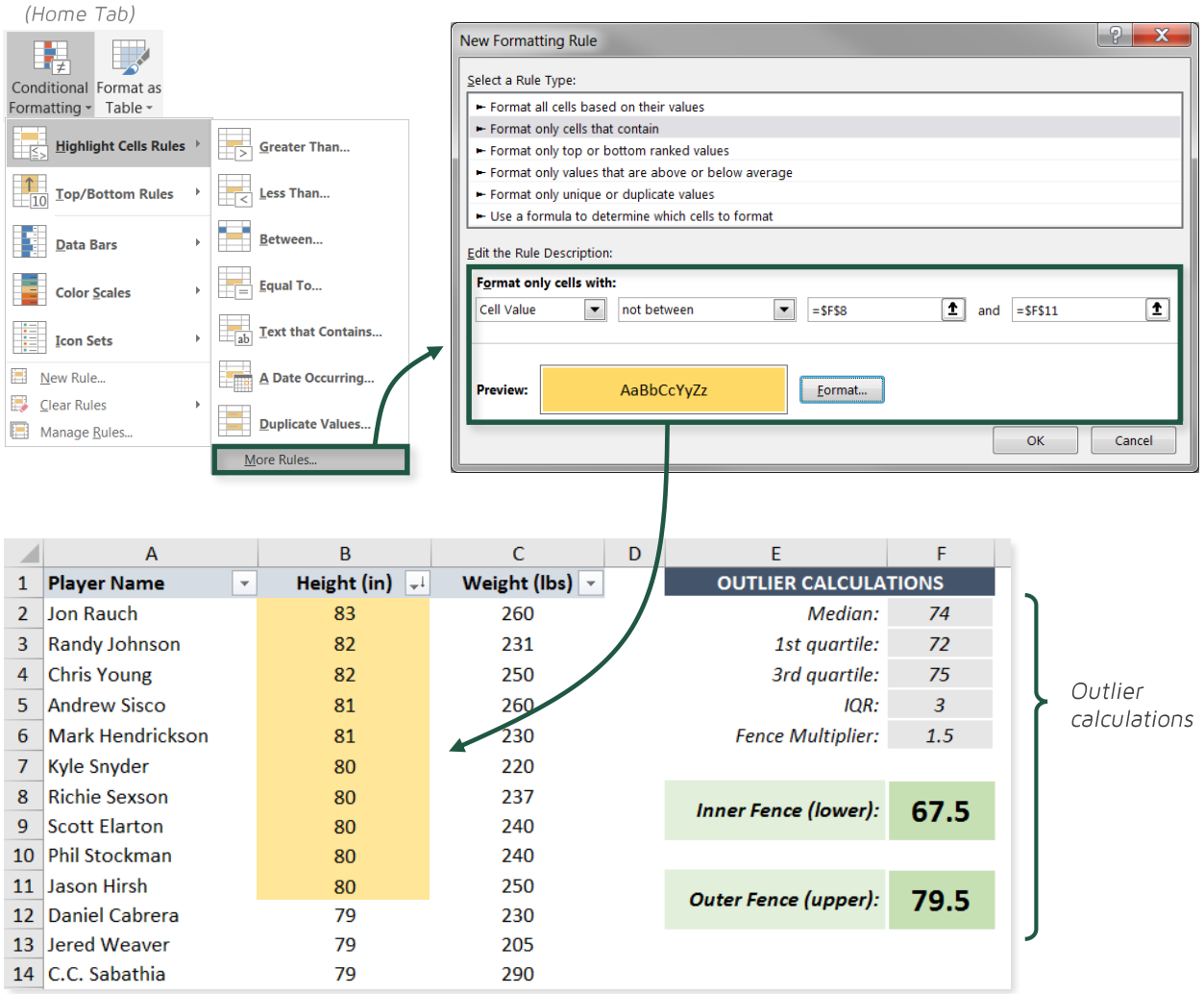
COMMON USE CASES:

- *Predicting future values such as interest rates or stock returns*
- *Calculating an expected range of future outcomes based on a given level of confidence*

PRO TIP

FIND OUTLIERS WITH STATS FUNCTIONS & FORMATTING

- Use functions like **MEDIAN** and **QUARTILE** to **calculate statistical outliers**, and **conditional formatting** to highlight them
  - Outliers are commonly defined using a "**fence**" based on how far a value falls outside of the **interquartile range** (*IQR*)
- **TIP**: For simple "***Top N***" or "***Top %***" calculations, use basic conditional formats



	A	B	C	D	E	F
1	**Player Name**	**Height (in)**	**Weight (lbs)**		**OUTLIER CALCULATIONS**	
2	Jon Rauch	83	260		*Median:*	*74*
3	Randy Johnson	82	231		*1st quartile:*	*72*
4	Chris Young	82	250		*3rd quartile:*	*75*
5	Andrew Sisco	81	260		*IQR:*	*3*
6	Mark Hendrickson	81	230		*Fence Multiplier:*	*1.5*
7	Kyle Snyder	80	220			
8	Richie Sexson	80	237		***Inner Fence (lower):***	**67.5**
9	Scott Elarton	80	240			
10	Phil Stockman	80	240			
11	Jason Hirsh	80	250		***Outer Fence (upper):***	**79.5**
12	Daniel Cabrera	79	230			
13	Jered Weaver	79	205			
14	C.C. Sabathia	79	290			

Outlier calculations

COMMON USE CASES:

- *Identifying statistical anomalies in a dataset, based on custom criteria*
- *Finding and removing values that may have been incorrectly entered*

ANALYTICS

4 STARS (ADVANCED)

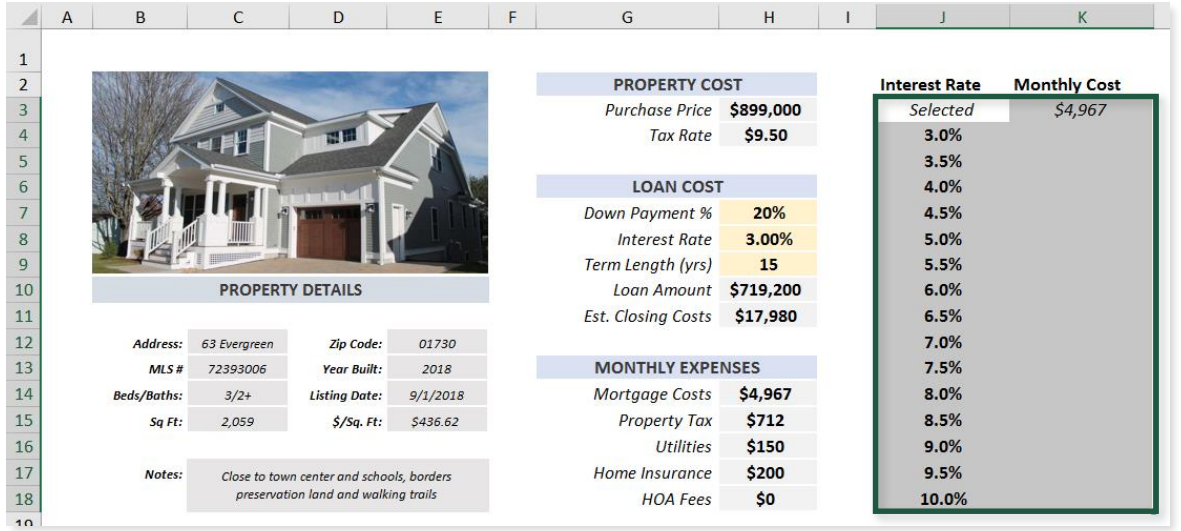
PRO TIP

EVALUATE VARIABLE INPUTS WITH DATA TABLES

- **Data Tables** allow you to calculate an array of results based on a range of input values (***Data > What-If Analysis > Data Table***)
  - **Data Tables** can be used to evaluate formula results based on changes to either a *single* input variable or *multiple* variables
- **TIP**: Change the calculation mode to "*Automatic Except for Data Tables*" to prevent them from constantly recalculating

ANALYTICS

4 STARS (ADVANCED)



PROPERTY DETAILS

Address:	63 Evergreen	Zip Code:	01730
MLS #	72393006	Year Built:	2018
Beds/Baths:	3/2+	Listing Date:	9/1/2018
Sq Ft:	2,059	$/Sq. Ft:	$436.62

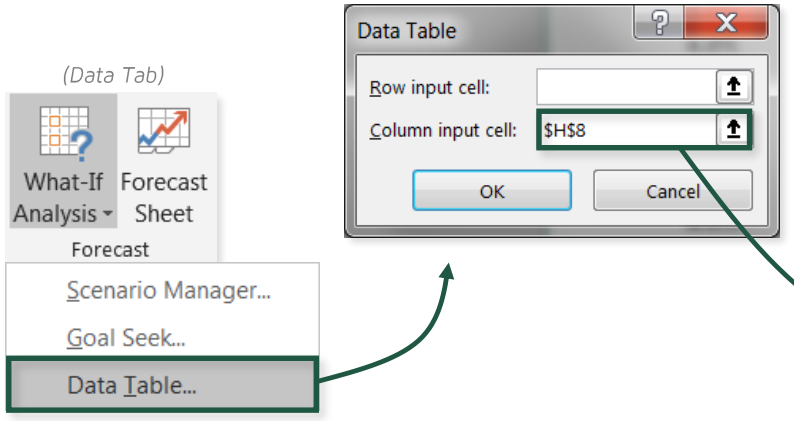
Notes: Close to town center and schools, borders preservation land and walking trails

PROPERTY COST	
Purchase Price	$899,000
Tax Rate	$9.50

LOAN COST	
Down Payment %	20%
Interest Rate	3.00%
Term Length (yrs)	15
Loan Amount	$719,200
Est. Closing Costs	$17,980

MONTHLY EXPENSES	
Mortgage Costs	$4,967
Property Tax	$712
Utilities	$150
Home Insurance	$200
HOA Fees	$0

Interest Rate	Monthly Cost
Selected	$4,967
3.0%	
3.5%	
4.0%	
4.5%	
5.0%	
5.5%	
6.0%	
6.5%	
7.0%	
7.5%	
8.0%	
8.5%	
9.0%	
9.5%	
10.0%	



Interest Rate	Monthly Cost
Selected	$4,967
3.0%	$4,967
3.5%	$5,141
4.0%	$5,320
4.5%	$5,502
5.0%	$5,687
5.5%	$5,876
6.0%	$6,069
6.5%	$6,265
7.0%	$6,464
7.5%	$6,667
8.0%	$6,873
8.5%	$7,082
9.0%	$7,295
9.5%	$7,510
10.0%	$7,729

COMMON USE CASES:

- *Calculating a matrix of results based on combinations of input values (i.e. monthly payments based interest rates and down payments)*
- *Identifying the optimal outcome given multiple variable inputs*

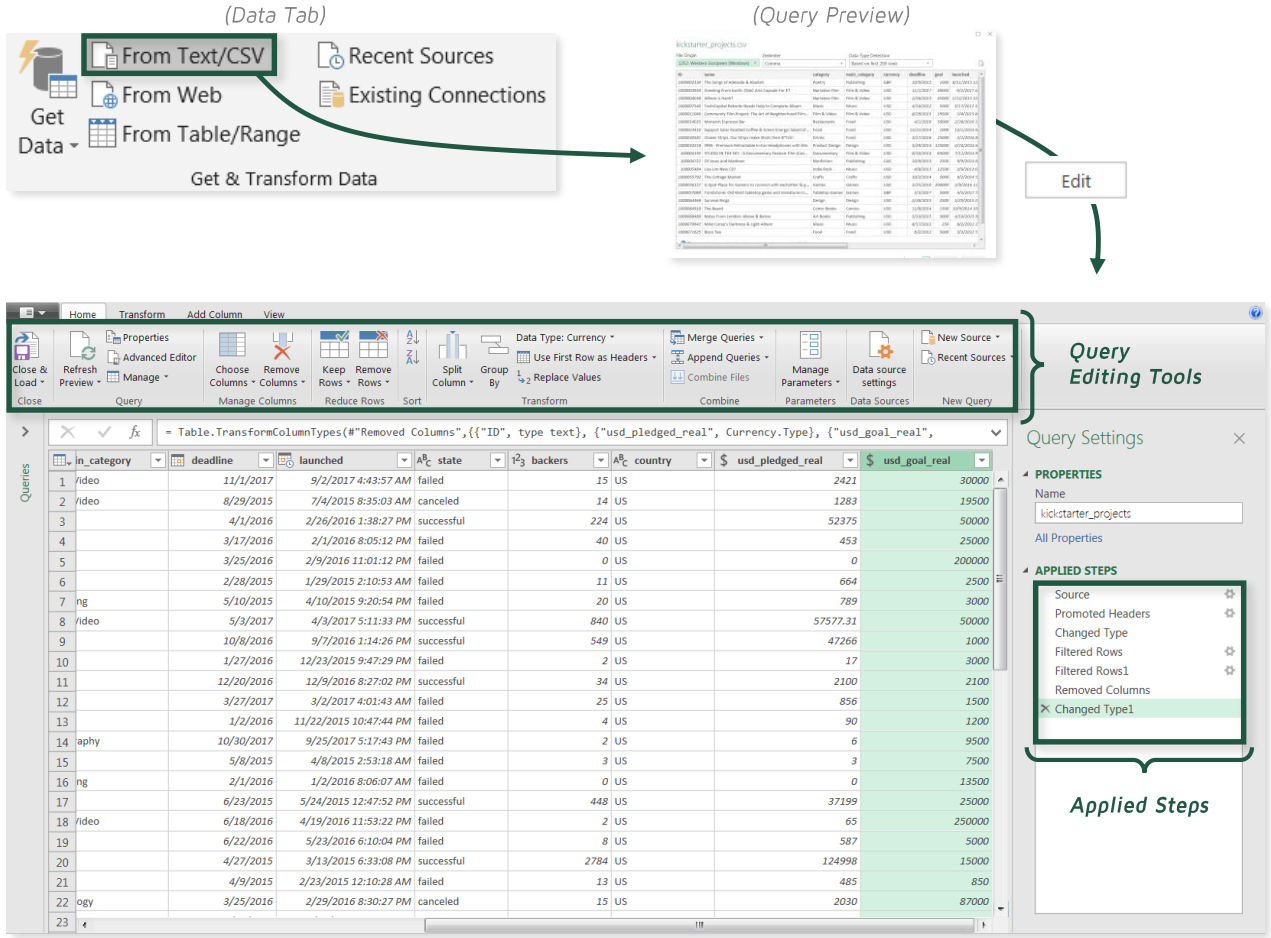
PRO TIP

CONNECT & SHAPE DATA WITH POWER QUERY

- **Power Query** (*aka "**Get Data**" or "**Get & Transform**"*) is used to connect, transform, and load data from external sources
  - All operations are saved with the query as "**Applied Steps**", and repeated each time the connection is refreshed (*like a macro*)
- Data can be loaded to worksheets or to the **Data Model**, where you can store much more data and build relational models to connect tables from multiple sources

ANALYTICS

4 STARS (ADVANCED)



COMMON USE CASES:

- *Connecting to flat files or database sources, and transforming or filtering the data before loading into Excel for further analysis*
- *Creating an automated ETL (**extract**, **transform**, **load**) process that can be refreshed as new data becomes available*

PRO TIP

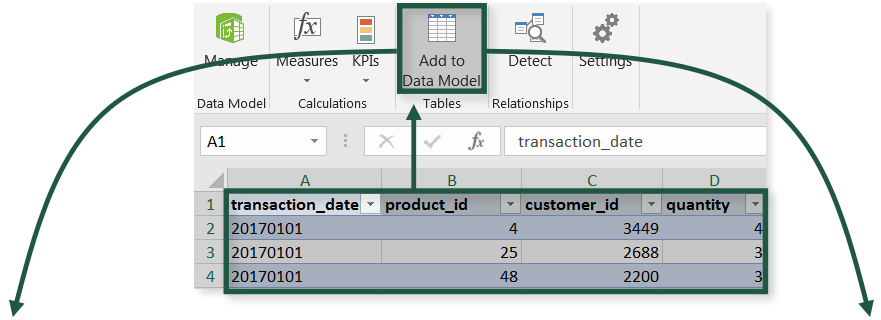
BUILD RELATIONAL DATA MODELS IN EXCEL

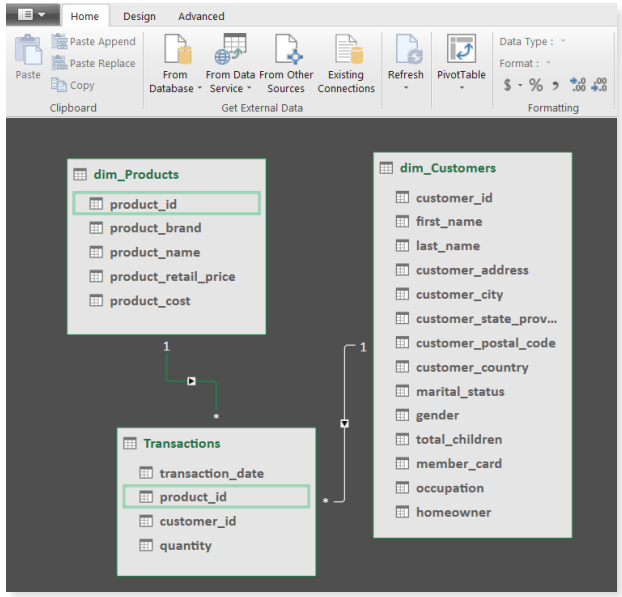
- Use Excel's **Data Model** to compress large amounts of data, create table relationships, and add calculated measures with DAX
  - Once table relationships are defined, use **Power Pivot** to explore and analyze data from multiple sources in a single view
- **NOTE**: Some versions of Excel may not have access to data modeling tools; if you don't see Power Pivot, check (***File > Options > Add-Ins > COM Add-Ins***)

ANALYTICS

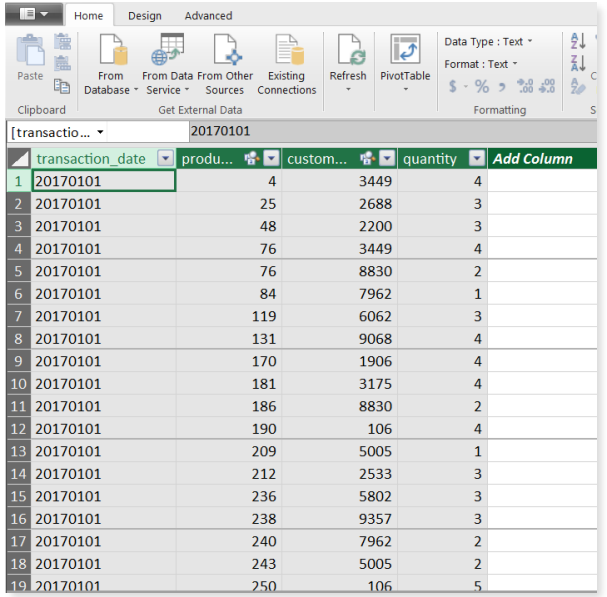
4 STARS (ADVANCED)

(Power Pivot Tab)





*(Data Model **Diagram View**)*



*(Data Model **Data View**)*

COMMON USE CASES:

- *Combining information from multiple sources without actually "stitching" the data together by merging or using LOOKUP/INDEX functions*
- *Building robust business intelligence solutions that integrate and blend data across many sources (sales, HR, finance, marketing, etc.)*

PRO TIP

EXPLORE DATA MODELS WITH CUBE FUNCTIONS

- Use **CUBE functions** (vs. Pivots) to pull values from a Data Model *directly into cells*:
  - **CUBESET:** A collection of items or members (*i.e. one column from a table in the model*)
  - **CUBEMEMBER:** A single item within a cubeset (*i.e. one item from a table column*)
  - **CUBERANKEDMEMBER**: A single item within a cubeset, based on an ordered rank
  - **CUBEVALUE:** An aggregated numerical value based on a set of member expressions
- **MORE INFO:** http://bit.ly/2PYgdoO

ANALYTICS

5 STARS (EXPERT)

CUBEMEMBERS *(shaded BLUE)*

CUBESET *(shaded ORANGE)*

CUBERANKEDMEMBERS *(shaded GREEN)*

CUBEVALUES *(shaded YELLOW)*

	A	B	C	D
1				
2			Sales by Member Type	
3			Transactions	Quantity
4		Golden	5,313	16,727
5		Silver	3,558	10,967
6		Bronze	23,305	71,167
7		Normal	9,776	30,040
8				
9			Sales by Product (*Top 5*)	
10	Rank	Brand:	Transactions	Quantity
11	2	Hermanos	1,334	4,163
12	3	Tell Tale	1,259	3,869
13	4	Ebony	1,181	3,563
14	5	Tri-State	1,109	3,419
15	6	High Top	1,118	3,393

COMMON USE CASES:

- *Building spreadsheet-based reports or dashboards from data in the Data Model without relying on PivotTables*
- *Documenting all of the sets and members within a data model*

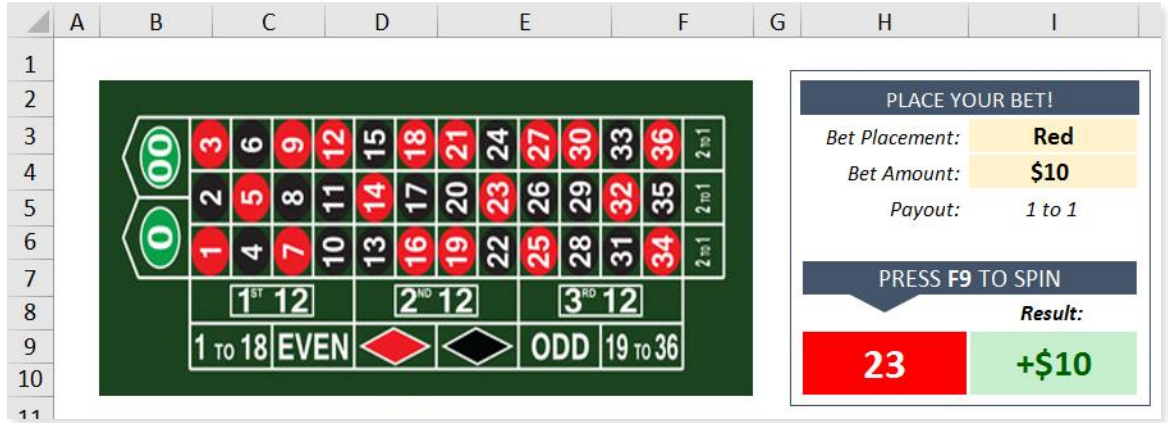
PRO TIP

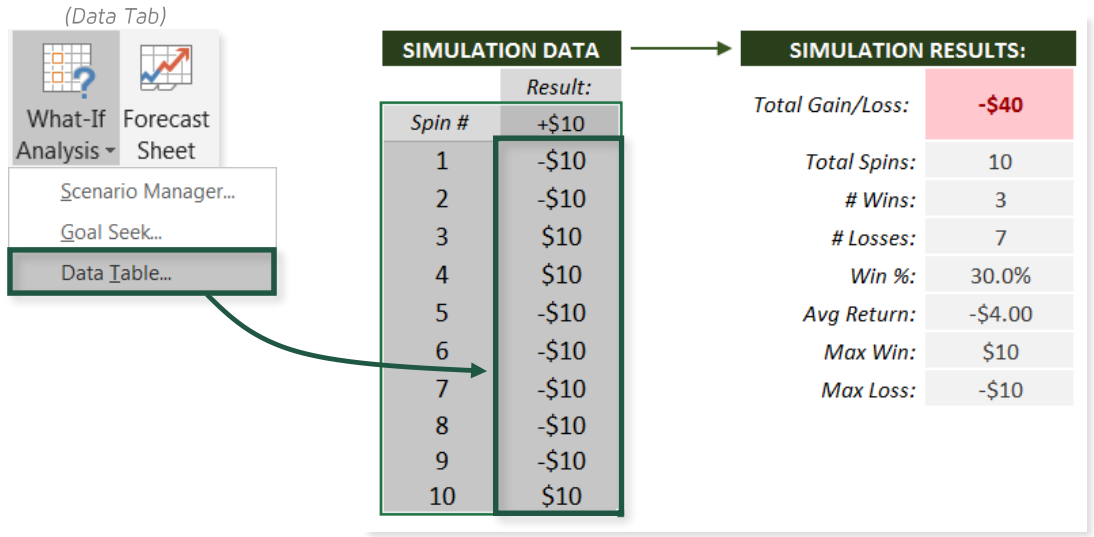
BUILD A MONTE CARLO SIMULATION MODEL

- **Monte Carlo simulation** is commonly used to predict the probability of given outcomes based on repeated random sampling
- In Excel, use **RAND** or **RANDBETWEEN** to randomize formula inputs, and **Data Tables** to generate an array of results
  - Based on the resulting array, you can use stats functions like **FREQUENCY** or **COUNTIF** to calculate outcome probabilities
  - **TIP:** Referencing a **blank cell** as the Data Table input allows each row to randomize

ANALYTICS

5 STARS (EXPERT)





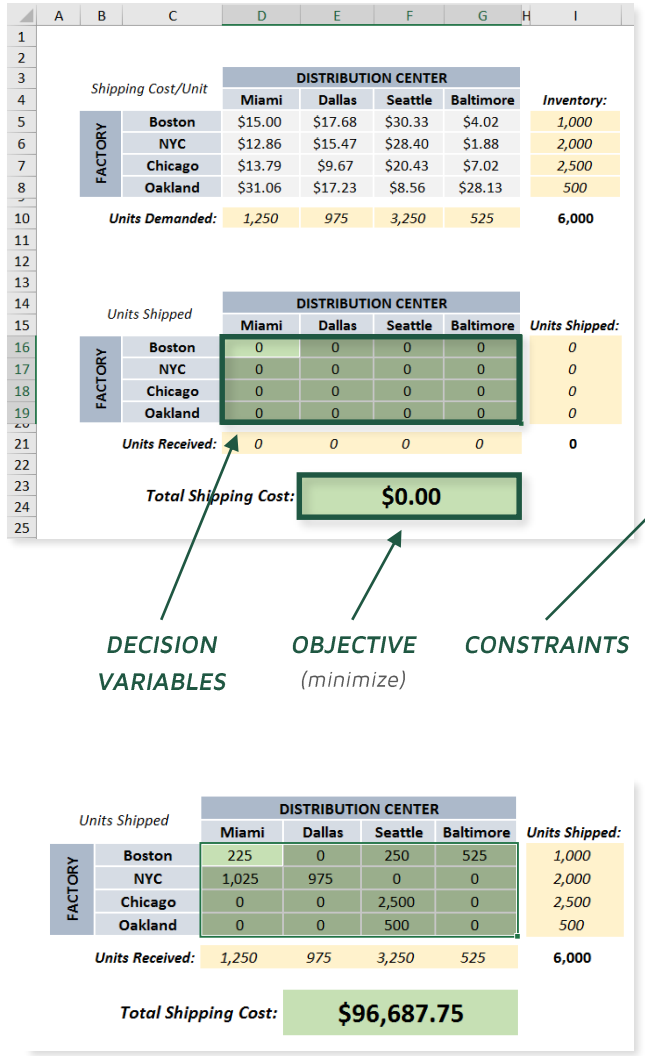
COMMON USE CASES:

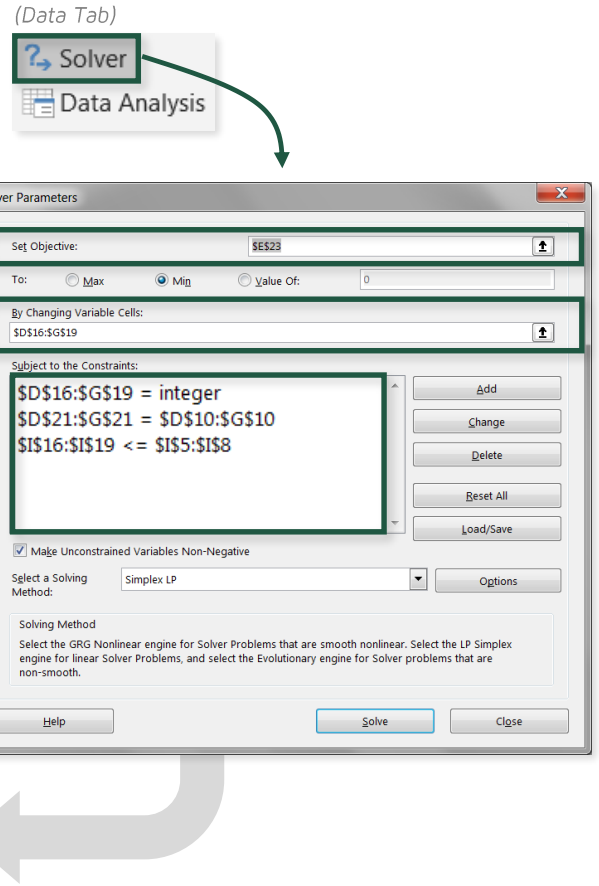
- *Randomly simulating a model thousands of times in order to understand the probability of certain outcomes (i.e. probability of a profit vs. loss)*
- *Building predictive models that account for a given degree of uncertainty for one of more inputs (i.e. future interest rates, supply costs, etc.)*

PRO TIP

OPTIMIZE COMPLEX MODELS WITH SOLVER

- Use **Solver** to solve complex optimization problems that require multiple decision variables and constraints
  - Unlike Goal Seek, **Solver** allows you to *minimize*, *maximize*, or *target* an objective value by changing multiple input cells, subject to given constraints
  - Select **Simplex LP** for linear optimizations, and **GRG** or **Evolutionary** for non-linear
- **NOTE**: To use Solver, activate the plug-in (***File > Options > Add-Ins > Excel Add-Ins***)





COMMON USE CASES:

- *Determining optimal results subject to real-world constraints (i.e. limited inventory, price floors, integer values, etc.)*

ANALYTICS

5 STARS (EXPERT)

PRO TIP

EXPLORE DATA WITH THE ANALYSIS TOOLPAK

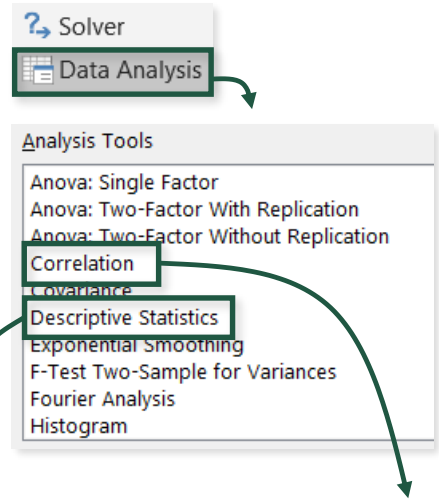
- **Analysis ToolPak** is a built-in Excel plug-in that supports a range of advanced **data analysis** and **statistical** methods (*Anova, Covariance, Regression, T-Tests, etc.*)
  - It can also be used to quickly generate key **descriptive statistics**, such as mean, mode, variance, range, count, skewness, etc.
- **NOTE**: You'll need to activate the plug-in (***File > Options > Add-Ins > Excel Add-Ins***)

ANALYTICS

5 STARS (EXPERT)

A	B	C	D
Athlete Name	**Age**	**Height (cm)**	**Weight (kg)**
Th Anh	20	165	58
Th Ngn Thng	23	147	47
Th Ngn Thng	19	147	47
A Lamusi	23	170	60
A. Joshua "Josh" West	31	207	105
Aadam Ismaeel Khamis	19	172	67
Aarik Wilson	25	191	88
Aarn Sarmiento Padilla	25	180	72
Aarn Sarmiento Padilla	21	180	72
Aaron Arthur Cook	25	183	80
Aaron Arthur Cook	17	183	80
Aaron Blunck	17	180	78
Aaron Brown	24	198	79
Aaron Brown	20	198	79
Aaron Feltham	26	190	94
Aaron Gate	25	181	71
Aaron Gate	21	181	71
Aaron J. "AJ" Bear	29	170	82
Aaron James Ramsey	21	178	70
Aaron James Scott	22	177	72

(Data Tab)



Age		*Height*		*Weight*	
Mean	26.17	Mean	176.621	Mean	72.29102
Standard Error	0.027019	Standard Error	0.054766	Standard Error	0.079224
Median	26	Median	176	Median	70
Mode	25	Mode	180	Mode	70
Standard Deviation	5.368748	Standard Deviation	10.88208	Standard Deviation	15.74192
Sample Variance	28.82346	Sample Variance	118.4196	Sample Variance	247.8081
Kurtosis	2.579227	Kurtosis	-0.07802	Kurtosis	1.991954
Skewness	1.019345	Skewness	0.168582	Skewness	0.954482
Range	59	Range	93	Range	186
Minimum	12	Minimum	133	Minimum	28
Maximum	71	Maximum	226	Maximum	214
Sum	1033244	Sum	6973350	Sum	2854194
Count	39482	Count	39482	Count	39482
Largest(1)	71	Largest(1)	226	Largest(1)	214
Smallest(1)	12	Smallest(1)	133	Smallest(1)	28
Confidence Level(95	0.052958	Confidence Level(95	0.107343	Confidence Level(95	0.155281

	Age	*Height*	*Weight*
Age	100%		
Height	**11%**	100%	
Weight	**18%**	**78%**	100%

COMMON USE CASES:

- *Quickly generating descriptive statistics without using cell formulas*
- *Exploring or analyzing data using more advanced statistical methods or tools (analyzing variance, building predictive models, etc.)*

RESOURCES & NEXT STEPS

RESOURCES & NEXT STEPS

1. Round out your skills with the full **Excel Maven** course stack:

   - Advanced Excel Formulas & Functions
   - Data Visualization with Excel Charts & Graphs
   - Data Analysis with Excel PivotTables
   - Intro to Power Query, Power Pivot & DAX

2. Check out **support.office.com** for resources & documentation:

   - **PC Shortcuts**: http://bit.ly/2Cx0gCa
   - **Mac Shortcuts**: http://bit.ly/2QwyqZQ
   - **Custom number formats**: http://bit.ly/2A1JFUD
   - **CUBE functions**: http://bit.ly/2PYgdo0
   - **Solver:** http://bit.ly/2JcZfQu
   - **Analysis ToolPak:** http://bit.ly/2NXT17D

3. Remember to **leave a review** if you enjoyed the course!

EXCEL MAVEN


THANK YOU